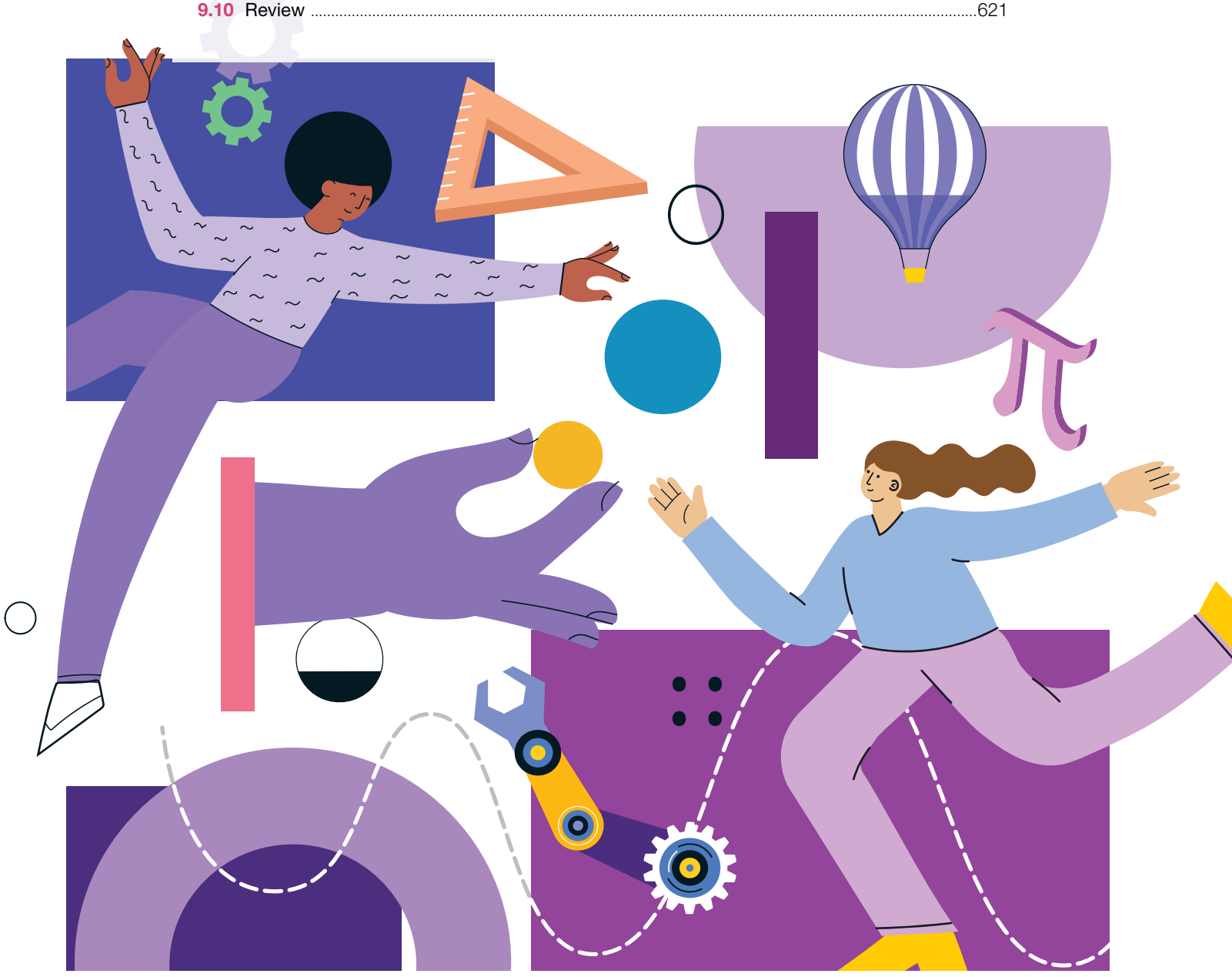
9 Non-linear relationships

LESSON SEQUENCE

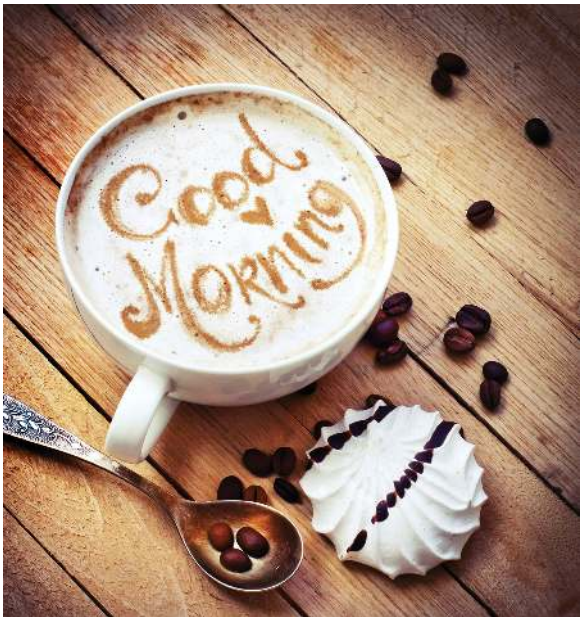
9.1 Overview 556
9.2 Plotting parabolas from a table of values 560
9.3 Sketching parabolas using transformations 569
9.4 Sketching parabolas using turning point form 580
9.5 Sketching parabolas in expanded form 589
9.6 Exponential graphs 596
9.7 Inverse proportion 604
9.8 Sketching the graph of a reciprocal 609
9.9 Sketching the circle 615
9.10 Review 621



LESSON
9.1 Overview

Why learn this?

So far, throughout high school, much of the focus of algebraic sketching has been on linear graphs. A linear graph is the graph of a straight line; therefore, a non-linear graph is any graph of a curve that is not straight. This means that non-linear graphing is a huge field of mathematics, encompassing many topics and areas of study.



If we think of the purpose of sketching graphs, to some extent it is to model relationships between real-life variables. Yet how often do we actually come up across a linear relationship? Throwing a ball through the air, the speed of a car as it accelerates from rest, the path of a runner around a track, the temperature of coffee as it cools — none of these relationships are linear. Even the path of light from a star in the night sky may not be a straight line through space to Earth, due to the curvature of space itself based on gravity!

Non-linear graphs, whether they be parabolas, hyperbolas, exponentials or circles, are some of the more common types of graphs that are used to model phenomena in everyday life. Thus, it is important to study these relationships and their graphs so we can use them to help us model concepts such as exponential growth of a colony of bacteria, or an inversely proportional relationship such as the decay of radioactive material over time.

Hey students! Bring these pages to life online

Watch videos

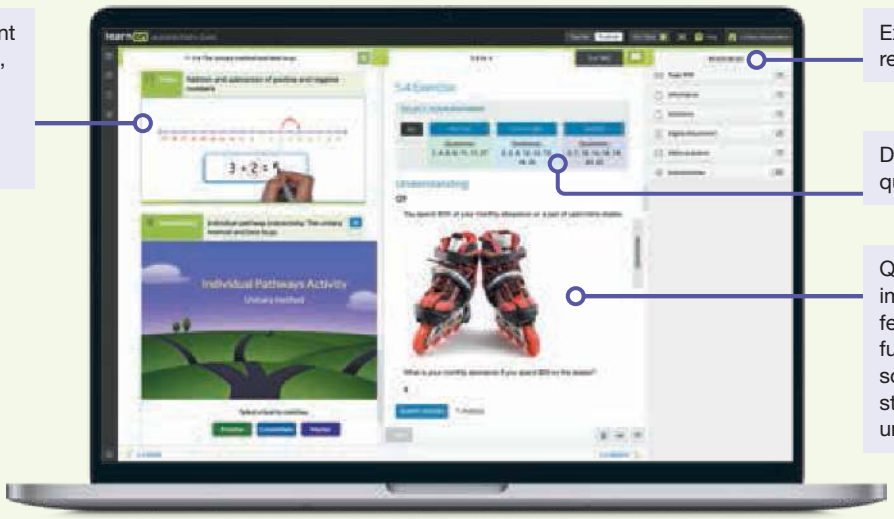
Engage with interactivities

Answer questions and check solutions

Find all this and MORE in jacPLUS



Reading content and rich media, including interactivities and videos for every concept



Extra learning resources

Differentiated question sets

Questions with immediate feedback, and fully worked solutions to help students get unstuck

Exercise 9.1 Pre-test

learnon

1. Complete the table of values for the equation $y = \frac{1}{2}(x-2)^2 - 3$.

x	-3	-2	-1	0
y				

2. For the graph of $y = -3(x+1)^2 - 2$, state the equation of the axis of symmetry.

3. MC For the graph of $y = 2(x+5)^2 + 8$, the turning point is:
   A. $(-5, -8)$
   B. $(-5, 8)$
   C. $(5, 8)$
   D. $(8, 5)$

4. MC The graph of $y = -(x-1)^2 - 3$ has:
   A. a maximum turning point
   B. a minimum turning point
   C. no turning point
   D. two maximum turning points.

5. MC The x-intercepts of $y = 2(x-2)^2 - 8$ are:
   A. $x = 2$ or $x = 8$
   B. $x = 0$ or $x = 8$
   C. $x = 10$ or $x = -6$
   D. $x = 0$ or $x = 4$

6. Calculate the coordinates of the turning point for the graph of $y = (x+3)(x+5)$.
   Write your answer in the form (a, b).

7. The equation $y = 2x^2 + bx - 1200$ has x-intercepts of $(-30, 0)$ and $(20, 0)$. Determine the value of b.

8. MC The radius r, of the circle $4x^2 + 4y^2 = 16$ is:
   A. $r = 16$
   B. $r = 8$
   C. $r = 4$
   D. $r = 2$

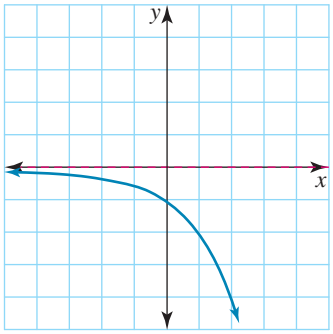
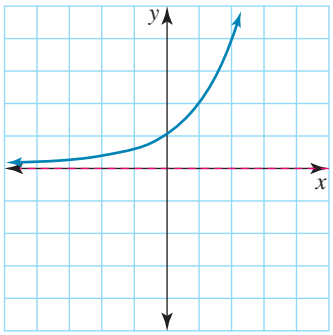
9. MC The center of the circle with equation $x^2 + 4x + y^2 - 6y + 9 = 0$ is:
   A. $(-4, -3)$
   B. $(4, -6)$
   C. $(-4, 6)$
   D. $(-2, 3)$

10. Calculate the points of intersection between the parabola $y = x^2$ and the circle $x^2 + y^2 = 1$, correct to two decimal places.

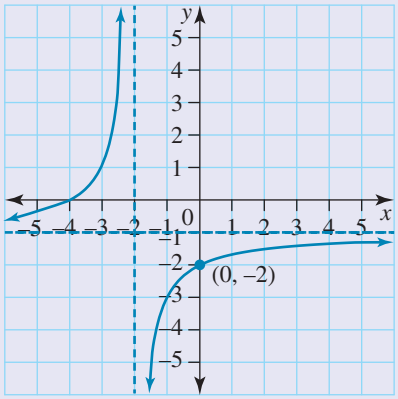
11. MC The horizontal asymptote for $y = 3^{-x} + 1$ is:
    A. $y = 0$
    B. $y = 1$
    C. $x = -1$
    D. $x = 0$

12. MC For the equation of a hyperbola $y = \frac{2}{x+3} - 1$, the vertical and horizontal asymptotes are:
    A. $x = \frac{2}{3}$ and $y = -1$
    B. $x = 3$ and $y = -1$
    C. $x = -3$ and $y = -1$
    D. $x = -1$ and $y = -3$

13. Match each graph with its correct equation.

Equation	Graph
a. $y = 2^x$	**1.** 
b. -2^x	**2.** 

14. MC From the graph of the hyperbola, the equation is:



A. $y = \dfrac{1}{x+2} - 1$

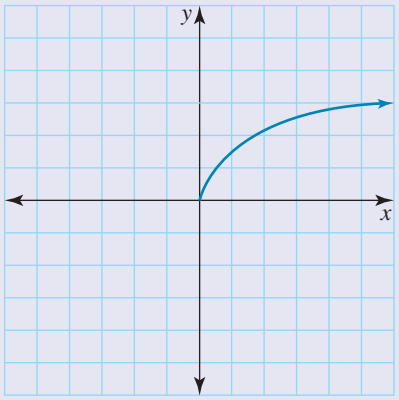
B. $y = \dfrac{-1}{x+2} - 1$

C. $y = \dfrac{2}{x+2} - 1$

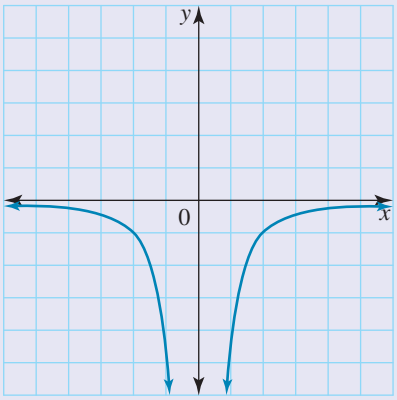
D. $y = \dfrac{-2}{x+2} - 1$

15. MC The graph of the truncus with equation $y = \frac{1}{x^2}$ reflected in the x-axis looks like:

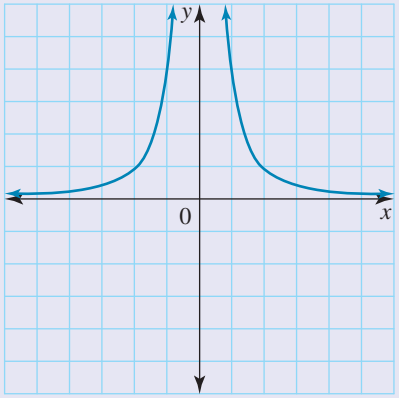
A.



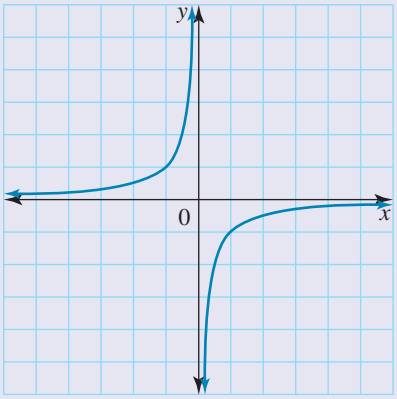
B.



C.



D.



LESSON
9.2 Plotting parabolas from a table of values

LEARNING INTENTIONS

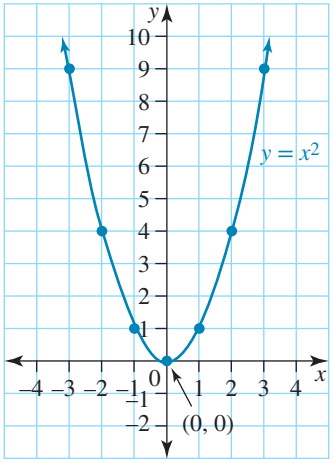
At the end of this lesson you should be able to:

- create a table of values and use this to sketch the graph of a parabola
- identify the axis of symmetry, turning point and y-intercept of a parabola.

9.2.1 Plotting parabolas

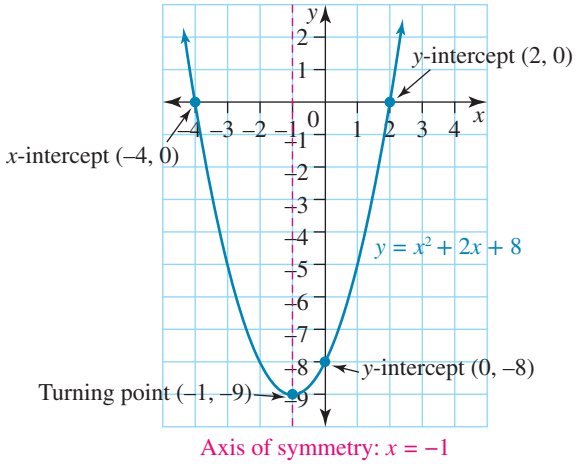
eles-4864

- The graphs of all quadratic relationships are called **parabolas**.
- If the equation of the parabola is given, a table of values can be produced by substituting x-values into the equation to obtain the corresponding y-values. These x-and y-values provide the coordinates for points that can be plotted and joined to form the shape of the graph.
- The graph of $y = x^2$ shown has been produced by generating a table of values.



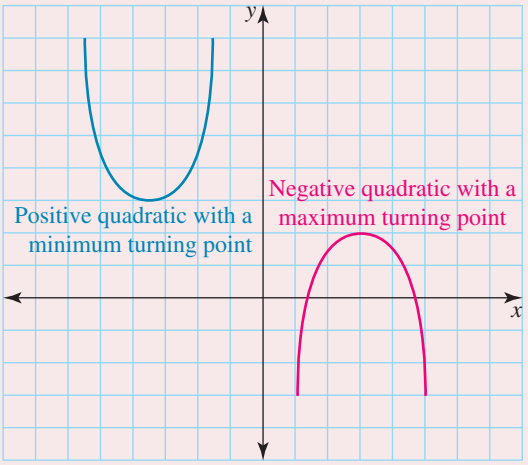
x	−3	−2	−1	0	1	2	3
y	9	4	1	0	1	4	9

- Parabolas are symmetrical; in other words, they have an **axis of symmetry**. In the parabola shown the axis of symmetry is the y-axis, also called the line $x = 0$.
- A parabola has a **vertex** or **turning point**. In this case the vertex is at the origin and is called a 'minimum turning point'.
- The ***y*-intercept** of a quadratic is the coordinate where the parabola cuts the y-axis. This can be found from a table of value by looking for the point where $x = 0$.
- The ***x*-intercept** of a quadratic is the coordinate where the parabola cuts the x-axis. This can be found from a table of value by looking for the point where $y = 0$.
- Consider the key features of the equation $y = x^2 + 2x + 8$.



Shapes of parabolas

- **Parabolas with the shape ∪ are said to be 'concave up' and have a minimum turning point.**
- **Parabolas with the shape ∩ are said to be 'concave down' and have a maximum turning point.**



Parabolas in the world around us

- Parabolas abound in the world around us. Here are some examples.



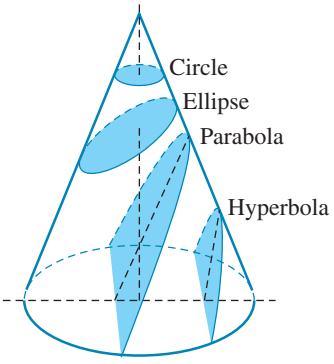
Satellite dishes



Water droplets from a hose



The cables from a suspension bridge



A cone when sliced parallel to its edge reveals a parabola.

WORKED EXAMPLE 1 Plotting parabolas using a table of values

Plot the graph of each of the following equations. In each case, use the values of x shown as the values in your table. State the equation of the axis of symmetry and the coordinates of the turning point.

a. $y = 2x^2$ **for** $-3 \leq x \leq 3$

b. $y = \frac{1}{2}x^2$ **for** $-3 \leq x \leq 3$

THINK

a. 1. Write the equation.

2. Produce a table of values using x-values from -3 to 3.

3. Draw a set of clearly labelled axes, plot the points and join them with a smooth curve. The scale would be from -2 to 20 on the y-axis and -4 to 4 on the x-axis.

4. Label the graph.

5. Write the equation of the axis of symmetry that divides the parabola exactly in half.

6. Write the coordinates of the turning point.

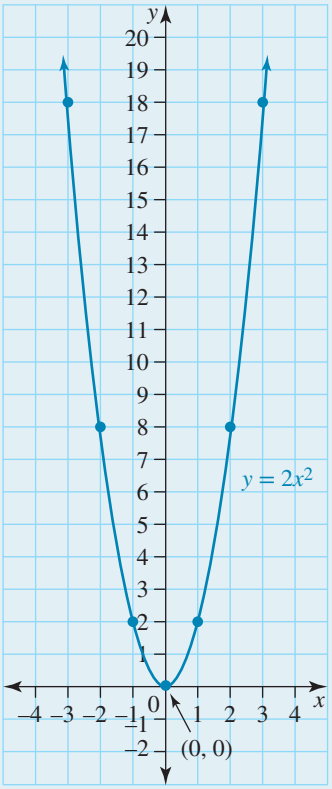
b. 1. Write the equation.

2. Produce a table of values using x-values from -3 to 3.

WRITE/DRAW

a. $y = 2x^2$

x	-3	-2	-1	0	1	2	3
y	18	8	2	0	2	8	18



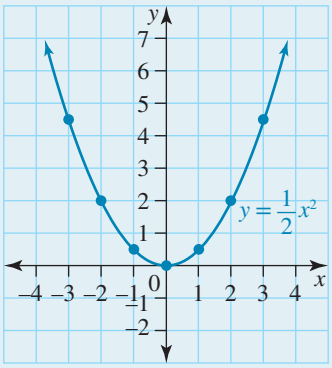
The equation of the axis of symmetry is $x = 0$.

The turning point is $(0, 0)$.

b. $y = \frac{1}{2}x^2$

x	-3	-2	-1	0	1	2	3
y	4.5	2	0.5	0	0.5	2	4.5

3. Draw a set of clearly labelled axes, plot the points and join them with a smooth curve. The scale would be from −2 to 6 on the *y*-axis and −4 to 4 on the *x*-axis.
4. Label the graph.



5. Write the equation of the line that divides the parabola exactly in half.

The equation of the axis of symmetry is $x = 0$.

6. Write the coordinates of the turning point.

The turning point is $(0, 0)$.

WORKED EXAMPLE 2 Determining the key features of a quadratic equation

Plot the graph of each of the following equations. In each case, use the values of *x* shown as the values in your table. State the equation of the axis of symmetry, the coordinates of the turning point and the *y*-intercept for each one.

a. $y = x^2 + 2$ **for** $-3 \leq x \leq 3$

b. $y = (x + 3)^2$ **for** $-6 \leq x \leq 0$

c. $y = -x^2$ **for** $-3 \leq x \leq 3$

THINK

a. 1. Write the equation.

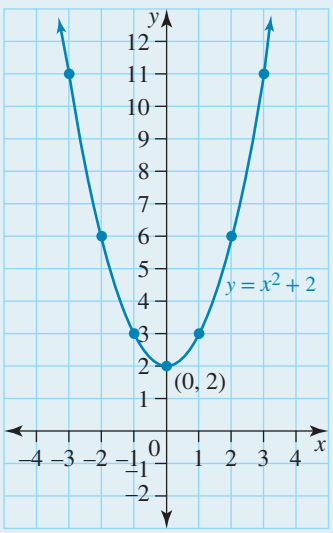
2. Produce a table of values.

WRITE/DRAW

a. $y = x^2 + 2$

x	−3	−2	−1	0	1	2	3
y	11	6	3	2	3	6	11

3. Draw a set of clearly labelled axes, plot the points and join them with a smooth curve. The scale on the *y*-axis would be from −2 to 12 and −4 to 4 on the *x*-axis.
4. Label the graph.



5. Write the equation of the line that divides the parabola exactly in half.

The equation of the axis of symmetry is $x = 0$.

6. Write the coordinates of the turning point.

The turning point is $(0, 2)$.

7. Determine the y-coordinate of the point where the graph crosses the y-axis.

The y-intercept is 2.

b. 1. Write the equation.

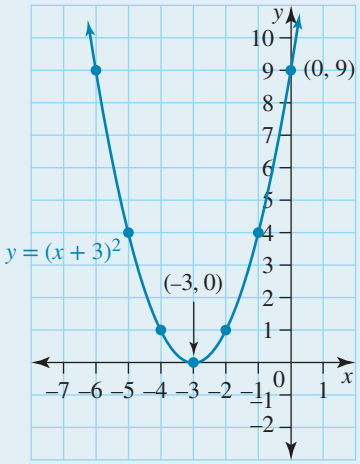
b. $y = (x + 3)^2$

2. Produce a table of values.

x	−6	−5	−4	−3	−2	−1	0
y	9	4	1	0	1	4	9

3. Draw a set of clearly labelled axes, plot the points and join them with a smooth curve. The scale on the y-axis would be from −2 to 10 and −7 to 1 on the x-axis.

4. Label the graph.



5. Write the equation of the line that divides the parabola exactly in half.

The equation of the axis of symmetry is $x = -3$.

6. Write the coordinates of the turning point.

The turning point is $(-3, 0)$.

7. Determine the y-coordinate of the point where the graph crosses the y-axis.

The y-intercept is 9.

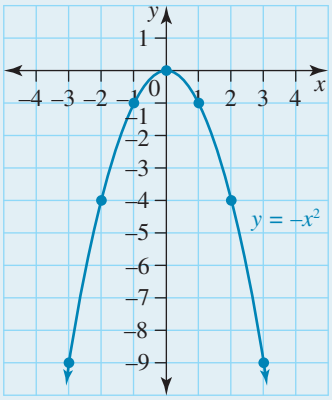
c. 1. Write the equation.

c. $y = -x^2$

2. Produce a table of values.

x	−3	−2	−1	0	1	2	3
y	−9	−4	−1	0	−1	−4	−9

3. Draw a set of clearly labelled axes, plot the points and join them with a smooth curve. The scale on the y-axis would be from -10 to 1 and from -4 to 4 on the x-axis.
4. Label the graph.



5. Write the equation of the line that divides the parabola exactly in half.

The equation of the axis of symmetry is $x = 0$.

6. Write the coordinates of the turning point.

The turning point is $(0, 0)$.

7. Determine the y-coordinate of the point where the graph crosses the y-axis.

The y-intercept is 0.

DISCUSSION

What x-values can a parabola have? What y-values can a parabola have?

on Resources

eWorkbook Topic 9 Workbook (worksheets, code puzzle and a project) (ewbk-2035)

Interactivities Individual pathway interactivity: Plotting parabolas (int-4605)
Plotting quadratic graphs (int-6150)
Parabolas in the world around us (int-7539)

Exercise 9.2 Plotting parabolas from a table of values

learnon

9.2 Quick quiz on | 9.2 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 3, 5, 11, 13, 15, 19, 22, 26, 29	2, 6, 8, 10, 16, 17, 20, 23, 27, 30	4, 7, 9, 12, 14, 18, 21, 24, 25, 28, 31

Fluency

1. WE1 Plot the graph of each of the following equations. In each case, use the values of x shown as the values in your table. State the equation of the axis of symmetry and the coordinates of the turning point.

   a. $y = 3x^2$ for $-3 \leq x \leq 3$

   b. $y = \frac{1}{4}x^2$ for $-3 \leq x \leq 3$

2. Compare the graphs drawn for question 1 with that of $y = x^2$. Explain how placing a number in front of x^2 affects the graph obtained.

3. WE2a Plot the graph of each of the following for values of x between -3 and 3. State the equation of the axis of symmetry, the coordinates of the turning point and the y-intercept for each one.

   a. $y = x^2 + 1$ b. $y = x^2 + 3$ c. $y = x^2 - 3$ d. $y = x^2 - 1$

4. Compare the graphs drawn for question 3 with the graph of $y = x^2$. Explain how adding to or subtracting from x^2 affects the graph obtained.

WE2b For questions 5 to 8, plot the graph of each of the following equations. In each case, use the values of x shown as the values in your table. State the equation of the axis of symmetry and the coordinates of the turning point and the y-intercept for each one.

5. $y = (x + 1)^2$ $-5 \leq x \leq 3$
6. $y = (x - 2)^2$ $-1 \leq x \leq 5$
7. $y = (x - 1)^2$ $-2 \leq x \leq 4$
8. $y = (x + 2)^2$ $-6 \leq x \leq 2$
9. Compare the graphs drawn for questions 5 to 8 with that for $y = x^2$. Explain how adding to or subtracting from x before squaring affects the graph obtained.

WE2c For questions 10 to 13, plot the graph of each of the following equations. In each case, use the values of x shown as the values in your table. State the equation of the axis of symmetry, the coordinates of the turning point and the y-intercept for each one.

10. $y = -x^2 + 1$ $-3 \leq x \leq 3$
11. $y = -(x + 2)^2$ $-5 \leq x \leq 1$
12. $y = -x^2 - 3$ $-3 \leq x \leq 3$
13. $y = -(x - 1)^2$ $-2 \leq x \leq 4$
14. Compare the graphs drawn for questions 10 to 13 with that for $y = x^2$. Explain how a negative sign in front of x^2 affects the graph obtained. Also compare the graphs obtained in questions 10 to 13 with those in questions 3 and 5 to 8. State which graphs have the same turning point. Describe how are they different.

Understanding

For questions **15** to **20**:

a. plot the graph
b. state the equation of the axis of symmetry
c. state the coordinates of the turning point and whether it is a maximum or a minimum
d. state the y-intercept.

15. $y = (x-5)^2 + 1 \qquad 0 \le x \le 6$

16. $y = 2(x+2)^2 - 3 \qquad -5 \le x \le 1$

17. $y = -(x-3)^2 + 4 \qquad 0 \le x \le 6$

18. $y = -3(x-1)^2 + 2 \qquad -2 \le x \le 4$

19. $y = x^2 + 4x - 5 \qquad -6 \le x \le 2$

20. $y = -3x^2 - 6x + 24 \qquad -5 \le x \le 3$

21. Use the equation $y = a(x-b)^2 + c$ to answer the following.

a. Explain how you can determine whether a parabola has a minimum or maximum turning point by looking only at its equation.
b. Explain how you can determine the coordinates of the turning point of a parabola by looking only at the equation.
c. Explain how you can obtain the equation of the axis of symmetry by looking only at the equation of the parabola.

22. MC For the graph of $y = (x-2)^2 + 5$, the turning point is:

A. $(5, 2)$ **B.** $(2, -5)$ **C.** $(2, 5)$ **D.** $(-2, -5)$

23. MC For the graph of $y = 3(x-1)^2 + 12$, the turning point is:

A. $(3, 12)$ **B.** $(1, 12)$ **C.** $(-1, 12)$ **D.** $(-3, 12)$

24. MC For the graph of $y = (x+2)^2 - 7$, the y-intercept is:

A. -2 **B.** -7 **C.** -3 **D.** -11

25. MC Select which of the following is true for the graph of $y = -(x-3)^2 + 4$.

A. Turning point $(3, 4)$, y-intercept -5
B. Turning point $(3, 4)$, y-intercept 5
C. Turning point $(-3, 4)$, y-intercept -5
D. Turning point $(-3, 4)$, y-intercept 5

Reasoning

26. A ball is thrown into the air. The height, h metres, of the ball at any time, t seconds, can be found by using the equation $h = -(t-4)^2 + 16$.

a. Plot the graph for values of t between 0 and 8.
b. Use the graph to determine:
i. the maximum height of the ball
ii. how long it takes for the ball to fall back to the ground from the moment it is thrown.



27. From a crouching position in a ditch, an archer wants to fire an arrow over a horizontal tree branch, which is 15 metres above the ground. The height, in metres (h), of the arrow t seconds after it has been fired is given by the equation $h = -8t(t-3)$.



a. Plot the graph for $t = 0, 1, 1.5, 2, 3$.

b. From the graph, determine:

i. the maximum height the arrow reaches

ii. whether the arrow clears the branch and the distance by which it clears or falls short of the branch

iii. the time it takes to reach maximum height

iv. how long it takes for the arrow to hit the ground after it has been fired.

28. There are 0, 1, 2 and infinite possible points of intersection for two parabolas.

a. Illustrate these on separate graphs.

b. Explain why infinite points of intersection are possible. Give an example.

c. Determine how many points of intersection are possible for a parabola and a straight line. Illustrate these.

Problem solving

29. The area of a rectangle in cm^2 is given by the equation $A = \frac{-8w}{5}(w-6)$, where w is the width of the rectangle in centimetres.

a. Complete a table of values for $-1 \leq w \leq 7$.

b. Explain which of the values for w from part **a** should be discarded and why.

c. Sketch the graph of A for suitable values of w.

d. Evaluate the maximum possible area of the rectangle. Show your working.

e. Determine the dimensions of the rectangle that produce the maximum area found in part **d**.

30. The path taken by a netball thrown by a rising Australian player is given by the quadratic equation $y = -x^2 + 3.2x + 1.8$, where y is the height of the ball and x is the horizontal distance from the player's upstretched hand.

a. Complete a table of values for $-1 \leq x \leq 4$.

b. Plot the graph.

c. Explain what values of x are 'not reasonable'.

d. Evaluate the maximum height reached by the netball.

e. Assuming that nothing hits the netball, determine how far away from the player the netball will strike the ground.

31. The values of a, b and c in the equation $y = ax^2 + bx + c$ can be calculated using three points that lie on the parabola. This requires solving triple simultaneous equations by algebra. This can also be done using a CAS calculator. If the points $(0, 1)$, $(1, 0)$ and $(2, 3)$ all lie on one parabola, determine the equation of the parabola.

LESSON
9.3 Sketching parabolas using transformations

LEARNING INTENTIONS

At the end of this lesson you should be able to:

- be familiar with the key features of the basic graph of a quadratic, $y = x^2$
- determine whether a dilation has made the graph of quadratic narrower or wider
- determine whether a translation has moved the graph of a quadratic function left/right or up/down
- sketch the graph of a quadratic equation that has undergone any of the following transformations: dilations, a reflection, or translations.

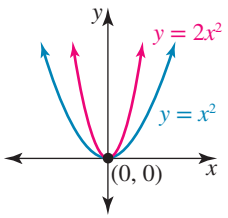
9.3.1 Sketching parabolas

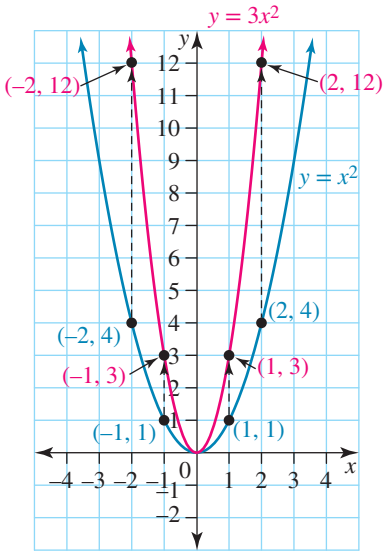
eles-4865

- A sketch graph of a parabola does not show a series of plotted points, but it does accurately locate important features such as x- and y-intercepts and turning points.
- The basic quadratic graph has the equation $y = x^2$. Transformations or changes in the features of the graph can be observed when the equation changes. These transformations include:
  - dilation
  - translation
  - reflection.

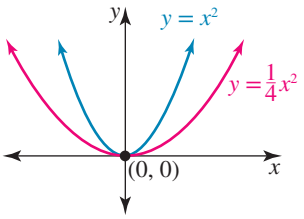
Dilation

- A dilation stretches a graph away from an axis. A dilation of factor 3 from the x-axis triples the distance of each point from the x-axis. This means the point $(2, 4)$ would become $(2, 12)$.
- Compare the graph of $y = 2x^2$ with that of $y = x^2$. This graph is thinner or closer to the y-axis and has a dilation factor of 2 from the x-axis. As the magnitude (or size) of the coefficient of x^2 increases, the graph becomes **narrower** and closer to the y-axis.





- The turning point has not changed under the transformation and is still $(0, 0)$.
- Compare the graph $y = \frac{1}{4}x^2$ with that of $y = x^2$.



The graph is **wider** or closer to the x-axis and has a dilation factor of factor $\frac{1}{4}$.

The turning point has not changed and is still $(0, 0)$. As the coefficient of x^2 decreases (but remains positive), the graph becomes wider or closer to the x-axis.

WORKED EXAMPLE 3 Determining dilation and the turning point

State whether each of the following graphs is wider or narrower than the graph of $y = x^2$ and state the coordinates of the turning point of each one.

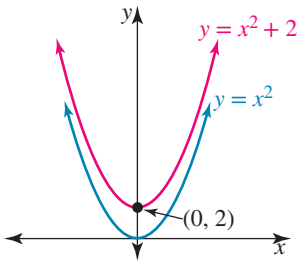
a. $y = \frac{1}{5}x^2$ **b.** $y = 4x^2$

THINK	WRITE
a. 1. Write the equation.	**a.** $y = \frac{1}{5}x^2$
2. Look at the coefficient of x^2 and decide whether it is greater than or less than 1.	$\frac{1}{5} < 1$, so the graph is wider than that of $y = x^2$.
3. The dilation doesn't change the turning point.	The turning point is $(0, 0)$.
b. 1. Write the equation.	**b.** $y = 4x^2$
2. Look at the coefficient of x^2 and decide whether it is greater than or less than 1.	$4 > 1$, so the graph is narrower than that of $y = x^2$.
3. The dilation doesn't change the turning point.	The turning point is $(0, 0)$.

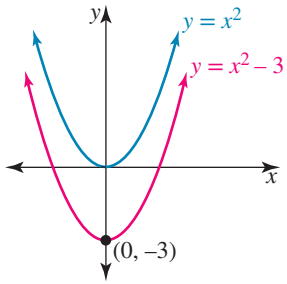
9.3.2 Vertical translation

eles-4866

- Compare the graph of $y = x^2 + 2$ with that of $y = x^2$.
  The whole graph has been moved or translated 2 units upwards. The turning point has become $(0, 2)$.



- Compare the graph of $y = x^2 - 3$ with that of $y = x^2$.
  The whole graph has been moved or translated 3 units downwards. The turning point has become $(0, -3)$.



WORKED EXAMPLE 4 Determining the vertical translation and the turning point

State the vertical translation and the coordinates of the turning point for the graphs of the following equations when compared to the graph of $y = x^2$.

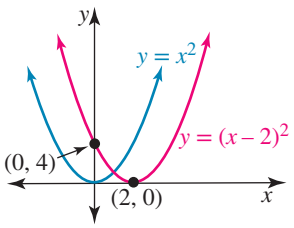
a. $y = x^2 + 5$ **b.** $y = x^2 - 4$

THINK	WRITE
a. 1. Write the equation.	**a.** $y = x^2 + 5$
2. +5 means the graph is translated upwards 5 units.	Vertical translation of 5 units up.
3. Translate the turning point of $y = x^2$ which is $(0, 0)$. The x-coordinate of the turning point remains 0, and the y-coordinate has 5 added to it.	The turning point becomes $(0, 5)$.
b. 1. Write the equation.	**b.** $y = x^2 - 4$
2. −4 means the graph is translated downwards 4 units.	Vertical translation of 4 units down.
3. Translate the turning point of $y = x^2$, which is $(0, 0)$. The x-coordinate of the turning point remains 0, and the y-coordinate has 4 subtracted from it.	The turning point becomes $(0, -4)$.

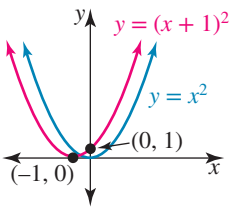
9.3.3 Horizontal translation

eles-4867

- Compare the graph of $y = (x - 2)^2$ with that of $y = x^2$.
  The whole graph has been moved or translated 2 units to the right. The turning point has become $(2, 0)$.



- Compare the graph of $y = (x + 1)^2$ with that of $y = x^2$.
  The whole graph has been moved or translated 1 unit left. The turning point has become $(-1, 0)$.



- *Note:* Horizontal translations appear to cause the graph to move in the opposite direction to the sign inside the brackets.

WORKED EXAMPLE 5 Determining the horizontal translation and the turning point

State the horizontal translation and the coordinates of the turning point for the graphs of the following equations when compared to the graph of $y = x^2$.

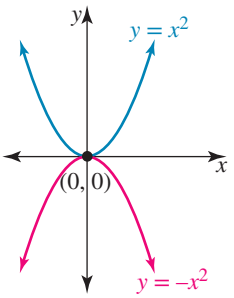
a. $y = (x - 3)^2$ **b.** $y = (x + 2)^2$

THINK	WRITE
a. 1. Write the equation.	**a.** $y = (x - 3)^2$
2. -3 means the graph is translated to the right 3 units.	Horizontal translation of 3 units to the right
3. Translate the turning point of $y = x^2$, which is $(0, 0)$. The y-coordinate of the turning point remains 0, and the x-coordinate has 3 added to it.	The turning point becomes $(3, 0)$.
b. 1. Write the equation.	**b.** $y = (x + 2)^2$
2. $+2$ means the graph is translated to the left 2 units.	Horizontal translation of 2 units to the left
3. Translate the turning point of $y = x^2$, which is $(0, 0)$. The y-coordinate of the turning point remains 0, and the x-coordinate has 2 subtracted from it.	The turning point becomes $(-2, 0)$.

9.3.4 Reflection

eles-4868

- Compare the graph of $y = -x^2$ with that of $y = x^2$.



In each case the axis of symmetry is the line $x = 0$ and the turning point is $(0, 0)$. The only difference between the equations is the negative sign in $y = -x^2$, and the difference between the graphs is that $y = x^2$ 'sits' on the x-axis and $y = -x^2$ 'hangs' from the x-axis. (One is a reflection or mirror image of the other.) The graph of $y = x^2$ has a minimum turning point, and the graph of $y = -x^2$ has a maximum turning point.

Shapes of quadratic graphs

Any quadratic graph where x^2 is positive has a ∪ shape and is said to be upright. Conversely, if x^2 is negative the graph has a ∩ shape and is said to be inverted.

WORKED EXAMPLE 6 Identifying the coordinates of the turning point

For each of the following graphs, identify the coordinates of the turning point and state whether it is a maximum or a minimum.

a. $y = -(x-7)^2$ **b.** $y = 5 - x^2$

THINK	WRITE
a. 1. Write the equation.	**a.** $y = -(x-7)^2$
2. It is a horizontal translation of 7 units to the right, so 7 units is added to the x-coordinate of $(0, 0)$.	The turning point is $(7, 0)$.
3. The sign in front of the x^2 term is negative, so it is inverted.	Maximum turning point.
b. 1. Write the equation.	**b.** $y = 5 - x^2$
2. Rewrite the equation so that the x^2 term is first.	$y = -x^2 + 5$
3. The vertical translation is 5 units up, so 5 units is added to the y-coordinate of $(0, 0)$.	The turning point is $(0, 5)$.
4. The sign in front of the x^2 term is negative, so the graph is inverted.	Maximum turning point.

TI | THINK

a.

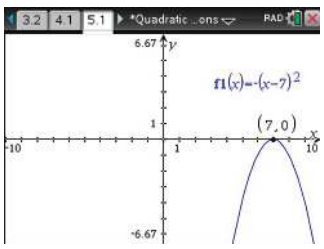
In a new problem, on a Graphs page, complete the function entry line as:
$f1(x) = -(x-7)^2$
Then press ENTER.
To locate the turning point, press:

- MENU
- 6: Analyze Graph
- 3: Maximum

Drag the dotted line to the left of the turning point (the lower bound), click ENTER and then drag the dotted line to the right of the turning point (the upper bound) and press ENTER.
The turning point will be shown.

DISPLAY/WRITE

a.



The turning point $(7, 0)$ is a maximum.

CASIO | THINK

a.

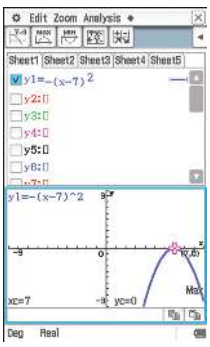
On the Graph & Table screen, complete the function entry line as:
$y1 = -(x-7)^2$
Tap the graphing icon and the graph will be displayed.
To locate the turning point, tap:

- Analysis
- G-Solve
- Max

The turning point will be shown.

DISPLAY/WRITE

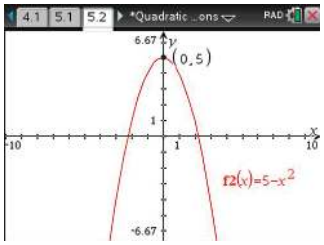
a.



The turning point $(7, 0)$ is a maximum.

b.
In a new problem, on a new Graphs page, complete the function entry line as:
$f1(x) = 5 - x^2$
Then press ENTER.
To locate the turning point, press:
- MENU
- 6: Analyze Graph
- 3: Maximum

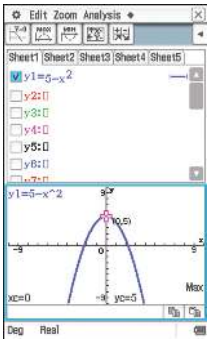
Drag the dotted line to the left of the turning point (the lower bound), click ENTER and then drag the dotted line to the right of the turning point (the upper bound) and press ENTER.
The turning point will be shown.

b.


The turning point $(0, 5)$ is a maximum.

b.
On the Graph & Table screen, complete the function entry line as:
$y1 = 5 - x^2$
Tap the graphing icon and the graph will be displayed.
To locate the turning point, press:
- Analysis
- G-Solve
- Max

The turning point will be shown.

b.


The turning point $(0, 5)$ is a maximum.

9.3.5 Combining transformations

eles-4869

- Often, multiple transformations will be applied to the equation $y = x^2$ to produce a new graph.
- We can determine the transformations applied by looking at the equation of the resulting quadratic.

Combining transformations

A quadratic of the form $y = a(x - h)^2 + k$ has been:
- **dilated by a factor of $|a|$ from the *x-axis*, where $|a|$ is the magntiude (or size) of a.**
- **reflected in the *x-axis* if $a < 0$**
- **translated *h units* horizontally:**
  - **if $h > 0$, the graph is translated to the *right***
  - **if $h < 0$, the graph is translated to the *left***
- **translated *k units* vertically:**
  - **if $k > 0$, the graph is translated *upwards***
  - **if $k < 0$, the graph is translated *downwards*.**

WORKED EXAMPLE 7 Determining transformations

For each of the following quadratic equations:
- **i. state the appropriate dilation, reflection and translation of the graph of $y = x^2$ needed to obtain the graph**
- **ii. state the coordinates of the turning point**
- **iii. hence, sketch the graph.**

a. $y = (x + 3)^2$ **b.** $y = -2x^2$

THINK

a. **1.** Write the quadratic equation.

2. Identify the transformation needed — horizontal translation only, no dilation or reflection.

WRITE/DRAW

a. $y = (x + 3)^2$

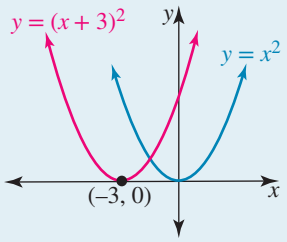
i. Horizontal translation of 3 units to the left

3. State the turning point.

4. Sketch the graph of $y = (x + 3)^2$. You may find it helpful to lightly sketch the graph of $y = x^2$ on the same set of axes first.

ii. The turning point is $(-3, 0)$.

iii.



b. 1. Write the quadratic equation.

2. Identify the transformations needed — dilation (2 in front of x^2) and reflection (negative in front of x^2), no translation.

3. The turning point remains the same as there is no translation.

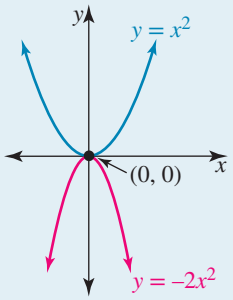
4. Sketch the graph of $y = -2x^2$. You may find it helpful to lightly sketch the graph of $y = x^2$ on the same set of axes first.

b. $y = -2x^2$

i. This is a reflection, so the graph is inverted. As $2 > 1$, the graph is narrower than that of $y = x^2$.

ii. The turning point is $(0, 0)$.

iii.



TI | THINK

a.

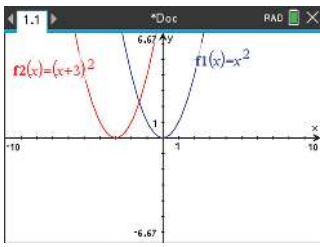
1. On a Graphs page, complete the function entry lines as:
   $f1(x) = x^2$
   $f2(x) = (x + 3)^2$
   The graphs will be displayed.

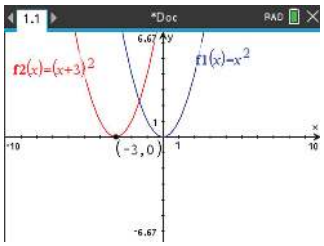
2. To locate the turning point, press:
   - MENU
   - 6: Analyze Graph
   - 2: Minimum

   Drag the dotted line to the left of the turning point (the lower bound), click ENTER and then drag the dotted line to the right of the turning point (the upper bound) and press ENTER.
   The turning point will be shown.

DISPLAY/WRITE

a.





CASIO | THINK

a.

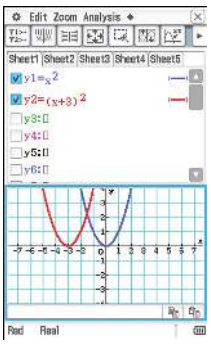
1. On a Graph & Table screen, complete the function entry lines as:
   $y1 = x^2$
   $y2 = (x + 3)^2$
   Tap the graphing icon and the graphs will be displayed.

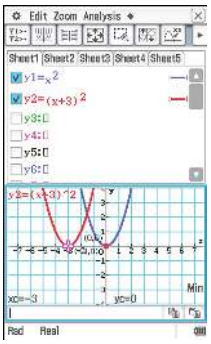
2. To locate the turning point, press:
   - Analysis
   - G-Solve
   - Min

   Select the graph you want using the up and down arrow keys and then press EXE.
   The turning point will be shown.

DISPLAY/WRITE

a.





b.

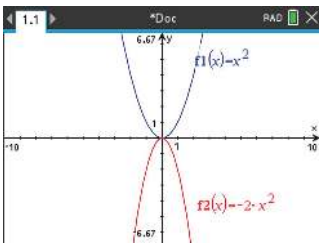
1. On a Graphs page, complete the function entry lines as:

$f1(x) = x^2$

$f2(x) = -2x^2$

The graphs will be displayed.

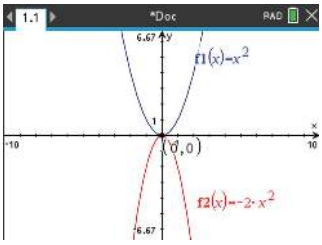
b.



2. To locate the turning point, press:

- MENU
- 6: Analyze Graph
- 2: Minimum

Drag the dotted line to the left of the turning point (the lower bound), click ENTER and then drag the dotted line to the right of the turning point (the upper bound) and press ENTER.
The turning point will be shown.



b.

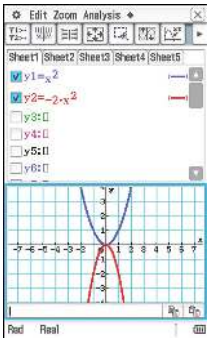
1. On a Graph & Table screen, complete the function entry lines as:

$y1 = x^2$

$y2 = -2x^2$

Tap the graphing icon and the graphs will be displayed.

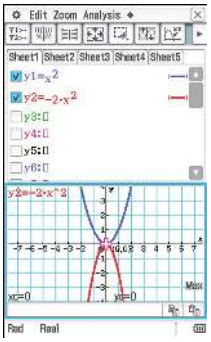
b.



2. To locate the point of intersection, press:

- Analysis
- G-Solve
- Max

Select the graph you want using the up and down arrow keys and then press EXE.
The turning point will be shown.



DISCUSSION

Determine the turning points of the graphs $y = x^2 + k$ and $y = (x - h)^2$.

on Resources

eWorkbook Topic 9 Workbook (worksheets, code puzzle and a project) (ewbk-2035)

Interactivities Individual pathway interactivity: Sketching parabolas (int-4606)
Horizontal translations of parabolas (int-6054)
Vertical translations of parabolas (int-6055)
Dilation of parabolas (int-6096)
Reflection of parabolas (int-6151)

Exercise 9.3 Sketching parabolas using transformations

learnon

9.3 Quick quiz on | 9.3 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 4, 7, 10, 13, 16, 17, 22, 25	2, 5, 8, 11, 14, 18, 19, 23, 26	3, 6, 9, 12, 15, 20, 21, 24, 27, 28

Fluency

WE3 For questions **1** to **3**, state whether each of the following graphs is wider or narrower than the graph of $y = x^2$ and state the coordinates of the turning point of each one.

1. a. $y = 5x^2$ **b.** $y = \frac{1}{3}x^2$

2. a. $y = 7x^2$ **b.** $y = 10x^2$ **c.** $y = \frac{2}{5}x^2$

3. a. $y = 0.25x^2$ **b.** $y = 1.3x^2$ **c.** $y = \sqrt{3}x^2$

WE4 For questions **4** to **6**, state the vertical translation and the coordinates of the turning point for the graphs of each of the following equations when compared to the graph of $y = x^2$.

4. a. $y = x^2 + 3$ **b.** $y = x^2 - 1$

5. a. $y = x^2 - 7$ **b.** $y = x^2 + \frac{1}{4}$ **c.** $y = x^2 - \frac{1}{2}$

6. a. $y = x^2 - 0.14$ **b.** $y = x^2 + 2.37$ **c.** $y = x^2 + \sqrt{3}$

WE5 For questions **7** to **9**, state the horizontal translation and the coordinates of the turning point for the graphs of the following equations when compared to the graph of $y = x^2$.

7. a. $y = (x - 1)^2$ **b.** $y = (x - 2)^2$

8. a. $y = (x + 10)^2$ **b.** $y = (x + 4)^2$ **c.** $y = \left(x - \frac{1}{2}\right)^2$

9. a. $y = \left(x + \frac{1}{5}\right)^2$ **b.** $y = (x + 0.25)^2$ **c.** $y = (x + \sqrt{3})^2$

WE6 For questions **10** to **12**, for each of the following graphs identify the coordinates of the turning point and state whether it is a maximum or a minimum.

10. a. $y = -x^2 + 1$ **b.** $y = x^2 - 3$

11. a. $y = -(x + 2)^2$ **b.** $y = 3x^2$ **c.** $y = 4 - x^2$

12. a. $y = -2x^2$ **b.** $y = (x - 5)^2$ **c.** $y = 1 + x^2$

For questions **13** to **15**, in each of the following state whether the graph is wider or narrower than $y = x^2$ and whether it has a maximum or a minimum turning point.

13. **a.** $y = 3x^2$ **b.** $y = -3x^2$

14. **a.** $y = \frac{1}{2}x^2$ **b.** $y = -\frac{1}{5}x^2$ **c.** $y = -\frac{4}{3}x^2$

15. **a.** $y = 0.25x^2$ **b.** $y = \sqrt{3}x^2$ **c.** $y = -0.16x^2$

Understanding

WE7 For questions **16** to **21**:

i. state the appropriate dilation, reflection and translation of the graph of $y = x^2$ needed to obtain the graph

ii. state the coordinates of the turning point

iii. hence, sketch the graph.

16. **a.** $y = (x + 1)^2$ **b.** $y = -3x^2$ **c.** $y = x^2 + 1$

17. **a.** $y = \frac{1}{3}x^2$ **b.** $y = x^2 - 3$

18. **a.** $y = (x - 4)^2$ **b.** $y = -\frac{2}{5}x^2$ **c.** $y = 5x^2$

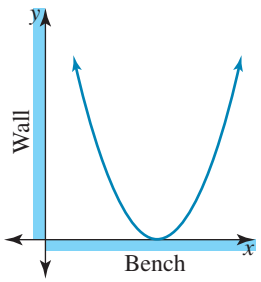
19. **a.** $y = -x^2 + 2$ **b.** $y = -(x - 6)^2$

20. **a.** $y = -x^2 - 4$ **b.** $y = 2(x + 1)^2 - 4$ **c.** $y = \frac{1}{2}(x - 3)^2 + 2$

21. **a.** $y = -\frac{1}{3}(x + 2)^2 + \frac{1}{4}$ **b.** $y = -\frac{7}{4}(x - 1)^2 - \frac{3}{2}$

Reasoning

22. A vase 25 cm tall is positioned on a bench near a wall as shown. The shape of the vase follows the curve $y = (x - 10)^2$, where y cm is the height of the vase and x cm is the distance of the vase from the wall.



a. Identify how far the base of the vase is from the wall.

b. Determine the shortest distance from the top of the vase to the wall.

c. If the vase is moved so that the top just touches the wall, determine the new distance from the wall to the base.

d. Determine the new equation that follows the shape of the vase.

23. Tom is standing at the start of a footpath at $(0, 0)$ that leads to the base of a hill. The height of the hill is modelled by the equation $h = -\frac{1}{10}(d - 25)^2 + 40$, where h is the height of the hill in metres and d is the horizontal distance from the start of the path.

a. Calculate how tall the hill is.

b. Determine how far the base of the hill is from the beginning of the footpath.

c. If the footpath is to be extended so the lead in to the hill is 50 m, determine the new equation that models the height of the hill.

d. The height of the hill has been incorrectly measured and should actually be 120 m. Adjust the equation from part **c** to correct this error and state the transformation applied.

24. A ball is thrown vertically upwards. Its height in metres after t seconds is given by $h = 7t - t^2$.



a. Sketch the graph of the height of the ball against time.

b. Evaluate the highest point reached by the ball. Show your full working.

A second ball is thrown vertically upwards. Its height in metres after t seconds is given by $h = 10t - t^2$.

c. On the same set of axes used for part **a**, sketch the graph of the height of the second ball against time.

d. State the difference in the highest point reached by the two balls.

Problem solving

25. Consider the quadratic equation $y = x^2 - 4x + 7$.

a. Determine the equivalent inverted equation of the quadratic that just touches the one above at the turning point.

b. Confirm your result graphically.

26. Consider the equation $y = 3(x - 2)^2 - 7$.

a. State the coordinates of the turning point and y-intercept.

b. State a sequence of transformations that when applied to the graph of $y = \frac{-3}{2}(x - 2)^2 - 7$ will produce the graph of $y = x^2$.

27. A parabola has the equation $y = -\frac{1}{2}(x - 3)^2 + 4$. A second parabola has an equation defined by $Y = 2(y - 1) - 3$.

a. Determine the equation relating Y to x.

b. State the appropriate dilation, reflection and translation of the graph of $Y = x^2$ required to obtain the graph of $Y = 2(y - 1) - 3$.

c. State the coordinates of the turning point $Y = 2(y - 1) - 3$.

d. Sketch the graph of $Y = 2(y - 1) - 3$.

28. A ball shot at a certain angle to the horizontal follows a parabolic path. It reaches a maximum height of 200 m when its horizontal distance from its starting point is 10 m. When the ball's horizontal distance from the starting point was 1 m, the ball had reached a height of 38 m.

Determine an equation to model the ball's flight, clearly defining your chosen pronumerals.



LESSON
9.4 Sketching parabolas using turning point form

LEARNING INTENTIONS

At the end of this lesson you should be able to:
- determine the axis of symmetry and turning point of a quadratic in turning point form
- calculate the y-intercept and any x-intercepts of a quadratic in turning point form.

eles-4870

9.4.1 Turning point form

- When a quadratic equation is expressed in the form $y = a(x-h)^2 + k$:
  - the turning point is the point (h, k)
  - the axis of symmetry is $x = h$
  - the x-intercepts are calculated by solving $a(x-h)^2 + k = 0$.
- Changing the values of a, h and k in the equation transforms the shape and position of the parabola when compared with the parabola $y = x^2$.

$y = a(x-h)^2 + k$

Reflects and dilates | Translates left and right | Translates up and down

Turning point form

A quadratic of the form $y = a(x-h)^2 + k$ has:
- **a turning point at the coordinate (h, k)**
  - **the turning point will be a minimum if $a > 0$**
  - **the turning point will be a maximum if $a < 0$**
- **an axis of symmetry of $x = h$**
- **a y-intercept of $(0, ah^2 + k)$.**

The number of x-intercepts depends on the values of a, h and k. Changing the value of a does not change the position of the turning point, only h and k.

WORKED EXAMPLE 8 Determining the turning point from turning point form

For each of the following equations, state the coordinates of the turning point of the graph and whether it is a maximum or a minimum.

a. $y = (x-6)^2 - 4$ **b.** $y = -(x+3)^2 + 2$

THINK	WRITE
a. 1. Write the equation.	**a.** $y = (x-6)^2 - 4$
2. Identify the transformations — horizontal translation of 6 units to the right and a vertical translation of 4 units down. State the turning point.	The turning point is $(6, -4)$.
3. As a is positive $(a = 1)$, the graph is upright with a minimum turning point.	Minimum turning point.
b. 1. Write the equation	**b.** $y = -(x+3)^2 + 2$
2. Identify the transformations — horizontal translation of 3 units to the left and a vertical translation of 2 units up. State the turning point.	The turning point is $(-3, 2)$.
3. As a is negative $(a = -1)$, the graph is inverted with a maximum turning point.	Maximum turning point.

9.4.2 x- and y-intercepts of quadratic graphs

eles-4871

- Other key features such as the x- and y-intercepts can also be determined from the equation of a parabola.
- The point(s) where the graph cuts or touches the x-axis are called the x-intercept(s). At these points, $y = 0$.
- The point where the graph cuts the y-axis is called the y-intercept. At this point, $x = 0$.

WORKED EXAMPLE 9 Determining the axial intercepts from turning point form

For the parabolas with the following equations:

i. determine the y-intercept

ii. determine the x-intercepts (where they exist).

a. $y = (x + 3)^2 - 4$ **b.** $y = 2(x - 1)^2$ **c.** $y = -(x + 2)^2 - 1$

THINK	WRITE
a. 1. Write the equation.	**a.** $y = (x + 3)^2 - 4$
2. Calculate the y-intercept by substituting $x = 0$ into the equation.	y-intercept: when $x = 0$, $y = (0 + 3)^2 - 4$ $= 9 - 4$ $= 5$ The y-intercept is 5.
3. Calculate the x-intercepts by substituting $y = 0$ into the equation and solving for x. Add 4 to both sides of the equation. Take the square root of both sides of the equation. Subtract 3 from both sides of the equation. Solve for x.	x-intercepts: when $y = 0$, $(x + 3)^2 - 4 = 0$ $(x + 3)^2 = 4$ $(x + 3) = +2$ or -2 $x = 2 - 3$ or $x = -2 - 3$ $x = -1$ $x = -5$ The x-intercepts are -5 and -1.
b. 1. Write the equation.	**b.** $y = 2(x - 1)^2$
2. Calculate the y-intercept by substituting $x = 0$ into the equation.	y-intercept: when $x = 0$, $y = 2(0 - 1)^2$ $= 2 \times 1$ $= 2$ The y-intercept is 2.
3. Calculate the x-intercepts by substituting $y = 0$ into the equation and solving for x. Note that there is only one solution for x and so there is only one x-intercept. (The graph touches the x-axis.)	x-intercepts: when, $y = 0$, $2(x - 1)^2 = 0$ $(x - 1)^2 = 0$ $x - 1 = 0$ $x = 0 + 1$ $x = 1$ The x-intercept is 1.
c. 1. Write the equation.	**c.** $y = -(x + 2)^2 - 1$
2. Calculate the y-intercept by substituting $x = 0$ into the equation.	y-intercept: when $x = 0$, $y = -(0 + 2)^2 - 1$ $= -4 - 1$ $= -5$ The y-intercept is -5.

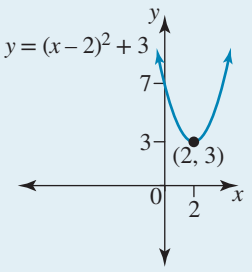
3. Calculate the x-intercepts by substituting $y = 0$ into the equation and solving for x. We cannot take the square root of -1 to obtain real solutions; therefore, there are no x-intercepts.	x-intercepts: when $y = 0$, $-(x+2)^2 - 1 = 0$ $(x+2)^2 = -1$ There are no real solutions, so there are no x-intercepts.

WORKED EXAMPLE 10 Sketching a quadratic in turning point form

For each of the following:

i. **write the coordinates of the turning point**
ii. **state whether the graph has a maximum or a minimum turning point**
iii. **state whether the graph is wider, narrower or the same width as the graph of $y = x^2$**
iv. **calculate the y-intercept**
v. **calculate the x-intercepts**
vi. **sketch the graph.**

a. $y = (x-2)^2 + 3$ **b.** $y = -2(x+1)^2 + 6$

THINK	WRITE/DRAW
a. 1. Write the equation.	**a.** $y = (x-2)^2 + 3$
2. State the coordinates of the turning point from the equation. Use (h, k) as the equation is in the turning point form of $y = a(x-h)^2 + k$ where $a = 1, h = 2$ and $k = 3$.	The turning point is $(2, 3)$.
3. State the nature of the turning point by considering the sign of a.	The graph has a minimum turning point as the sign of a is positive.
4. Specify the width of the graph by considering the magnitude of a.	The graph has the same width as $y = x^2$ since $a = 1$.
5. Calculate the y-intercept by substituting $x = 0$ into the equation.	y-intercept: when $x = 0$, $y = (0-2)^2 + 3$ $= 4 + 3$ $= 7$ y-intercept is 7.
6. Calculate the x-intercepts by substituting $y = 0$ into the equation and solving for x. As we have to take the square root of a negative number, we cannot solve for x.	x-intercepts: when $y = 0$, $(x-2)^2 + 3 = 0$ $(x-2)^2 = -3$ There are no real solutions, and hence no x-intercepts.
7. Sketch the graph, clearly showing the turning point and the y-intercept. 8. Label the graph.	

b. **1.** Write the equation.

b. $y = -2(x+1)^2 + 6$

2. State the coordinates of the turning point from the equation. Use (h, k) as the equation is in the turning point form of $y = a(x-h)^2 + k$ where $a = -2, h = -1$ and $k = 6$.

The turning point is $(-1, 6)$.

3. State the nature of the turning point by considering the sign of a.

The graph has a maximum turning point as the sign of a is negative.

4. Specify the width of the graph by considering the magnitude of a.

The graph is narrower than $y = x^2$ since $|a| > 1$.

5. Calculate the y-intercept by substituting $x = 0$ into the equation.

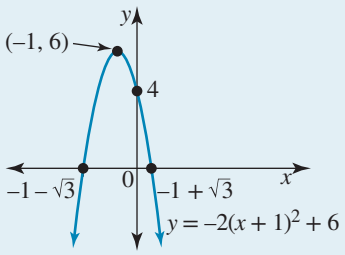
y-intercept: when $x = 0$,
$y = -2(0+1)^2 + 6$
$= -2 \times 1 + 6$
$= 4$
The y-intercept is 4.

6. Calculate the x-intercepts by substituting $y = 0$ into the equation and solving for x.

x-intercepts: when $y = 0$,
$-2(x+1)^2 + 6 = 0$
$2(x+1)^2 = 6$
$(x+1)^2 = 3$
$x + 1 = \sqrt{3}$ or $x + 1 = -\sqrt{3}$
$x = -1 + \sqrt{3}$ or $x = -1 - \sqrt{3}$
The x-intercepts are $-1 - \sqrt{3}$ and $-1 + \sqrt{3}$ (or approximately -2.73 and 0.73).

7. Sketch the graph, clearly showing the turning point and the x- and y-intercepts.

8. Label the graph.



Note: **Unless otherwise stated, exact values for the intercepts should be shown on sketch graphs.**

DISCUSSION

Does a in the equation $y = a(x-h)^2 + k$ have any impact on the turning point?

Resources

eWorkbook	Topic 9 Workbook (worksheets, code puzzle and a project) (ewbk-2035)
Video eLessons	Sketching quadratics in turning point form (eles-1926) Solving quadratics in turning point form (eles-1941)
Interactivities	Individual pathway interactivity: Sketching parabolas in turning point form (int-4607) Quadratic functions (int-2562)

Exercise 9.4 Sketching parabolas using turning point form

learnon

9.4 Quick quiz on | **9.4 Exercise**

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 4, 7, 9, 12, 14, 17, 20, 23, 27	2, 5, 8, 10, 13, 15, 18, 21, 24, 28	3, 6, 11, 16, 19, 22, 25, 26, 29

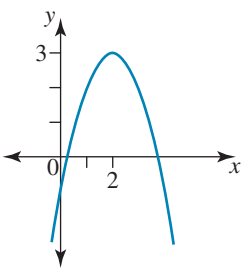
Fluency

WE8 For questions 1 to 3, for each of the following equations, state the coordinates of the turning point of the graph and whether it is a maximum or a minimum.

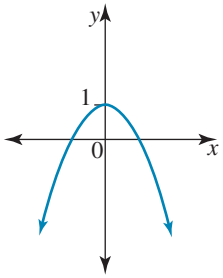
1. a. $y=(x-1)^2+2$ b. $y=(x+2)^2-1$ c. $y=(x+1)^2+1$
2. a. $y=-(x-2)^2+3$ b. $y=-(x-5)^2+3$ c. $y=(x+2)^2-6$
3. a. $y=\left(x-\frac{1}{2}\right)^2-\frac{3}{4}$ b. $y=\left(x-\frac{1}{3}\right)^2+\frac{2}{3}$ c. $y=(x+0.3)^2-0.4$
4. For each of the following, state:
   i. the coordinates of the turning point
   ii. whether the graph has a maximum or a minimum turning point
   iii. whether the graph is wider, narrower or the same width as that of $y=x^2$.
   a. $y=2(x+3)^2-5$ b. $y=-(x-1)^2+1$
5. For each of the following, state:
   i. the coordinates of the turning point
   ii. whether the graph has a maximum or a minimum turning point
   iii. whether the graph is wider, narrower or the same width as that of $y=x^2$.
   a. $y=-5(x+2)^2-4$ b. $y=\frac{1}{4}(x-3)^2+2$
6. For each of the following, state:
   i. the coordinates of the turning point
   ii. whether the graph has a maximum or a minimum turning point
   iii. whether the graph is wider, narrower or the same width as that of $y=x^2$.
   a. $y=-\frac{1}{2}(x+1)^2+7$ b. $y=0.2\left(x-\frac{1}{5}\right)^2-\frac{1}{2}$

7. Select the equation that best suits each of the following graphs.

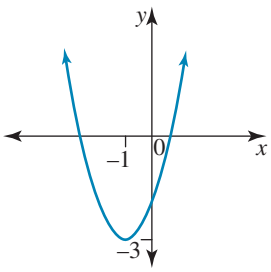
i.



ii.



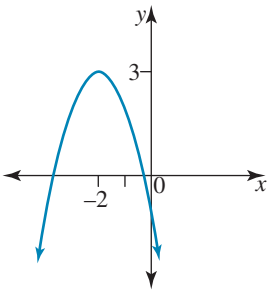
iii.



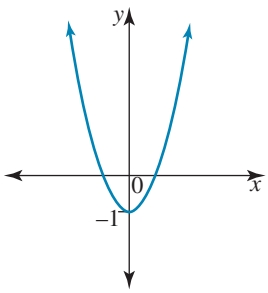
a. $y = (x+1)^2 - 3$ **b.** $y = -(x-2)^2 + 3$ **c.** $y = -x^2 + 1$

8. Select the equation that best suits each of the following graphs.

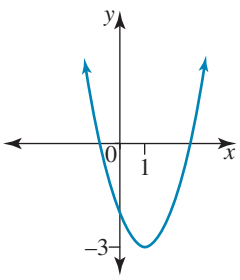
i.



ii.



iii.



a. $y = (x-1)^2 - 3$ **b.** $y = -(x+2)^2 + 3$ **c.** $y = x^2 - 1$

9. MC The translations required to change $y = x^2$ into $y = \left(x - \frac{1}{2}\right)^2 + \frac{1}{3}$ are:

A. right $\frac{1}{2}$, up $\frac{1}{3}$

B. left $\frac{1}{2}$, down $\frac{1}{3}$

C. right $\frac{1}{2}$, down $\frac{1}{3}$

D. left $\frac{1}{2}$, up $\frac{1}{3}$

10. MC For the graph $\frac{1}{4}\left(x - \frac{1}{2}\right)^2 + \frac{1}{3}$, the effect of the $\frac{1}{4}$ on the graph is:

A. no effect
B. to make the graph narrower
C. to make the graph wider
D. to invert the graph

11. MC Compared to the graph of $y = x^2$, $y = -2(x+1)^2 - 4$ is:

A. inverted and wider
B. inverted and narrower
C. upright and wider
D. upright and narrower

12. MC A graph that has a minimum turning point $(1, 5)$ and that is narrower than the graph of $y = x^2$ is:

A. $y = (x-1)^2 + 5$ **B.** $y = \frac{1}{2}(x+1)^2 + 5$

C. $y = 2(x-1)^2 + 5$ **D.** $y = 2(x+1)^2 + 5$

13. MC Compared to the graph of $y = x^2$, the graph of $y = -3(x-1)^2 - 2$ has the following features.

A. Maximum TP at $(-1, -2)$, narrower
B. Maximum TP at $(1, -2)$, narrower
C. Maximum TP at $(1, 2)$, wider
D. Minimum TP at $(1, -2)$, narrower

WE9 For questions **14** to **16**, for the parabolas with the following equations:

i. determine the y-intercept
ii. determine the x-intercepts (where they exist).

14. a. $y = (x+1)^2 - 4$ **b.** $y = 3(x-2)^2$

15. a. $y = -(x+4)^2 - 2$ **b.** $y = (x-2)^2 - 9$

16. a. $y = 2x^2 + 4$ **b.** $y = (x+3)^2 - 5$

Understanding

17. WE10 For each of the following:

i. write the coordinates of the turning point
ii. state whether the graph has a maximum or a minimum turning point
iii. state whether the graph is wider, narrower or the same width as the graph of $y = x^2$
iv. calculate the y-intercept
v. calculate the x-intercepts
vi. sketch the graph.

a. $y = (x-4)^2 + 2$ **b.** $y = (x-3)^2 - 4$ **c.** $y = (x+1)^2 + 2$

18. For each of the following:

i. write the coordinates of the turning point
ii. state whether the graph has a maximum or a minimum turning point
iii. state whether the graph is wider, narrower or the same width as the graph of $y = x^2$
iv. calculate the y-intercept
v. calculate the x-intercepts
vi. sketch the graph.

a. $y = (x+5)^2 - 3$ **b.** $y = -(x-1)^2 + 2$ **c.** $y = -(x+2)^2 - 3$

19. For each of the following:

i. write the coordinates of the turning point
ii. state whether the graph has a maximum or a minimum turning point
iii. state whether the graph is wider, narrower or the same width as the graph of $y = x^2$
iv. calculate the y-intercept
v. calculate the x-intercepts
vi. sketch the graph.

a. $y = -(x+3)^2 - 2$ **b.** $y = 2(x-1)^2 + 3$ **c.** $y = -3(x+2)^2 + 1$

20. Consider the equation $2x^2 - 3x - 8 = 0$.

- **a.** Complete the square.
- **b.** Use the result to determine the exact solutions to the original equation.
- **c.** Determine the turning point of $y = 2x^2 - 3x - 8$ and indicate its type.

21. Answer the following questions.

- **a.** Determine the equation of a quadratic that has a turning point of $(-4, 6)$ and has an x-intercept at $(-1, 0)$.
- **b.** State the other x-intercept (if any).

22. Write the new equation for the parabola $y = x^2$ that has been:

- **a.** reflected in the x-axis
- **b.** dilated by a factor of 7 away from the x-axis
- **c.** translated 3 units in the negative direction of the x-axis
- **d.** translated 6 units in the positive direction of the y-axis
- **e.** dilated by a factor of $\frac{1}{4}$ from the x-axis, reflected in the x-axis, and translated 5 units in the positive direction of the x-axis and 3 units in the negative direction of the y-axis.

Reasoning



23. The price of shares in fledgling company 'Lollies'r'us' plunged dramatically one afternoon, following the breakout of a small fire on the premises. However, Ms Sarah Sayva of Lollies Anonymous agreed to back the company, and share prices began to rise.
Sarah noted at the close of trade that afternoon that the company's share price followed the curve:
$P = 0.1(t - 3)^2 + 1$ where $\$P$ is the price of shares t hours after noon.

- **a.** Sketch a graph of the relationship between time and share price to represent the situation.
- **b.** Determine the initial share price.
- **c.** Determine the lowest price of shares that afternoon.
- **d.** Evaluate the time when the price was at its lowest.
- **e.** Determine the final price of 'Lollies'r'us' shares as trade closed at 5 pm.

24. Rocky is practising for a football kicking competition. After being kicked, the path that the ball follows can be modelled by the quadratic relationship:

$$h = -\frac{1}{30}(d - 15)^2 + 8$$

where h is the vertical distance the ball reaches (in metres), and d is the horizontal distance (in metres).

- **a.** Determine the initial vertical height of the ball.
- **b.** Determine the exact maximum horizontal distance the ball travels.
- **c.** Write down both the maximum height and the horizontal distance when the maximum height is reached.

25. Answer the following questions.

- **a.** If the turning point of a particular parabola is $(2, 6)$, suggest a possible equation for the parabola.
- **b.** If the y-intercept in part **a** is $(0, 4)$, determine the exact equation for the parabola.

26. Answer the following questions.

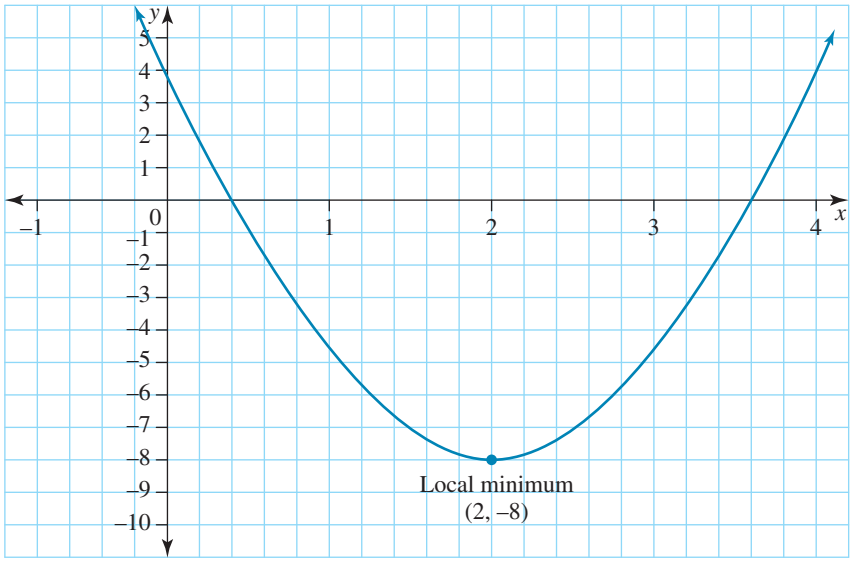
- **a.** If the turning point of a particular parabola is (p, q), suggest a possible equation for the parabola.
- **b.** If the y-intercept in part **a** is $(0, r)$, determine the exact equation for the parabola.

Problem solving

27. Use the completing the square method to write each of the following in turning point form and sketch the parabola for each part.

a. $y = x^2 - 8x + 1$ **b.** $y = x^2 + 4x - 5$ **c.** $y = x^2 + 3x + 2$

28. Use the information given in the graph shown to answer the following questions.



a. Determine the equation of the parabola shown.
b. State the dilation and translation transformations that have been applied to $y = x^2$ to achieve this parabola.
c. This graph is reflected in the x-axis. Determine the equation of the reflected graph.
d. Sketch the graph of the reflected parabola.

29. The graph of a quadratic equation has a turning point at $(-3, 8)$ and passes through the point $(-1, 6)$.

a. Determine the equation of this parabola.
b. State the transformations that have been applied to $y = x^2$ to produce this parabola.
c. Calculate the x and y-intercepts of this parabola.
d. The graph is reflected in the x-axis, dilated by a factor of 2 from the x-axis, and then reflected in the y-axis. Sketch the graph of this new parabola.

LESSON
9.5 Sketching parabolas in expanded form

LEARNING INTENTIONS

At the end of this lesson you should be able to:

- determine the x-intercept/s, y-intercept and turning point of a quadratic equation and sketch its graph by first factorising the equation
- determine the x-intercept/s, y-intercept and turning point of a quadratic equation and sketch its graph using the quadratic formula.

9.5.1 Parabolas of the form $y = ax^2 + bx + c$

eles-4872

- The general form of a quadratic equation is $y = ax^2 + bx + c$ where a, b and c are constants.
- The x-intercepts can be found by letting $y = 0$, factorising and using the Null Factor Law to solve for x.
- The y-intercept can be found by letting $x = 0$ and solving for y.
- The x-coordinate of the turning point lies midway between the x-intercepts.
- Once the midpoint of the x-intercepts is found, this value can be substituted into the original equation to find the y-coordinate of the turning point.
- Once these intercepts and turning point have been found, it is possible to sketch the parabola.
- If an equation is not written in turning point form, and cannot be readily factorised, then we will need to use the quadratic formula to help find all key points.

Quadratic formula

A quadratic of the form $y = ax^2 + bx + c$ has:

- **a y-intercept at the coordinate $(0, c)$.**
- **x-intercepts that can be found using the quadratic formula; that is, when the equation $ax^2 + bx + c = 0$**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- **a turning point at the coordinate $\left(-\frac{b}{2a}, c - \frac{b^2}{4a}\right)$.**

For example, given the equation $y = x^2 + 4x - 6$:

- In this example we have $a = 1$, $b = 4$ and $c = -6$.
- The y-intercept is $(0, c) = (0, -6)$
- The x-intercepts are given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(4) \pm \sqrt{(4)^2 - 4 \times (1) \times (-6)}}{2 \times (1)} = \frac{-4 \pm \sqrt{40}}{2} = \frac{-4 \pm 2\sqrt{10}}{2} = -2 \pm \sqrt{10}$$

- Therefore, the x-intercepts are $\left(-2 + \sqrt{10}, 0\right)$ and $\left(-2 - \sqrt{10}, 0\right)$.
- The turning point is given by $\left(-\frac{b}{2a}, c - \frac{b^2}{4a}\right) = \left(-\frac{(4)}{2 \times (1)}, -6 - \frac{(4)^2}{4 \times (1)}\right) = (-2, -10)$.

Note: Do not convert answer to decimals unless specified by the question. It is always best practice to leave coordinates in exact form.

WORKED EXAMPLE 11 Sketching a factorised quadratic equation

Sketch the graph of $y = (x - 3)(x + 2)$.

THINK

1. The equation is in factorised form. To calculate the x-intercepts, let $y = 0$ and use the Null Factor Law.

2. The x-coordinate of the turning point is midway between the x-intercepts. Calculate the average of the two x-intercepts to determine the midpoint between them.

3. • To calculate the y-coordinate of the turning point, substitute x_{TP} into the equation.
   • State the turning point.

4. • To calculate the y-intercept, let $x = 0$ and substitute.
   • State the y-intercept.

5. • Sketch the graph, showing all the important features.
   • Label the graph.

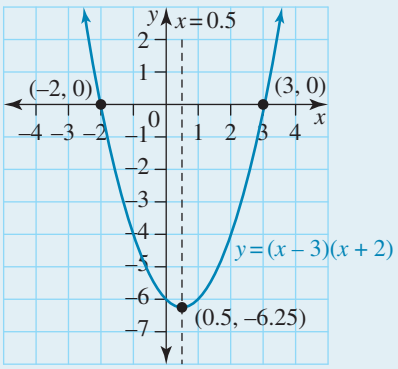
WRITE/DRAW

$y = (x - 3)(x + 2)$
$0 = (x - 3)(x + 2)$
$x - 3 = 0 \quad \text{or} \quad x + 2 = 0 \text{ (NFL)}$
$x = 3 \quad \text{or} \quad x = -2$
x-intercepts: $(3, 0)$ $(-2, 0)$

$x_{TP} = \dfrac{3 + (-2)}{2}$
$= 0.5$

$y = (x - 3)(x + 2)$
$y_{TP} = (0.5 - 3)(0.5 + 2)$
$= -6.25$
Turning point: $(0.5, -6.25)$

$y = (0 - 3)(0 + 2)$
$= -6$
y-intercept: $(0, -6)$



WORKED EXAMPLE 12 Sketching a quadratic in expanded form

Sketch the graph of $y = 2x^2 - 6x - 6$.

THINK

1. The equation is not in factorised form, but there is a common factor of 2. Take out the common factor of 2.

WRITE/DRAW

$y = 2x^2 - 6x - 6$
$= 2(x^2 - 3x - 3)$

2. The equation cannot be factorised (no factors of -3 add to -3), so use completing the square to write the equation in turning point form.
- Halve and then square the coefficient of x.
- Add this and then subtract it from the equation.
- Collect the terms for and write the perfect square.
- Simplify the brackets to write the equation in turning point form.
- Identify the coordinates of the turning point (h, k).

$$y = 2\left(x^2 - 3x + \left(\frac{3}{2}\right)^2 - \left(\frac{3}{2}\right)^2 - 3\right)$$
$$= 2\left(\left[x - \frac{3}{2}\right]^2 - \left(\frac{3}{2}\right)^2 - 3\right)$$
$$= 2\left(\left[x - \frac{3}{2}\right]^2 - \frac{9}{4} - 3\right)$$
$$= 2\left(\left[x - \frac{3}{2}\right]^2 - \frac{21}{4}\right)$$
$$= 2\left(x - \frac{3}{2}\right)^2 - 2 \times \frac{21}{4}$$
$$= 2\left(x - \frac{3}{2}\right)^2 - \frac{21}{2}$$
Turning point : $\left(\frac{3}{2}, \frac{-21}{2}\right)$

3.
- To calculate the x-intercepts, let $y = 0$. No factors of -3 add to -3, so use the quadratic formula to calculate the x-intercepts.

x-intercepts: let $y = 0$.
$0 = 2x^2 - 6x - 6$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
where $a = 2, b = -6, c = -6$
$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4 \times 2 \times (-6)}}{2(2)}$$
$$= \frac{6 \pm \sqrt{36 + 48}}{4}$$
$$= \frac{6 \pm \sqrt{84}}{4} = \frac{6 \pm 2\sqrt{21}}{4}$$

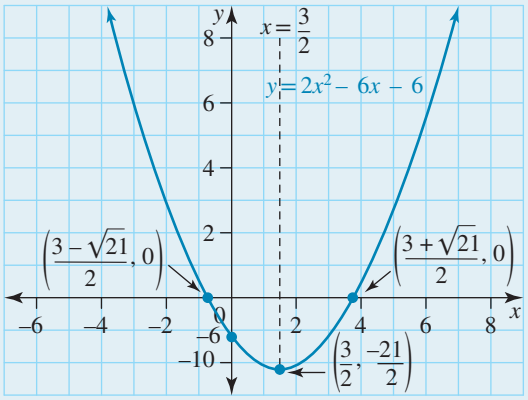
- State the x-intercepts.

The x-intercepts are:
$$x = \frac{3 + \sqrt{21}}{2} \text{ and } x = \frac{3 - \sqrt{21}}{2}$$
$x \approx 3.79$ $\quad$ $x \approx -0.79$

4.
- To calculate the y-intercepts, let $x = 0$ and substitute.
- State the y-intercept.

$y = 2x^2 - 6x - 6$
$y = 2(0)^2 - 6(0) - 6$
$= -6$
y-intercept: $(0, -6)$

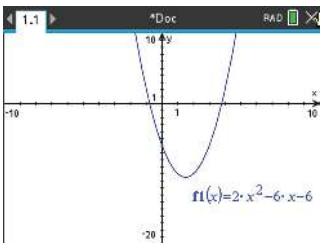
5.
- Sketch the graph, showing all the important features.
- Label the graph and show the exact values of the x-intercepts.



TI | THINK

1. On a Graphs page, complete the function entry line as:
   $f1(x) = 2x^2 - 6x - 6$
   The graph will be displayed.

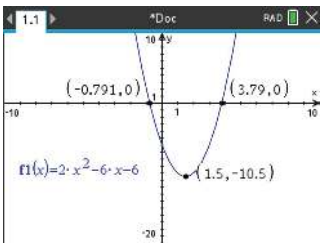
DISPLAY/WRITE



2. To locate the turning point, press:
   - MENU
   - 6: Analyze Graph
   - 2: Minimum

   Drag the dotted line to the left of the turning point (the lower bound), click ENTER and then drag the dotted line to the right of the turning point (the upper bound) and press ENTER.
   To locate the intercepts, press:
   - MENU
   - 6: Analyze Graph
   - 1: Zero

   Locate the points as described above.
   The points of interest will be shown.

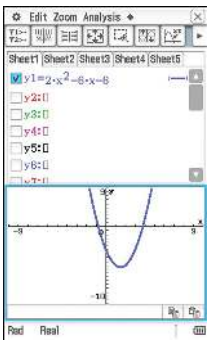


The graph is shown, along with the critical points.

CASIO | THINK

1. On a Graph & Table screen, complete the function entry line as:
   $y1 = 2x^2 - 6x - 6$
   Tap the graphing icon and the graph will be displayed.

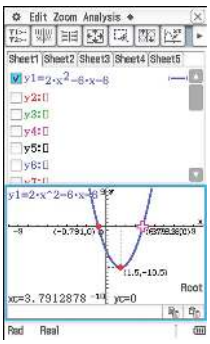
DISPLAY/WRITE



2. To locate the turning point, press:
   - Analysis
   - G-Solve
   - Min

   To locate the turning point, press:
   - Analysis
   - G-Solve
   - Root

   To locate the second root, tap the right arrow.
   The points of interest will be shown.



The turning point is (1.5, 10.5).
The x-intercepts are $(-0.791, 0)$ and $(3.791, 0)$.

DISCUSSION

What strategy can you use to remember all of the information necessary to sketch a parabola?

Resources

eWorkbook	Topic 9 Workbook (worksheets, code puzzle and a project) (ewbk-2035)
Video eLessons	Sketching quadratics in factorised form (eles-1927) Sketching parabolas using the quadratic formula (eles-1945)
Interactivity	Individual pathway interactivity: Sketching parabolas of the form $y = ax^2 + bx + c$ (int-4608)

Exercise 9.5 Sketching parabolas in expanded form

learnon

9.5 Quick quiz on | **9.5 Exercise**

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 2, 5, 8, 11, 14	3, 6, 9, 12, 15	4, 7, 10, 13, 16

Fluency

1. What information is necessary to be able to sketch a parabola?

WE11 For questions 2 to 4, sketch the graph of each of the following.

2. a. $y = (x - 5)(x - 2)$ b. $y = (x + 4)(x - 7)$
3. a. $y = (x + 3)(x + 5)$ b. $y = (2x + 3)(x + 5)$
4. a. $y = (4 - x)(x + 2)$ b. $y = \left(\frac{x}{2} + 3\right)(5 - x)$

WE12 For questions 5 to 7, sketch the graph of each of the following.

5. a. $y = x^2 + 4x + 2$ b. $y = x^2 - 4x - 5$ c. $y = 2x^2 - 4x - 3$
6. a. $y = -2x^2 + 11x + 5$ b. $y = -2x^2 + 12x$ c. $y = 3x^2 + 6x + 1$
7. a. $y = -3x^2 - 5x + 2$ b. $y = 2x^2 + 8x - 10$ c. $y = -3x^2 + 7x + 3$

Understanding



8. The path of a soccer ball kicked by the goal keeper can be modelled by the equation $y = -\frac{1}{144}(x^2 - 24x)$ where y is the height of the soccer ball and x is the horizontal distance from the goalie, both in metres.
   a. Sketch the graph.
   b. Calculate how far away from the player does the ball first bounce.
   c. Calculate the maximum height of the ball.

9. The monthly profit or loss, p, (in thousands of dollars) for a new brand of chicken loaf is given by $p = 3x^2 - 15x - 18$, where x is the number of months after its introduction (when $x = 0$).
   a. Sketch the graph.
   b. Determine during which month a profit was first made.
   c. Calculate the month in which the profit is $54 000.



10. The height, h metres, of a model rocket above the ground t seconds after launch is given by the equation $h = 4t(50 - t)$, where $0 \leq t \leq 50$.
    a. Sketch the graph of the rocket's flight.
    b. State the height of the rocket above the ground when it is launched.
    c. Calculate the greatest height reached by the rocket.
    d. Determine how long the rocket takes to reach its greatest height.
    e. Determine how long the rocket is in the air.

Reasoning

11. The equation $y = x^2 + bx + 7500$ has x-intercepts of $(-150, 0)$ and $(-50, 0)$. Determine the value of b in the equation. Justify your answer.

12. The equation $y = x^2 + bx + c$ has x-intercepts of m and n. Determine the value of b in the equation. Justify your answer.

13. A ball thrown from a cliff follows a parabolic path of the form $y = ax^2 + bx + c$. The ball is released at the point (0, 9), reaches a maximum height at (2, 11) and passes through the point (6, 3) on its descent.
    Determine the equation of the ball's path. Show full working.

Problem solving

14. A ball is thrown upwards from a building and follows the path given by the formula $h = -x^2 + 4x + 21$. The ball is h metres above the ground when it is a horizontal distance of x metres from the building.
    a. Sketch the graph of the path of the ball.
    b. Determine the maximum height of the ball.
    c. Determine how far the ball is from the wall when it reaches the maximum height.
    d. Determine how far from the building the ball lands.

15. During an 8-hour period, an experiment is done in which the temperature of a room follows the relationship $T = h^2 - 8h + 21$, where T is the temperature in degrees Celsius h hours after starting the experiment.



a. Sketch the graph of this quadratic.
b. Identify the initial temperature.
c. Determine if the temperature is increasing or decreasing after 3 hours.
d. Determine if the temperature is increasing or decreasing after 5 hours.
e. Determine the minimum temperature and when it occurred.
f. Determine the temperature after 8 hours.

16. A ball is thrown out of a window, passing through the points $(0, 7)$, $(6, 18)$ and $(18, 28)$. A rule for the height of the ball in metres is given by $h = ax^2 + bx + c$ where x is the horizontal distance, in metres, covered by the ball.

a. Determine the values of a, b and c in the rule for the height of the ball.
b. Calculate the height that the ball was thrown from.
c. Evaluate the maximum height reached by the ball.
d. Determine horizontal distance covered by the ball when it hits the ground.
e. Sketch the flight path of the ball, making sure you show all key points.

LESSON
9.6 Exponential graphs

LEARNING INTENTIONS

At the end of this lesson you should be able to:

- determine the asymptote of an exponential equation and sketch a graph of the equation
- sketch the graph of an exponential graph after it has undergone a number of transformations that may include dilations, a reflection in the x- or y-axis, or translations.

9.6.1 Exponential functions

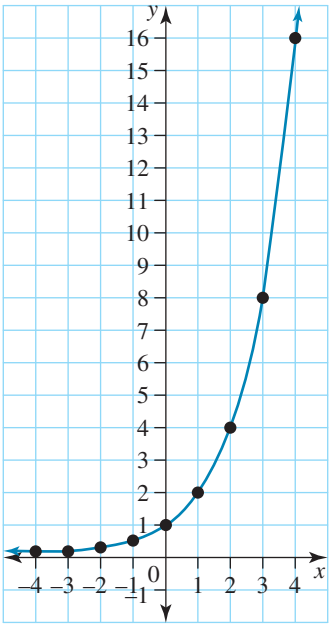
eles-4873

- Relationships of the form $y = a^x$ are called **exponential functions** with base a, where a is a real number not equal to 1, and x is the index power or exponent.
- The term 'exponential' is used, as x is an exponent (or index).
  For example, the graph of the exponential function $y = 2^x$ can be plotted by completing a table of values.

  Remember that $2^{-3} = \frac{1}{2^3}$

  $= \frac{1}{8}$, and so on.

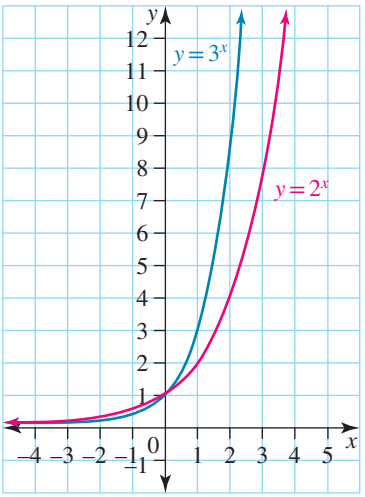
x	-4	-3	-2	-1	0	1	2	3	4
y	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8	16



- For an exponential function, the ratio of consecutive y-values is always constant. In the table above the ratio of consecutive y-values is 2.
- The graph has many significant features.
  - The y-intercept is 1.
  - The value of y is always greater than zero.
  - As x decreases, y gets closer to but never reaches zero. So the graph gets closer to but never reaches the x-axis. The x-axis (or the line $y = 0$) is called an **asymptote.**
  - As x increases, y becomes very large.

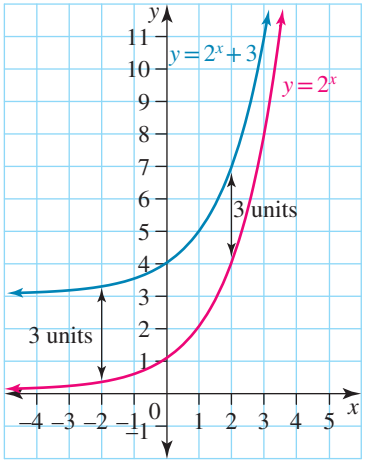
Comparing exponential graphs

- The diagram at right shows the graphs of $y = 2^x$ and $y = 3^x$.
- The graphs both pass through the point (0, 1).
- The graph of $y = 3^x$ climbs more steeply than the graph of $y = 2^x$.
- $y = 0$ is an asymptote for both graphs.



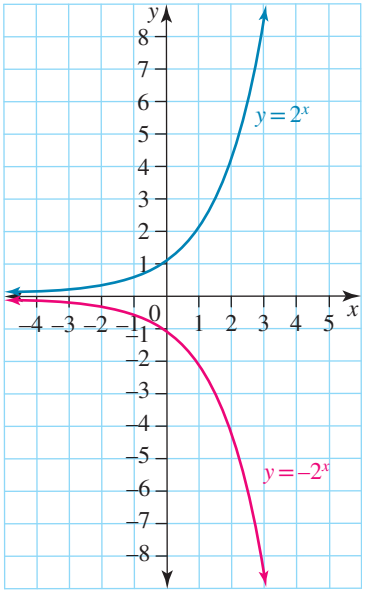
Translation and reflection

Vertical translation



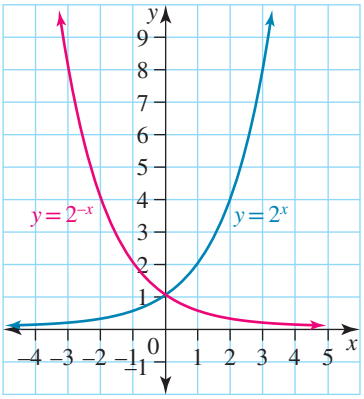
- The diagram shows the graphs of $y = 2^x$ and $y = 2^x + 3$.
- The graphs have identical shape.
- Although they appear to get closer to each other, the graphs are constantly 3 vertical units apart.
- As x becomes very small, the graph of $y = 2^x + 3$ approaches but never reaches the line $y = 3$, so $y = 3$ is the horizontal asymptote.
- When the graph of $y = 2^x$ is translated 3 units upward, it becomes the graph of $y = 2^x + 3$.

Reflection about the x-axis



- The diagram shows the graphs of $y = 2^x$ and $y = -2^x$.
- The graphs have identical shape.
- The graph of $y = -2^x$ is a reflection about the x-axis of the graph of $y = 2^x$.
- The x-axis ($y = 0$) is an asymptote for both graphs.
- In general, the graph of $y = -a^x$ is a reflection about the x-axis of the graph of $y = a^x$.

Reflection about the y-axis



- The diagram shows the graphs of $y = 2^x$ and $y = 2^{-x}$.
- The graphs have identical shape.
- The graph of $y = 2^{-x}$ is a reflection about the y-axis of the graph of $y = 2^x$.
- Both graphs pass through the point (0, 1).
- The x-axis ($y = 0$) is an asymptote for both graphs.
- In general, the graph of $y = a^{-x}$ is a reflection about the y-axis of the graph of $y = a^x$.

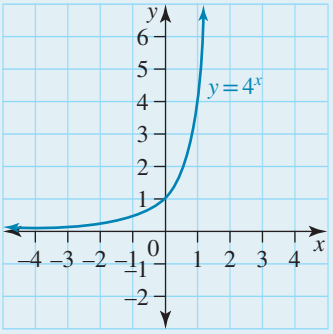
WORKED EXAMPLE 13 Sketching exponential graphs

Given the graph of $y = 4^x$, sketch on the same axes the graphs of:

a. $y = 4^x - 2$

b. $y = -4^x$

c. $y = 4^{-x}$.



THINK

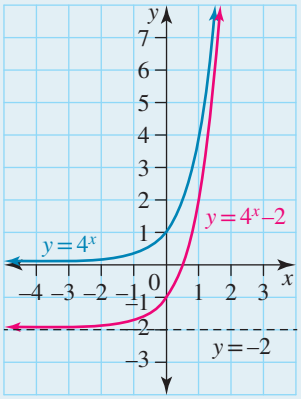
a. The graph of $y = 4^x$ has already been drawn. It has a y-intercept of 1 and a horizontal asymptote at $y = 0$. The graph of $y = 4^x - 2$ has the same shape as $y = 4^x$ but is translated 2 units vertically down. It has a y-intercept of -1 and a horizontal asymptote at $y = -2$.

b. $y = -4^x$ has the same shape as $y = 4^x$ but is reflected about the x-axis. It has a y-intercept of -1 and a horizontal asymptote at $y = 0$.

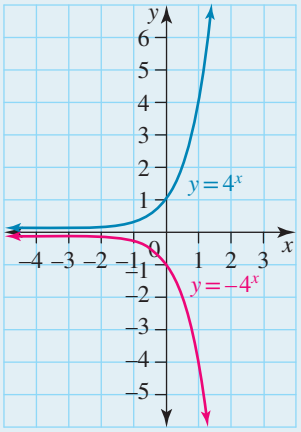
c. $y = 4^{-x}$ has the same shape as $y = 4^x$ but is reflected about the y-axis. The graphs have the same y-intercept and the same horizontal asymptote ($y = 0$).

DRAW

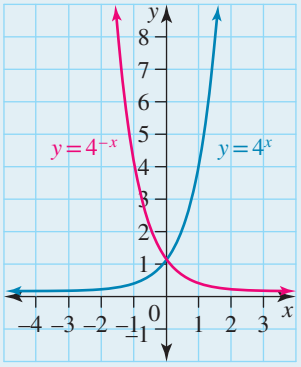
a.



b.



c.



9.6.2 Combining transformations of exponential graphs

eles-4874

- It is possible to combine translations, dilations and reflections in one graph.

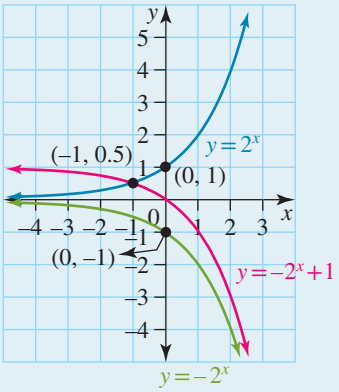
WORKED EXAMPLE 14 Sketching exponentials with multiple transformations

By considering transformations to the graph of $y = 2^x$, sketch the graph of $y = -2^x + 1$.

THINK

1. Start by sketching $y = 2^x$.
   It has a y-intercept of 1 and a horizontal asymptote at $y = 0$.
2. Sketch $y = -2^x$ by reflecting $y = 2^x$ about the x-axis.
   It has a y-intercept of -1 and a horizontal asymptote at $y = 0$.
3. Sketch $y = -2^x + 1$ by translating $y = -2^x$ upwards by 1 unit.
   The graph has a y-intercept of 0 and a horizontal asymptote at $y = 1$.

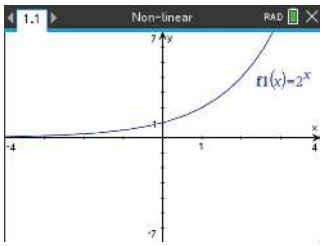
DRAW



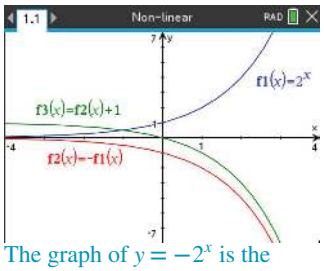
TI | THINK

1. In a new problem, on a Graphs page, complete the function entry line as:
   $f1(x) = 2^x$
   Then press ENTER.

DISPLAY/WRITE



2. Press TAB and complete the function entry line as:
   $f2(x) = -f1(x)$
   Then press ENTER.
   Press TAB and complete the function entry line as:
   $f3(x) = f2(x) + 1$
   Then press ENTER.

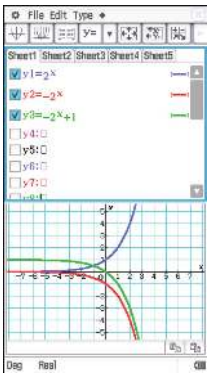


The graph of $y = -2^x$ is the reflection of the graph of $y = 2^x$ in the x-axis.
The graph of $y = -2^x + 1$ is the graph of $y = -2^x$ translated upwards by 1 unit.
The graph of $y = -2^x + 1$ passes through the origin and has a horizontal asymptote at $y = 1$.

CASIO | THINK

Open the Graph & Table screen and complete the function entry line as:
$y1 = 2^x$
$y2 = -2^x$
$y3 = -2^x + 1$
Then tap the graphing icon and the graphs will be displayed.

DISPLAY/WRITE



From $y1$ to $y2$ the graph has undergone a reflection about the x-axis. From $y2$ to $y3$ the graph has undergone a translation upwards by 1 unit.

DISCUSSION

Explain whether the graph of an exponential function will always have a horizontal asymptote.

Resources

eWorkbook Topic 9 Workbook (worksheets, code puzzle and a project) (ewbk-2035)

Interactivities Individual pathway interactivity: Exponential functions and graphs (int-4609)
Exponential functions (int-5959)

Exercise 9.6 Exponential graphs

learnon

9.6 Quick quiz | 9.6 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 6, 8, 10, 13, 15, 18	2, 4, 7, 11, 14, 16, 19	3, 5, 9, 12, 17, 20

Fluency

1. Complete the following table and use it to plot the graph of $y = 3^x$, for $-3 \leq x \leq 3$.

x	-3	-2	-1	0	1	2	3
y							

2. If $x = 1$, calculate the value of y when:

a. $y = 2^x$ **b.** $y = 3^x$ **c.** $y = 4^x$

3. If $x = 1$, calculate the value of y when:

a. $y = 10^x$ **b.** $y = a^x$.

4. Using a calculator or graphing program, sketch the graphs of $y = 2^x$, $y = 3^x$ and $y = 4^x$ on the same set of axes.

a. Describe the common features among the graphs.
b. Describe how the value of the base $(2, 3, 4)$ affects the graph.
c. Predict where the graph $y = 8^x$ would lie and sketch it in.

5. Using graphing technology, sketch the following graphs on one set of axes.
$y = 3^x$, $y = 3^x + 2$, $y = 3^x + 5$ and $y = 3^x - 3$

a. State what remains the same in all of these graphs.
b. State what is changed.
c. For the graph of $y = 3^x + 10$, write down:
i. the y-intercept
ii. the equation of the horizontal asymptote.

6. Using graphing technology, sketch the graphs of:

a. $y = 2^x$ and $y = -2^x$
b. $y = 3^x$ and $y = -3^x$
c. $y = 6^x$ and $y = -6^x$.
d. State the relationship between these pairs of graphs.

7. Using graphing technology, sketch the graphs of:

a. $y = 2^x$ and $y = 2^{-x}$

b. $y = 3^x$ and $y = 3^{-x}$

c. $y = 6^x$ and $y = 6^{-x}$

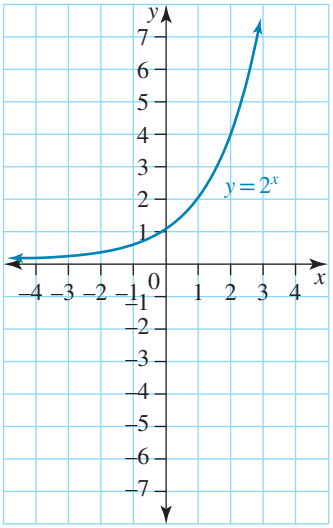
d. State the relationship between these pairs of graphs.

8. **WE13** Given the graph of $y = 2^x$, sketch on the same axes the graphs of:

a. $y = 2^x + 6$

b. $y = -2^x$

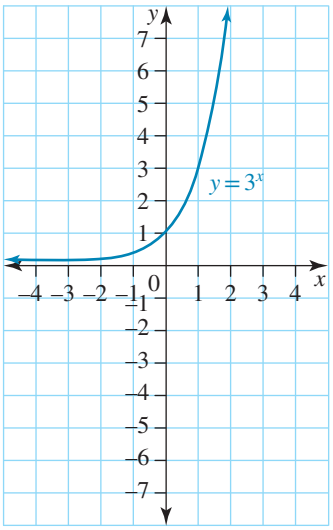
c. $y = 2^{-x}$.



9. Given the graph of $y = 3^x$, sketch on the same axes the graphs of:

a. $y = 3^x + 2$

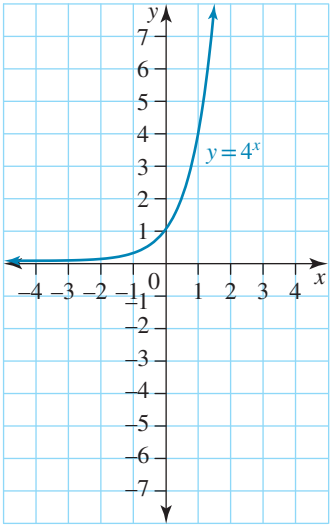
b. $y = -3^x$.



10. Given the graph of $y = 4^x$, sketch on the same axes the graphs of:

a. $y = 4^x - 3$

b. $y = 4^{-x}$.



Understanding

11. **WE14** By considering transformations of the graph of $y = 2^x$, sketch the following graphs on the same set of axes.

a. $y = 2^{-x} + 2$

b. $y = -2^x + 3$

12. By considering transformations of the graph of $y = 5^x$, sketch the following graphs on the same set of axes.

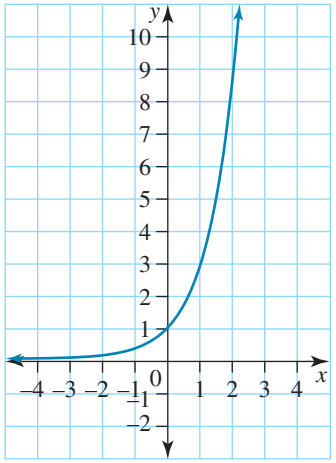
a. $y = -5^x + 10$

b. $y = 5^{-x} + 10$

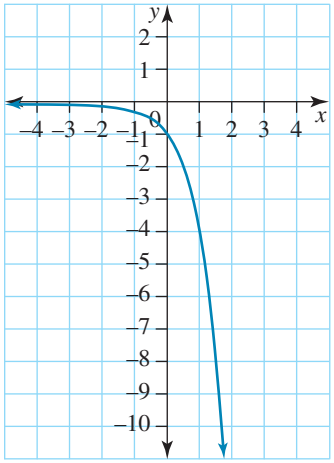
13. Match each equation with its correct graph.

A. $y = 2^x$ **B.** $y = 3^x$ **C.** $y = -4^x$ **D.** $y = 5^{-x}$

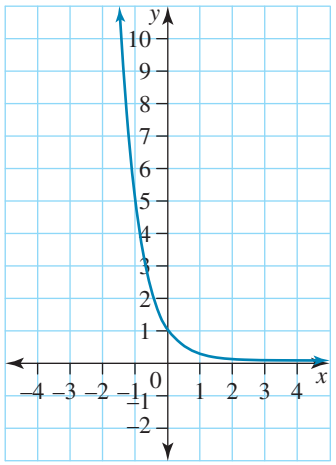
a.



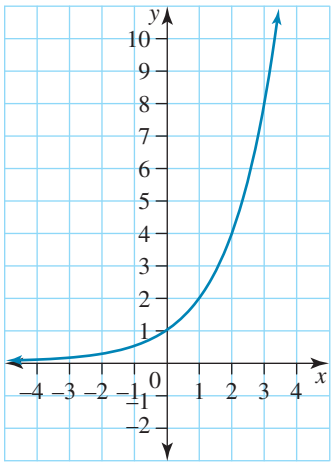
b.



c.



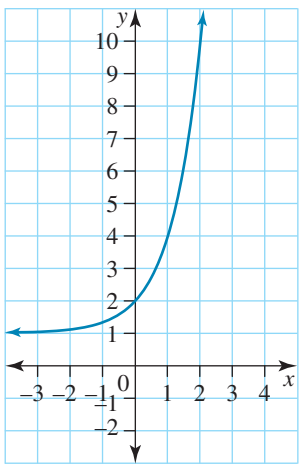
d.



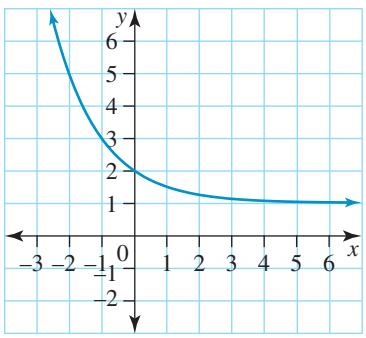
14. Match each equation with its correct graph. Explain your answer.

A. $y = 2^x + 1$ **B.** $y = 3^x + 1$ **C.** $y = -2^x + 1$ **D.** $y = 2^{-x} + 1$

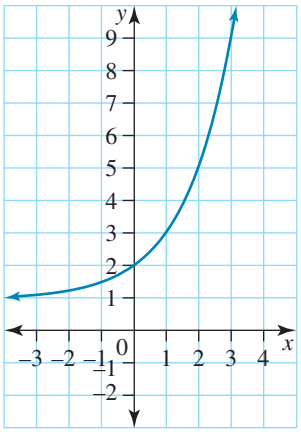
a.



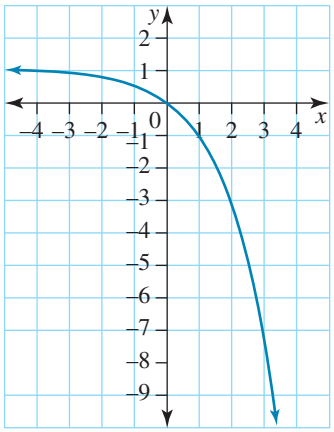
b.



c.



d.



Reasoning

15. By considering transformations of the graph of $y = 3^x$, sketch the graph of $y = -3^{-x} - 3$.

16. The graph of $f(x) = 16^x$ can be used to solve for x in the exponential equation $16^x = 32$. Sketch a graph of $f(x) = 16^x$ and use it to solve $16^x = 32$.

17. The graph of $f(x) = 6^{x-1}$ can be used to solve for x in the exponential equation $6^{x-1} = 36\sqrt{6}$. Sketch a graph of $f(x) = 6^{x-1}$ and use it to solve $6^{x-1} = 36\sqrt{6}$.

Problem solving

18. Sketch the graph of $y = -2^{-x} + 2$.

19. The number of bacteria, N, in a certain culture is reduced by a third every hour so

$$N = N_0 \times \left(\frac{1}{3}\right)^t$$



where t is the time in hours after 12 noon on a particular day. Initially there are 10 000 bacteria present.

a. State the value of N_0.

b. Calculate the number of bacteria, correct to the nearest whole number, in the culture when:

i. $t = 2$ **ii.** $t = 5$ **iii.** $t = 10$.

20. a. The table shows the population of a city between 1850 and 1930. Explain if the population growth is exponential.

Year	1850	1860	1870	1880	1890	1900	1910	1920	1930
Population (million)	1.0	1.3	1.69	2.197	2.856	3.713	4.827	6.275	8.157

b. Determine the common ratio in part **a**.

c. Evaluate the percentage increase every ten years.

d. Estimate the population in 1895.

e. Estimate the population in 1980.

LESSON
9.7 Inverse proportion

LEARNING INTENTIONS

At the end of this lesson you should be able to:

- calculate the constant of proportionality, k, between two variables that are inversely proportional to each other
- determine the rule between two inversely proportional variables and plot the graph of their relationship.

9.7.1 Inverse proportion

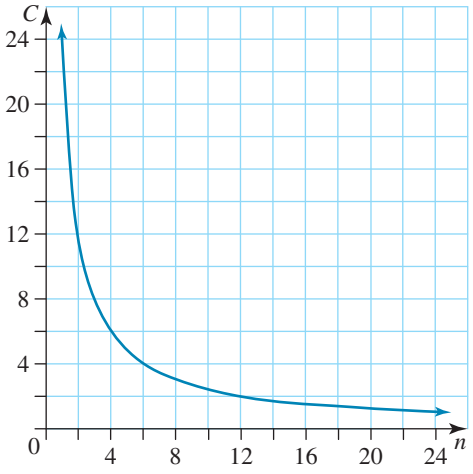
eles-4875

- If 24 sweets are shared between 4 children, then each child will receive 6 sweets. If the sweets are shared by 3 children, then each will receive 8 sweets.
- The relationship between the number of children (C) and the number sweets for each child (n) can be given in a table.



C	1	2	3	4	6	8	12
n	24	12	8	6	4	3	2

- As the number of children (C) increases, the number of sweets for each child (n) decreases. This is an example of **inverse proportion** or inverse variation.
- We say that 'n is inversely proportional to C' or 'n varies inversely as C'.
- This is written as $C \propto \frac{1}{n}$, or $C = \frac{k}{n}$, where k is a constant (the **constant of proportionality**). This formula can be rearranged to $Cn = k$. Note that multiplying any pair of values in the table (3×8, 12×2) gives the same result.
- The relationship has some important characteristics:
  - As C increases, n decreases, and vice versa.
  - The graph of the relationship is a **hyperbola**.



WORKED EXAMPLE 15 Calculating the constant of proportionality

y is inversely proportional to x and $y = 10$ when $x = 2$.

a. Calculate the constant of proportionality, k, and hence the rule relating x and y.

b. Plot a graph of the relationship between x and y, for values of x from 2 to 10.

THINK	WRITE/DRAW
a. 1. Write the relationship between the variables.	a. $y \propto \frac{1}{x}$
2. Rewrite as an equation using k, the constant of proportionality	$y = \frac{k}{x}$

3. Substitute $y = 10, x = 2$, into $y = \frac{k}{x}$ and solve for k

$$10 = \frac{k}{2}$$
$$k = 20$$

4. Write the rule by substituting $k = 20$ into $y = \frac{k}{x}$.

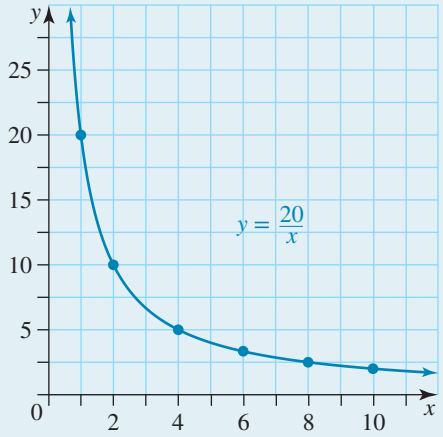
$$y = \frac{20}{x}$$

b. 1. Use the rule $y = \frac{20}{x}$ to set up a table of values for x and y, taking values for x which are positive factors of k so that only whole number values of y are obtained.
For example, $x = 4, y = \frac{20}{4} = 5$.

b.

x	2	4	6	8	10
y	10	5	3.3	2.5	2

2. Plot the points on a clearly labelled set of axes and join the points with a smooth curve. Label the graph.



WORKED EXAMPLE 16 Determining the rule for an inversely proportional relationship

When a wire is connected to a power source, the amount of electrical current (I) passing through the wire is inversely proportional to the resistance (R) of the wire. If a current of 0.2 amperes flows through a wire of resistance 60 ohms:
a. calculate the constant of proportionality
b. determine the rule relating R and I
c. calculate the resistance if the current equals 5 amperes
d. determine the current that will flow through a wire of resistance 20 ohms.



THINK

Summarise the information in a table.

$I \propto \frac{1}{R}$

Write the rule.

a. 1. Substitute $R = 60$, $I = 0.2$, into $I = \frac{k}{R}$.

2. Solve for k.

WRITE

R	60		20
I	0.2	5	

$$I = \frac{k}{R}$$

a. $$0.2 = \frac{k}{60}$$
$$0.2 \times 60 = k$$
$$k = 12$$

b. Write the rule using $k = 12$.

b. $I = \frac{12}{R}$

c. 1. Substitute $I = 5$ into $I = \frac{12}{R}$.

c. $5 = \frac{12}{R}$

$5R = 12$

$R = \frac{12}{5}$

2. Solve for R.

$= 2.4$

3. Write the answer in a sentence.

The resistance equals 2.4 ohms.

d. 1. Substitute $R = 20$ into $I = \frac{12}{R}$.

d. $I = \frac{12}{20}$

$= 0.6$

2. Write the answer in a sentence.

The current will be 0.6 amperes.

DISCUSSION

Explain what is meant by inverse proportion.

Resources

eWorkbook Topic 9 Workbook (worksheets, code puzzle and a project) (ewbk-2035)

Interactivities Individual pathway interactivity: Inverse proportion (int-4512)
Inverse proportion (int-6058)

Exercise 9.7 Inverse proportion

learnon

9.7 Quick quiz on | **9.7 Exercise**

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 3, 6, 10, 13	2, 4, 7, 11, 14	5, 8, 9, 12, 15

Fluency

1. Decide whether inverse proportion exists between each pair of variables. If it does exist, write an equation to describe the relationship.

a. The speed of a car (s) and the time (t) it takes to complete one lap of a race circuit.
b. The amount of money (D) that I have and the number (n) of cards that I can buy.
c. The time (t) that it takes to make a pair of jeans and the number of pairs (p) that can be made in one day.
d. The price (P) of petrol and the amount (L) that can be bought for $80.
e. The price (P) of petrol and the cost (C) of buying 80 L.
f. The number of questions (n) in a test and the amount of time (t) available to answer each one.

2. List three examples of inverse proportion.

3. **WE15** y varies inversely as x and $y = 100$ when $x = 10$.
   a. Calculate the constant of proportionality, k, and hence the rule relating x and y.
   b. Plot a graph of the relationship between x and y, for values of x that are positive factors of k less than 21.

4. p is inversely proportional to q and $p = 12$ when $q = 4$.
   a. Calculate the constant of proportionality, k, and hence the rule relating p and q.
   b. Plot a graph of the relationship between q and p, for values of q that are positive factors of k less than 11.

5. y varies inversely as x and $y = 42$ when $x = 1$.
   a. Calculate the constant of proportionality, k, and hence the rule relating x and y.
   b. Plot a graph of the relationship between x and y, for values of x from 1 to 10.

Understanding

6. **WE16** When a constant force is applied to an object, its acceleration is inversely proportional to its mass. When the acceleration of an object is 40 m/s^2, the corresponding mass is 100 kg.
   a. Calculate the constant of proportionality.
   b. Determine the rule relating mass and acceleration.
   c. Determine the acceleration of a 200 kg object.
   d. Determine the acceleration of a 1000 kg object.

7. The number of colouring pencils sold is inversely proportional to the price of each pencil.
   Two thousand pencils are sold when the price is $0.25 each.
   a. Calculate the constant of proportionality.
   b. Determine the number of pencils that could be sold for $0.20 each.
   c. Determine the number of pencils that could be sold for $0.50 each.



8. The time taken to complete a journey is inversely proportional to the speed travelled. A trip is completed in 4.5 hours travelling at 75 km per hour.
   a. Calculate the constant of proportionality.
   b. Determine how long, to the nearest minute, the trip would take if the speed was 85 km per hour.
   c. Determine the speed required to complete the journey in 3.5 hours, correct to 1 decimal place.
   d. Determine the distance travelled in each case.

9. The cost per person travelling in a charter plane is inversely proportional to the number of people in the charter group. It costs $350 per person when 50 people are travelling.
   a. Calculate the constant of variation.
   b. Determine the cost per person, to the nearest cent, if there are 75 people travelling.
   c. Determine how many people are required to reduce the cost to $250 per person.
   d. Determine the total cost of hiring the charter plane.



Reasoning

10. The electrical current in a wire is inversely proportional to the resistance of the wire to that current. There is a current of 10 amperes when the resistance of the wire is 20 ohms.

- **a.** Calculate the constant of proportionality.
- **b.** Determine the current possible when the resistance is 200 ohms.
- **c.** Determine the resistance of the wire when the current is 15 amperes.
- **d.** Justify your answer to parts **b** and **c** using a graph.

11. The pressure of an ideal gas is inversely proportional to the volume taken up by the gas. A balloon is filled with air so it takes up 3 L at a pressure of 5 atmospheres.



- **a.** Calculate the constant of proportionality.
- **b.** Determine the new volume of the balloon if the pressure was dropped to 0.75 atmospheres.
- **c.** Determine the pressure if the same amount of air took up a volume of 6 L.

12. Two equations relating the time of a trip, T, and the speed at which they travel, S, are given. For both cases the time is inversely proportional to the speed: $T_1 = \frac{5}{S_1}$ and $T_2 = \frac{7}{S_2}$. Explain what impact the different constants of proportionality have on the time of the trip.

Problem solving

13. The time it takes to pick a field of strawberries is inversely proportional to the number of pickers. It takes 2 people 5 hours to pick all of the strawberries in a field.



- **a.** Calculate the constant of proportionality.
- **b.** Determine the rule relating time (T) and the number of pickers (P).
- **c.** Determine the time spent if there are 6 pickers.

14. For a constant distance covered by a sprinter, the sprinter's speed is inversely proportional to their time. If a sprinter runs at a speed of 10.4 m/s, the corresponding time is 9.62 seconds.

- **a.** Calculate the constant of variation.
- **b.** Determine the rule relating speed (V) and time (T).
- **c.** Determine the time, correct to 2 decimal places, if they ran at a speed of 10.44 m/s.
- **d.** Determine the time, correct to 2 decimal places, if they ran at a speed of 6.67 m/s.

15. A holiday hostel is built to accommodate group bookings of up to 45 people. It is known that it would cost each of the individuals in a group of 20 people $67.50 per night to rent this venue. The cost of the venue will remain the same no matter how many people are part of the group booking.



- **a.** Write the rule for the cost per person (C) and the number of people in a group booking (n).
- **b.** If 6 people from the original group are no longer able to attend, determine the new cost per person.
- **c.** Calculate the cheapest possible cost per person.

LESSON
9.8 Sketching the graph of a reciprocal

LEARNING INTENTIONS

At the end of this lesson you should be able to:

- determine the equation of the vertical and horizontal asymptote of a hyperbola
- sketch the graph of a hyperbola, and understand the effect dilations and reflections have on the shape of the graph.

9.8.1 Hyperbolas

eles-4876

- The graph of a reciprocal function is called a hyperbola.
- A hyperbola is a function of the form $xy = k$ or $y = \frac{k}{x}$; that is, the product of the two variables is a fixed constant.

WORKED EXAMPLE 17 Creating a table of values in order to graph a hyperbola

Complete the table of values below and use it to plot the graph of $y = \frac{1}{x}$.

x	–3	–2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2	3
y									

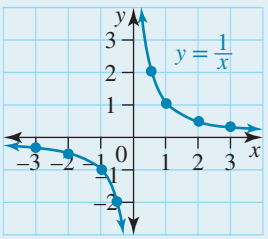
THINK

1. Substitute each x-value into the function $y = \frac{1}{x}$ to obtain the corresponding y-value.

WRITE/DRAW

x	–3	–2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2	3
y	$-\frac{1}{3}$	$-\frac{1}{2}$	-1	-2	Undef.	2	1	$\frac{1}{2}$	$\frac{1}{3}$

2. Draw a set of axes and plot the points from the table. Join them with a smooth curve.



- The graph in Worked example 17 has several important features.
  1. There is no function value (y-value) when $x = 0$. At this point the hyperbola is undefined. When this occurs, the line that the graph approaches ($x = 0$) is called a vertical asymptote.
  2. As x becomes larger and larger, the graph gets very close to but will never touch the x-axis. The same is true as x becomes smaller and smaller. The hyperbola also has a horizontal asymptote at $y = 0$.
  3. The hyperbola has two separate branches. It cannot be drawn without lifting your pen from the page and is an example of a discontinuous graph.
- Graphs of the form $y = \frac{k}{x}$ are the same basic shape as $y = \frac{1}{x}$ with y-values dilated by a factor of k.

General form of a hyperbola

The general form of a hyperbola with multiple transformations is given by the equation

$$y = \frac{a}{x-h} + k.$$

- **This graph has a horizontal asymptote with equation $y = k$. This occurs because the fraction $\frac{a}{x-h}$ cannot be made to equal 0, which means y can never be equal to k.**
- **The graph has a vertical asymptote with equation $x = h$. This is because the denominator of a fraction cannot be equal to 0, so x can never be equal to h.**
- **The graph will be reflected in the x-axis if $a < 0$.**

***Note:* Not all exponential graphs will have an x- or y-intercept. This will depend on the equation of the asymptotes.**

WORKED EXAMPLE 18 Identifying the asymptotes of a hyperbola

a. Plot the graph of $y = \frac{4}{x}$ for $-2 \leq x \leq 2$.

b. Write down the equation of each asymptote.

THINK

a. 1. Prepare a table of values taking x-values from -2 to 2. Fill in the table by substituting each x-value into the given equation to find the corresponding y-value.

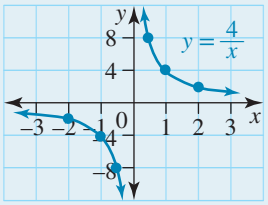
2. Draw a set of axes and plot the points from the table. Join them with a smooth curve.

b. Consider any lines that the curve approaches but does not cross.

WRITE/DRAW

a.

x	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2
y	-2	-4	-8	Undef.	8	4	2



b. Vertical asymptote is $x = 0$.
Horizontal asymptote is $y = 0$.

WORKED EXAMPLE 19 Sketching a reflected hyperbola

Plot the graph of $y = \frac{-3}{x}$ for $-3 \leq x \leq 3$.

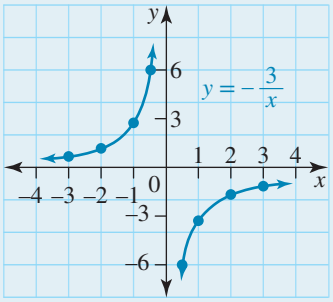
THINK

1. Draw a table of values and substitute each x-value into the given equation to find the corresponding y-value.

WRITE/DRAW

x	-3	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2	3
y	1	1.5	3	6	Undef.	-6	-3	-1.5	-1

2. Draw a set of axes and plot the points from the table. Join them with a smooth curve.



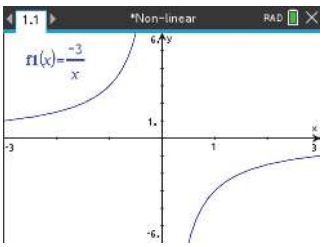
TI | THINK

1. In a new problem, open a Graphs page to sketch the graph, complete the entry line as:

$f1(x) = \dfrac{-3}{x}$

Then press ENTER.

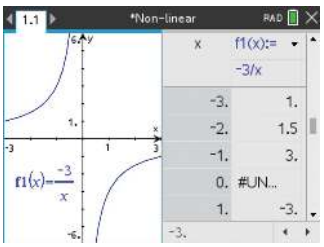
DISPLAY/WRITE



2. To see the table of values (for x and y values), press:

- Menu
- 7: Table
- 1: Split-screen Table (Ctrl+T)

Scroll up to see the negative x-values.



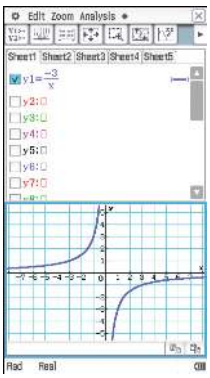
CASIO | THINK

1. Open the Graph & Table screen and complete the function entry line as:

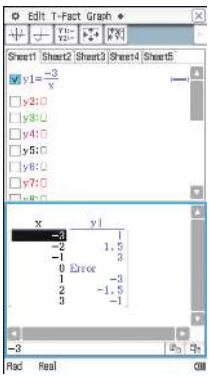
$f1(x) = \dfrac{-3}{x}$

Then tap the graphing icon and the graph will be displayed.

DISPLAY/WRITE



2. To see the table of values (for x and y values), tap on the table of values icon. The values displayed can be changed by editing the Table Input.



WORKED EXAMPLE 20 Determining the equation of the asymptotes of a hyperbola

Write the equation of the asymptotes of the following hyperbolas:

a. $y = \dfrac{3}{x+2}$ **b.** $y = -\dfrac{1}{x} + 4$ **c.** $y = \dfrac{5}{10-x} - 1$

THINK

a. Consider what value of x would make the denominator equal to 0, as well as any value added to the fraction.

b. Consider what value of x would make the denominator equal to 0, as well as any value added to the fraction.

c. Consider what value of x would make the denominator equal to 0, as well as any value added to the fraction.

WRITE

a. Vertical asymptote is $x = -2$
Horizontal asymptote is $y = 0$

b. Vertical asymptote is $x = 0$
Horizontal asymptote is $y = 4$

c. Vertical asymptote is $x = 10$
Horizontal asymptote is $y = -1$

WORKED EXAMPLE 21 Sketching a transformed hyperbola

Plot the graph of $y = -\dfrac{2}{x+1} + 3$.

THINK

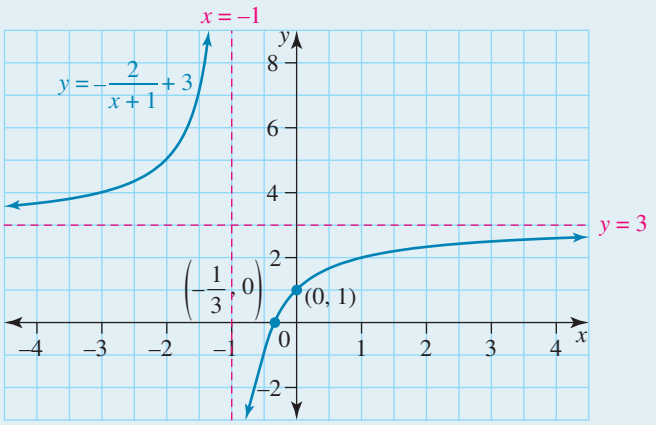
1. Consider what value of x would make the denominator equal to 0, as well as any value added to the fraction.
2. Calculate the y-intercept (when $x = 0$).
3. Calculate the x-intercept (when $y = 0$).
4. Sketch the graph with coordinates of both intercepts and both asymptotes labelled.

WRITE

Vertical asymptote is $x = -1$
Horizontal asymptote is $y = 3$

$$y = -\frac{2}{0+1} + 3 = 1$$
y-intercept is $(0, 1)$

$$0 = -\frac{2}{x+1} + 3$$
$$-3(x+1) = -2$$
$$x + 1 = \frac{-2}{-3}$$
$$x = \frac{2}{3} - 1 = -\frac{1}{3}$$
The x-intercept is $\left(-\frac{1}{3}, 0\right)$ $x = -1$



DISCUSSION

How could you summarise the effect of the transformations dealt with in this lesson on the shape of the basic hyperbola $y = \dfrac{1}{x}$?

on Resources

eWorkbook Topic 9 Workbook (worksheets, code puzzle and a project) (ewbk-2035)
Interactivities Individual pathway interactivity: The hyperbola (int-4610)
Hyperbolas (int-6155)

Exercise 9.8 Sketching the graph of a reciprocal

learnon

9.8 Quick quiz on | 9.8 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 4, 7, 10, 14, 17, 20	2, 5, 8, 11, 15, 18, 21	3, 6, 9, 12, 13, 16, 19, 22, 23

Fluency

1. WE17 Complete the table of values below and use it to plot the graph of $y = \frac{10}{x}$.

x	–5	–4	–3	–2	–1	0	1	2	3	4	5
y											

WE18 For questions 2 to 4, answer the following.

a. Plot the graph of each hyperbola.

b. Write down the equation of each asymptote.

2. $y = \frac{5}{x}$

3. $y = \frac{20}{x}$

4. $y = \frac{100}{x}$

5. Draw the graphs of $y = \frac{2}{x}$, $y = \frac{3}{x}$ and $y = \frac{4}{x}$, on the same set of axes.

6. Describe the effect of increasing the value of k on the graph of $y = \frac{k}{x}$.

7. WE19 Plot the graph of $y = \frac{-10}{x}$ for $-5 \leq x \leq 5$.

8. Draw the graphs of $y = \frac{6}{x}$ and $y = \frac{-6}{x}$, on the same set of axes.

9. Describe the effect of the negative in $y = \frac{-k}{x}$.

10. Complete the table of values below and use the points to plot $y = \frac{1}{x-1}$. State the equation of the vertical asymptote.

x	–3	–2	–1	0	1	2	3	4
y								

Understanding

11. Plot the graph of each hyperbola and label the vertical asymptote.

a. $y = \frac{1}{x-2}$

b. $y = \frac{1}{x-3}$

12. Plot the graph of $y = \frac{1}{x+1}$ and label the vertical asymptote.

13. Describe the effect of a in $y = \frac{1}{x-a}$.

For questions **14** to **16**:

i. **WE20** write the equations of the asymptotes of the following hyperbolas

ii. sketch the graphs of the following hyperbolas.

14. a. $y = \frac{-4}{x+1}$ **b.** $y = \frac{2}{x-1}$

15. a. $y = \frac{5}{x+2}$ **b.** $y = \frac{3}{x+2} - 2$

16. a. $y = -\frac{4}{x+2} + 1$ **b.** $y = \frac{7}{3-x} + 5$

Reasoning

17. Give an example of the equation of a hyperbola that has a vertical asymptote of:

a. $x = 3$ **b.** $x = -10$.

18. Give an example of a hyperbola that has the following key features.

a. Asymptotes of $x = 2$ and $y = 3$

b. Asymptotes of $x = -2$ and $y = 4$ and a y-intercept of -3

19. The graph of $y = \frac{1}{x}$ is reflected in the x-axis, dilated by a factor of 2 parallel to the y-axis or from the x-axis and translated 3 units to the left and down 1 unit. Determine the equation of the resulting hyperbola and give the equations of any asymptotes.

Problem solving

20. a. Complete the following table in order to graph the hyperbola defined by $y = \frac{1}{x^2}$.

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y						

b. This hyperbola is also known as a truncus. Give the equations of any asymptotes.

c. Determine the equation of the truncus which results when $y = \frac{1}{x^2}$ is reflected in the x-axis.

d. Determine the equation of the truncus which results when $y = \frac{1}{x^2}$ is reflected in the y-axis.

21. The temperature of a cup of coffee as it cools is modelled by the equation $T = \frac{780}{t+10} + 22$, where T represents the temperature in °C and t is the time in minutes since the coffee was first made.

a. State the initial temperature of the cup of coffee.

b. Calculate the temperature, to 1 decimal place, of the coffee after it has been left to cool for an hour.

c. A coffee will be too hot to drink unless its temperature has dropped below 50 °C. Determine how long someone would have to wait, to the nearest minute, before drinking the coffee.

d. Explain whether the coffee will ever cool to 0 °C. Justify your answer.

22. Consider the equation $y = \dfrac{6}{x^2 - 4} + 2$.

a. Calculate the x- and y-intercepts of the graph of the equation.

b. Use your knowledge of quadratics and hyperbolas to state the equation of any asymptotes of the above equation.

c. Using a table of values, or otherwise, sketch the graph of the equation.

23. Consider the truncus defined by $y = \dfrac{1}{x^2}$. This hyperbola is reflected in the x-axis, dilated by a factor of 3 parallel to the y-axis or from the x-axis and translated 1 unit to the left and up 2 units. Determine the equation of the resulting hyperbola and give the equations of any asymptotes.

LESSON
9.9 Sketching the circle

LEARNING INTENTIONS

At the end of this lesson you should be able to:

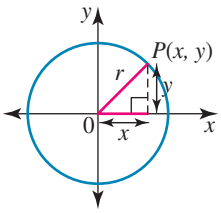
- identify the centre and radius of the graph of a circle from its equation and then sketch its graph
- use completing the square to turn an equation in expanded form into the form of an equation of a circle.

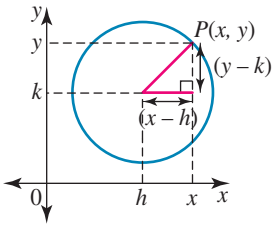
9.9.1 Circles

eles-5352

- A circle is the path traced out by a point at a constant distance (the radius) from a fixed point (the centre).
- Consider the circles shown. The first circle has its centre at the origin and radius r. Let $P(x, y)$ be a point on the circle.
  By Pythagoras: $x^2 + y^2 = r^2$.
  This relationship is true for all points, P, on the circle.
  The equation of a circle, with centre $(0, 0)$ and radius r is:

$$x^2 + y^2 = r^2$$





- If the circle is translated h units to the right, parallel to the x-axis, and k units upwards, parallel to the y-axis, then the centre of the circle will become (h, k). The radius will remain unchanged.

Equation of a circle

- **The equation of a circle, with centre $(0, 0)$ and radius r, is:**

$$x^2 + y^2 = r^2$$

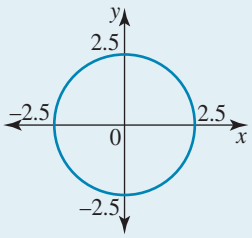
- **The equation of a circle, with centre (h, k) and radius r, is:**

$$(x - h)^2 + (y - k)^2 = r^2$$

Note: **We can produce an ellipse by dilating a circle from one or both of the axes.**

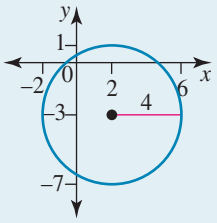
WORKED EXAMPLE 22 Identifying the centre and radius of a circle from its equation

Sketch the graph of $4x^2 + 4y^2 = 25$, stating the centre and radius.

THINK	WRITE/DRAW
1. Express the equation in general form by dividing both sides by 4.	$x^2 + y^2 = r^2$ $4x^2 + 4y^2 = 25$ $x^2 + y^2 = \frac{25}{4}$
2. State the coordinates of the centre.	Centre $(0, 0)$
3. Calculate the length of the radius by taking the square root of both sides. (Ignore the negative results.)	$r^2 = \frac{25}{4}$ $r = \frac{5}{2}$ Radius $= 2.5$ units
4. Sketch the graph.	

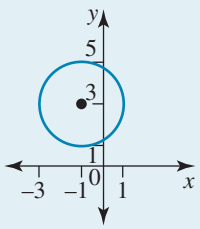
WORKED EXAMPLE 23 Sketching a circle from its equation

Sketch the graph of $(x - 2)^2 + (y + 3)^2 = 16$, clearly showing the centre and radius.

THINK	WRITE/DRAW
1. Express the equation in general form.	$(x - h)^2 + (y - k)^2 = r^2$ $(x - 2)^2 + (y + 3)^2 = 16$
2. State the coordinates of the centre.	Centre $(2, -3)$
3. State the length of the radius.	$r^2 = 16$ $r = 4$ Radius $= 4$ units
4. Sketch the graph.	

WORKED EXAMPLE 24 Completing the square to determine the equation of a circle

Sketch the graph of the circle $x^2 + 2x + y^2 - 6y + 6 = 0$.

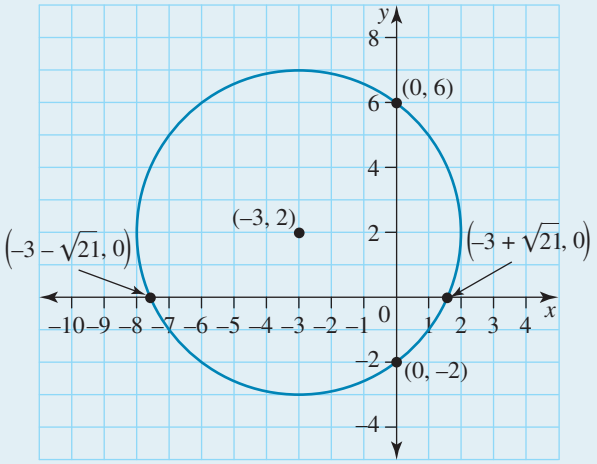
THINK	WRITE/DRAW
1. Express the equation in general form by completing the square on the x terms and again on the y terms.	$(x-h)^2 + (y-k)^2 = r^2$ $x^2 + 2x + y^2 - 6y + 6 = 0$ $(x^2 + 2x + 1) - 1 + (y^2 - 6y + 9) - 9 + 6 = 0$ $(x+1)^2 + (y-3)^2 - 4 = 0$ $(x+1)^2 + (y-3)^2 = 4$
2. State the coordinates of the centre.	Centre $(-1, 3)$
3. State the length of the radius.	$r^2 = 4$ $r = 2$ Radius $= 2$ units
4. Sketch the graph.	

WORKED EXAMPLE 25 Sketching a circle including the intercepts

Sketch the graph of the circle $(x+3)^2 + (y-2)^2 = 25$. Make sure to show all axial intercepts.

THINK	WRITE
1. State the coordinate of the centre.	Centre $(-3, 2)$
2. State the length of the radius.	$r^2 = 25$ $r = 5$ Radius $= 5$ units
3. To determine the y-intercepts let $x = 0$.	$(0+3)^2 + (y-2)^2 = 25$ $(y-2)^2 = 25 - 9$ $(y-2)^2 = 16$ $y = \pm 4$ $y = 4 + 2$ or $4 + 2$ The y-intercepts are $(0, 6)$ and $(0, -2)$
4. To determine the x-intercepts let $y = 0$.	$(x+3)^2 + (0-2)^2 = 25$ $(x+3)^2 = 25 - 4$ $x + 3 = \pm\sqrt{21}$ $x = -3 \pm \sqrt{21}$ The x-intercepts are $\left(-3 + \sqrt{21}, 0\right)$ and $\left(-3 - \sqrt{21}, 0\right)$.

5. Sketch the graph.



TI | THINK

On a Graphing page, press:

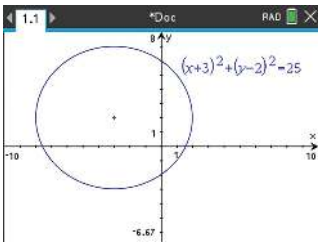
- MENU
- 3: Graph Entry/Edit
- 2: Relation

Then type:

$(x+3)^2+(y-2)^2=25$

Press ENTER and the graph will be displayed.

DISPLAY/WRITE



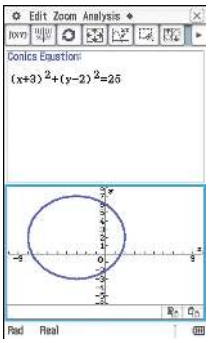
CASIO | THINK

On a Conics screen, type:

$(x+3)^2+(y-2)^2=25$

Press the graphing icon and the graph will be displayed.

DISPLAY/WRITE



DISCUSSION

How could you write equations representing a set of concentric circles (circles with the same centre, but different radii)?

Resources

eWorkbook Topic 9 Workbook (worksheets, code puzzle and a project) (ewbk-2035)

Interactivities Individual pathway interactivity: The circle (int-4611)
Graphs of circles (int-6156)

Exercise 9.9 Sketching the circle

learnon

9.9 Quick quiz on | 9.9 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 4, 7, 10, 13, 17	2, 5, 8, 11, 14, 15, 18	3, 6, 9, 12, 16, 19, 20

Fluency

WE22 For questions 1 to 3, sketch the graphs of the following, stating the centre and radius of each.

1. a. $x^2 + y^2 = 49$ b. $x^2 + y^2 = 4^2$
2. a. $x^2 + y^2 = 36$ b. $x^2 + y^2 = 81$
3. a. $2x^2 + 2y^2 = 50$ b. $9x^2 + 9y^2 = 100$

WE23 For questions 4 to 6, sketch the graphs of the following, clearly showing the centre and the radius.

4. a. $(x-1)^2 + (y-2)^2 = 5^2$ b. $(x+2)^2 + (y+3)^2 = 6^2$
5. a. $(x+3)^2 + (y-1)^2 = 49$ b. $(x-4)^2 + (y+5)^2 = 64$
6. a. $x^2 + (y+3)^2 = 4$ b. $(x-5)^2 + y^2 = 100$

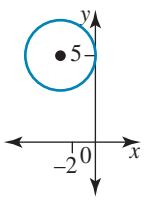
WE24 For questions 7 to 9, sketch the graphs of the following circles.

7. a. $x^2 + 4x + y^2 + 8y + 16 = 0$ b. $x^2 - 10x + y^2 - 2y + 10 = 0$
8. a. $x^2 - 14x + y^2 + 6y + 9 = 0$ b. $x^2 + 8x + y^2 - 12y - 12 = 0$
9. a. $x^2 + y^2 - 18y - 19 = 0$ b. $2x^2 - 4x + 2y^2 + 8y - 8 = 0$

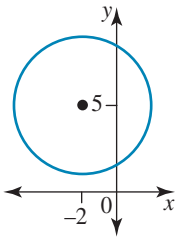
Understanding

10. **MC** The graph of $(x-2)^2 + (y+5)^2 = 4$ is:

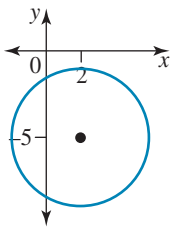
A.



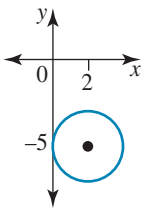
B.



C.



D.



11. MC The centre and radius of the circle $(x+1)^2+(y-3)^2=4$ is:

A. $(1,-3), 4$ **B.** $(-1,3), 2$

C. $(3,-1), 4$ **D.** $(1,-3), 2$

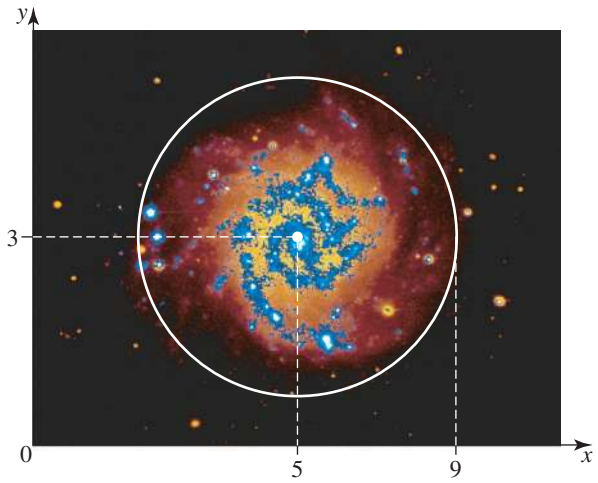
12. MC The centre and radius of the circle with equation $x^2+y^2+8x-10y=0$ is:

A. $(4,5), \sqrt{41}$ **B.** $(-4,5), 9$

C. $(4,-5), 3$ **D.** $(-4,5), \sqrt{41}$

Reasoning

13. Determine the equation representing the outer edge of the galaxy as shown in the photo below, using the astronomical units provided.



14. Circular ripples are formed when a water drop hits the surface of a pond.
If one ripple is represented by the equation $x^2+y^2=4$ and then 3 seconds later by $x^2+y^2=190$, where the length of measurements are in centimetres:

a. identify the radius (in cm) of the ripple in each case

b. calculate how fast the ripple is moving outwards.
(State your answers to 1 decimal place.)

15. Two circles with equations $x^2+y^2=4$ and $(x-1)^2+y^2=9$ intersect. Determine the point(s) of intersection. Show your working.

16. a. Graph the line $y=x$, the parabola $y=x^2$ and the circle $x^2+y^2=1$ on the one set of axes.

b. Evaluate algebraically the points of intersection of:

i. the line and the circle

ii. the line and the parabola

iii. the parabola and the circle.

Problem solving

17. Sketch the graph of $(x+6)^2+(y-3)^2=100$ showing all axial intercepts.

18. Determine the points of intersection between the quadratic equation $y=x^2-5$ and the circle given by $x^2+y^2=25$.

19. Determine the point(s) of intersection of the circles $x^2+y^2-2x-2y-2=0$ and $x^2+y^2-8x-2y+16=0$ both algebraically and graphically.

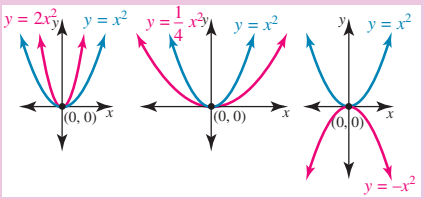
20. The general equation of a circle is given by $x^2+y^2+ax+by+c=0$. Determine the equation of the circle which passes through the points $(4,5)$, $(2,3)$ and $(0,5)$. State the centre of the circle and its radius.

LESSON
9.10 Review

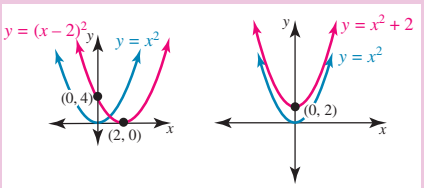
9.10.1 Topic summary

Transformations of the parabola

- The parabola with the equation $y = x^2$ has a turning point at (0,0).
- A **dilation** from the x-axis stretches the parabola.
- A **dilation** of factor a from the x-axis produces the equation $y = ax^2$ and:
  - will produce a narrow graph for $a > 0$
  - will produce a wider graph for $0 < a < 1$
  - For $a < 0$ (a is negative), the graph is **reflected** in the x-axis.



- The graph of $y = (x - h)^2$ has been translated left/right h units.
- The graph of $y = x^2 + k$ has been translated up/down k units.

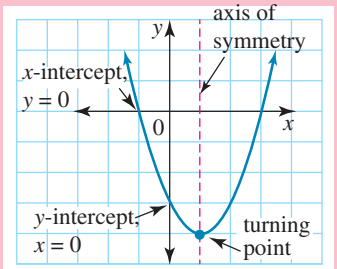


Inverse proportion and reciprocals

- The graph of a reciprocal function is called a hyperbola.
- If y and x are inversely proportional then $y = \frac{1}{x}$ or $y = \frac{k}{x}$, where k is the constant of proportionality.
- The graph of an equation in the form $y = \frac{k}{x}$ is called a hyperbola.
- The general form of a hyperbola is $y = \frac{a}{x - h} + k$ and has:
  - vertical asymptote at $x = h$
  - horizontal asymptote at $y = k$
  - y-intercept at $\left(0, \frac{-a}{h + k}\right)$
  - x-intercept at $\left(\frac{-a}{k + h}, 0\right)$.
- If $a < 0$ the graph is reflected in the x-axis.

The parabola

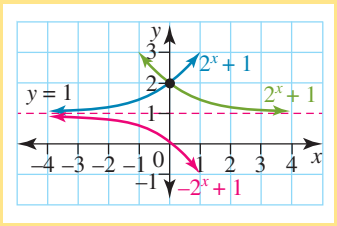
- The graph of a quadratic equation is called a parabola.



NON-LINEAR RELATIONSHIPS

Exponential graphs

- Equations $y = a^x$ are called exponential functions.
- The general form $y = a^{(x-h)} + k$ has:
  - a horizontal asymptote at $y = k$
  - a y-intercept at $(0, a^{-h} + k)$
- A negative in front of a results in a reflection in the x-axis.
- A negative in front of x results in a reflection in the y-axis.



Circles

- $x^2 + y^2 = r^2$ is a circle with centre (0, 0) and radius r.
- $(x - h)^2 + (y - k)^2 = r^2$ is a circle with centre (h, k) and radius r.
- Use completing the square to determine the equation of a circle:
  $x^2 + 6x + y^2 + 8x = 0$
  $x^2 + 6x + 9 + y^2 + 8x + 16 = 9 + 16$
  $(x + 3)^2 + (y + 4)^2 = 25$

Turning point form

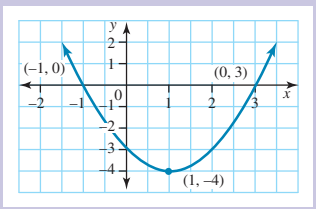
- Completing the square allows us to write a quadratic in turning point form.
- Turning point form:
  $y = a(x - h)^2 + k$
  - Turning point at (h, k)
  - The y-intercept is determined by setting $x = 0$ and solving for y.
  - x-intercept/s is determined by setting $y = 0$ and solving for x.

Expanded form

- The expanded form of a quadratic is $y = ax^2 + bx + c$.
- There are two methods for sketching: factorising or using the quadratic formula.

e.g. Consider $y = x^2 - 2x - 3$.

- To sketch, first factorise:
  $y = (x - 3)(x + 1)$
- Use the null-factor law to determine the x-intercepts:
  when $y = 0$, $0 = (x - 3)(x + 1)$
  so $x = 3$ or -1.
- Axis of symmetry: $x = \frac{3 + (-1)}{2} = 1$
- y-coordinate of the turning point:
  $x = 1$, $y = (1 - 3)(1 + 1) = -4$.
  Turning point is $(1, -4)$. The y-intercept when $x = 0$ is $(0, -3)$.



Quadratic formula

- For $y = ax^2 + bx + c$
  - axis of symmetry: $x = -\frac{b}{2a}$
  - turning point: $\left(-\frac{b}{2a}, c - \frac{b^2}{4a}\right)$
- x-intercepts can be calculated as: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Discriminant: $\Delta = b^2 - 4ac$ is used to determine the number of x-intercepts.
  - $\Delta > 0$, two x-intercepts
  - $\Delta = 0$, one x-intercept
  - $\Delta < 0$, no x-intercept

9.10.2 Success criteria

Tick the column to indicate that you have completed the lesson and how well you have understood it using the traffic light system.

(**Green:** I understand; **Yellow:** I can do it with help; **Red:** I do not understand)

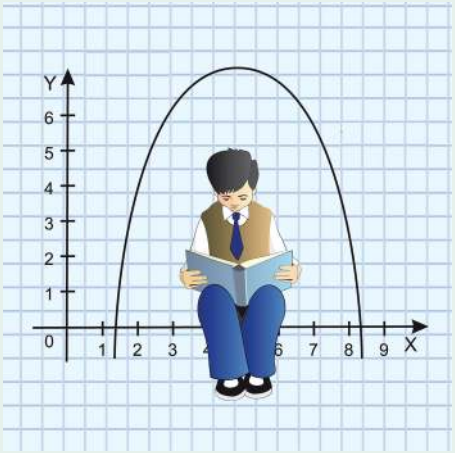
Lesson	Success criteria			
9.2	I can create a table of values and use this to sketch the graph of a parabola.			
	I can identify the axis of symmetry, turning point and y-intercept of a parabola.			
9.3	I am familiar with the key features of the basic quadratic $y = x^2$.			
	I can determine whether a dilation has made the graph of quadratic narrower or wider			
	I can determine whether a translation has moved the graph of a quadratic function left/right or up/down			
	I can sketch a quadratic equation that has undergone any of the following transformations: dilations, reflections or translations.			
9.4	I can determine the axis of symmetry and turning point of a quadratic expressed in turning point form.			
	I can calculate the y-intercept and any x-intercepts of a quadratic in turning point form.			
9.5	I can determine the x-intercept/s, y-intercept and turning point of a quadratic equation and sketch its graph by first factorising the equation.			
	I can determine the x-intercept/s, y-intercept and turning point of a quadratic equation and sketch its graph using the quadratic formula.			
9.6	I can determine the equation of the asymptote of an exponential equation and sketch a graph of the equation.			
	I can sketch the graph of an exponential equation after it has undergone any number of transformations that may include dilations, a reflection in the x- or y-axis, or translations.			
9.7	I can calculate the constant of proportionality, k, between two variables that are inversely proportional to each other.			
	I can determine the rule between two inversely proportional variables and plot the graph of their relationship.			
9.8	I can determine the equation of the vertical and horizontal asymptote of a hyperbola.			
	I can sketch the graph of a hyperbola, and understand the effect dilations and reflections have on the shape of the graph.			
9.9	I can identify the centre and radius of the graph of a circle from its equation and then sketch its graph.			
	I can use completing the square to turn an equation in expanded form into the form of an equation of a circle.			

9.10.3 Project

Parametric equations

You are familiar with the quadratic equation $y = x^2$ and its resulting graph. Let us consider an application of this equation by forming a relationship between x and y through a third variable, say, t.

$$x = t \text{ and } y = t^2$$



It is obvious that these two equations are equivalent to the equation $y = x^2$. This third variable t is known as a parameter, and the two equations are now called parametric equations. We cannot automatically assume that the resulting graph of these two parametric equations is the same as that of $y = x^2$ for all real values of x. It is dependent on the range of values of t.

Consider the parametric equations $x = t$ and $y = t^2$ for values of the parameter $t \geq 0$ for questions **1** to **3**.

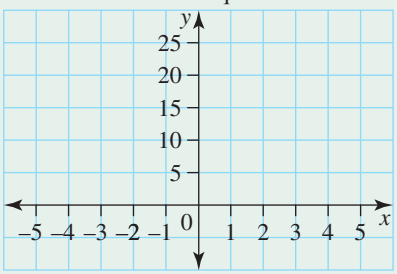
1. Complete the following table by calculating x- and y-values from the corresponding t-value.

t	x	y
0		
1		
2		
3		
4		
5		

2. Graph the x-values and corresponding y-values on this Cartesian plane. Join the points with a smooth curve and place an arrow on the curve to indicate the direction of increasing t-values.

3. Is there any difference between this graph and that of $y = x^2$? Explain your answer.

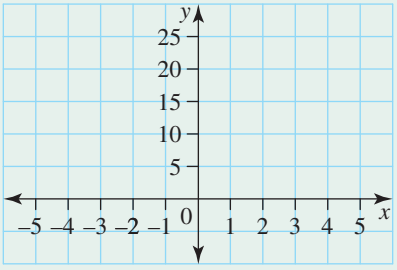
Parametric equations



4. Consider now the parametric equations $x = 1 - t$ and $y = (1 - t)^2$. These clearly are also equivalent to the equation $y = x^2$. Complete the table and draw the graph of these two equations for values of the parameter $t \geq 0$. Draw an arrow on the curve in the direction of increasing t-values.

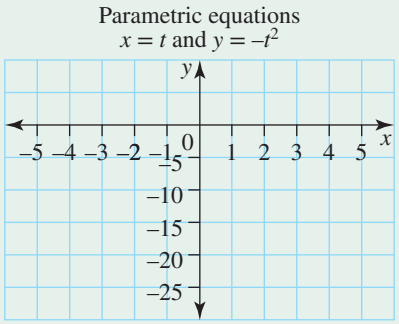
t	x	y
0		
1		
2		
3		
4		
5		

Parametric equations



Describe the shape of your resulting graph. What values of the parameter t would produce the same curve as that obtained in question **2**?

5. The graph of $y = -x^2$ is a reflection of $y = x^2$ in the x-axis. Construct a table and draw the graph of the parametric equations $x = t$ and $y = -t^2$ for parameter values $t \geq 0$. Remember to place an arrow on the curve in the direction of increasing t-values.
6. Without constructing a table, predict the shape of the graph of the parametric equations $x = 1 - t$ and $y = -(1 - t)^2$ for parameter values $t \geq 0$. Draw a sketch of the shape.



Resources

eWorkbook Topic 9 Workbook (worksheets, code puzzle and a project) (ewbk-2035)
Interactivities Crossword (int-2851)
Sudoku puzzle (int-3596)

Exercise 9.10 Review questions

learnon

Fluency

1. MC The turning point for the graph $y = 3x^2 - 4x + 9$ is:
   A. $\left(\frac{1}{3}, 1\frac{2}{3}\right)$
   B. $\left(\frac{1}{3}, \frac{2}{3}\right)$
   C. $\left(\frac{1}{6}, 1\frac{1}{6}\right)$
   D. $\left(\frac{2}{3}, 7\frac{2}{3}\right)$

2. MC Select which graph of the following equations has the x-intercepts closest together.
   A. $y = x^2 + 3x + 2$
   B. $y = x^2 + x - 2$
   C. $y = 2x^2 + x - 15$
   D. $y = 4x^2 + 27x - 7$

3. MC Select which graph of the equations below has the largest y-intercept.
   A. $y = 3(x - 2)^2 + 9$
   B. $y = 5(x - 1)^2 + 8$
   C. $y = 2(x - 1)^2 + 19$
   D. $y = 2(x - 5)^2 + 4$

4. MC The translation required to change $y = x^2$ into $y = (x - 3)^2 + \frac{1}{4}$ is:
   A. right 3, up $\frac{1}{4}$
   B. right 3, down $\frac{1}{4}$
   C. left 3, down $\frac{1}{4}$
   D. left 3, up $\frac{1}{4}$

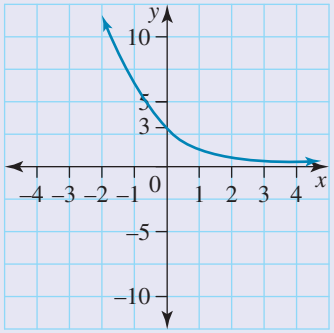
5. MC If y is inversely proportional to x, then select which of the following statements is true.
   A. $x + y$ is a constant value.
   B. $y \div x$ is a constant value.
   C. $y \times x$ is a constant value.
   D. $y - x$ is a constant value.

6. MC The number of calculators a company sells is inversely proportional to the selling price. If a company can sell 1000 calculators when the price is $22, determine how many they could sell if they reduced the price to $16.

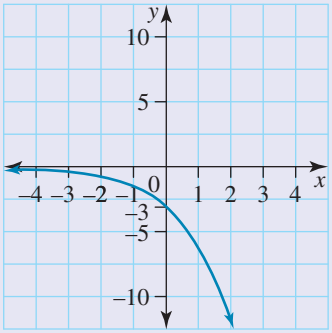
A. 2000 **B.** 727 **C.** 6000 **D.** 1375

7. MC The graph of $y = -3 \times 2^x$ is best represented by:

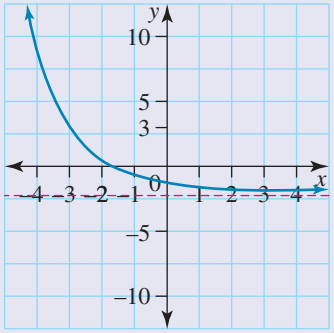
A.



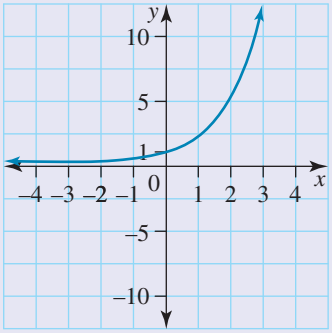
B.



C.



D.



8. Use the completing the square method to determine the turning point for each of the following graphs.

a. $y = x^2 - 8x + 1$

b. $y = x^2 + 4x - 5$

9. For the graph of the equation $y = x^2 + 8x + 7$, produce a table of values for the x-values between -9 and 1, and then plot the graph. Show the y-intercept and turning point. From your graph, state the x-intercepts.

10. For each of the following, determine the coordinates of the turning point and the x- and y-intercepts and sketch the graph.

a. $y = (x - 3)^2 + 1$

b. $y = 2(x + 1)^2 - 5$

11. For the equation $y = -x^2 - 2x + 15$, sketch the graph and determine the x- and y-intercepts and the coordinates of the turning point.

12. Draw the graph of $y = 10 \times 3^x$ for $-2 \leq x \leq 4$.

13. Draw the graph of $y = 10^{-x}$ for $-2 \leq x \leq 2$.

14. For the exponential function $y = 5^x$:

a. complete the table of values

x	y
−3	
−2	
−1	
0	
1	
2	
3	

b. plot the graph.

15. a. On the same axes, draw the graphs of $y = (1.2)^x$ and $y = (1.5)^x$.

b. Use your answer to part **a** to explain the effect of changing the value of a in the equation of $y = a^x$.

16. a. On one set of axes, draw the graphs of $y = 2 \times 3^x$, $y = 5 \times 3^x$ and $y = \frac{1}{2} \times 3^x$

b. Use your answer to part **a** to explain the effect of changing the value of k in the equation of $y = ka^x$.

17. a. On the same set of axes, sketch the graphs of $y = (2.5)^x$ and $y = (2.5)^{-x}$.

b. Use your answer to part **a** to explain the effect of a negative index on the equation $y = a^x$.

18. Sketch each of the following.

a. $y = \frac{4}{x}$

b. $y = -\frac{2}{x}$

19. Sketch $y = \frac{-3}{x-2}$.

20. Give an example of an equation of a hyperbola that has a vertical asymptote at $x = -3$.

21. Sketch each of these circles. Clearly show the centre and the radius.

a. $x^2 + y^2 = 16$

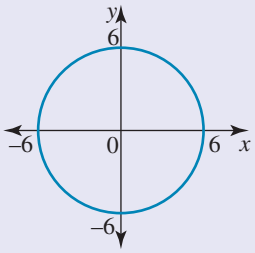
b. $(x-5)^2 + (y+3)^2 = 64$

22. Sketch the following circles. Remember to first complete the square.

a. $x^2 + 4x + y^2 - 2y = 4$

b. $x^2 + 8x + y^2 + 8y = 32$

23. Determine the equation of this circle.



Problem solving

24. The height, h, in metres of a golf ball t seconds after it is hit is given by the formula $h = 4t - t^2$.
- **a.** Sketch the graph of the path of the ball.
- **b.** Calculate the maximum height the golf ball reaches.
- **c.** Determine how long it takes for the ball to reach the maximum height.
- **d.** Determine how long is it before the ball lands on the ground after it has been hit.

25. A soccer ball is kicked upwards in the air. The height, h, in metres, t seconds after the kick is modelled by the quadratic equation $h = -5t^2 + 20t$.
- **a.** Sketch the graph of this relationship.
- **b.** Determine how many seconds the ball is in the air for.
- **c.** Determine how many seconds the ball is above a height of 15 m. That is, solve the quadratic inequation $-5t^2 + 20t > 15$.
- **d.** Calculate how many seconds the ball is above a height of 20 m.

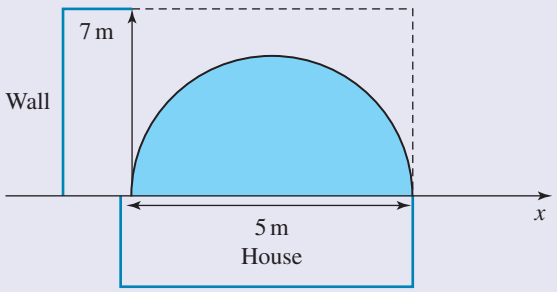
26. The height of the water level in a cave is determined by the tides. At any time, t, in hours after 9 am, the height, $h(t)$, in metres, can be modelled by the function $h(t) = t^2 - 12t + 32, 0 \leq t \leq 12$.
- **a.** Determine the values of t for which the model is valid. Write your answer in interval notation.
- **b.** State the initial height of the water.
- **c.** Bertha has dropped her keys onto a ledge which is 7 metres from the bottom of the cave. By using a graphics calculator, determine the times in which she would be able to climb down to retrieve her keys. Write your answers correct to the nearest minute.

27. When a drop of water hits the flat surface of a pool, circular ripples are made. One ripple is represented by the equation $x^2 + y^2 = 9$ and 5 seconds later, the ripple is represented by the equation $x^2 + y^2 = 225$, where the lengths of the radii are in cm.
- **a.** State the radius of each of the ripples.
- **b.** Sketch these graphs.
- **c.** Evaluate how fast the ripple are moving outwards.
- **d.** If the ripple continues to move at the same rate, determine when it will hit the edge of the pool which is 2 m from its centre.

28. Seventy grams of ammonium sulfate crystals are dissolved in 0.5 L of water.
- **a.** Determine the concentration of the solution in g/mL.
- **b.** Another 500 mL of water is added. Determine the concentration of the solution now.

29. Use technology to graph the equations $y = x^2$ and $y = 2^x$ on the same axes. Using the graphs, determine the solution to $x^2 = 2^x$. Give your answers correct to 2 decimal places. How can you be sure you have found all the solutions?

30. A grassed area is planted in a courtyard that has a width of 5 metres and length of 7 metres. The shape of the grassed area is described by the function $P = -x^2 + 5x$, where P is the distance, in metres, from the house and x is the distance, in metres from the side wall. The diagram represents this information on a Cartesian plane.



a. In terms of P, write down an inequality that describes the region where the grass has been planted.

b. Determine the maximum distance the grass area is planted from the house.

c. The owners of the house have decided that they would prefer all of the grass to be within a maximum distance of 3.5 metres from the house. The shape of the lawn following this design can be described by the equation $N(x) = ax^2 + bx + c$

i. Using algebra, show that this new design can be described by the function $N(x) = -0.56x(x - 5)$.

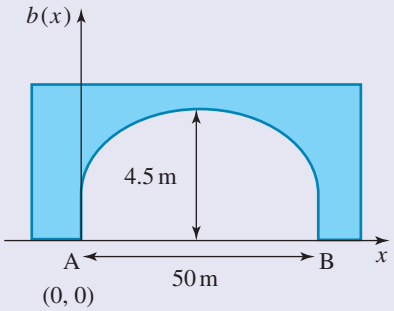
ii. Describe the transformation that maps $P(x)$ to $N(x)$.

d. If the owners decide on the first design, $P(x)$, the percentage of area within the courtyard without grass is 40.5%. By using any method, determine the approximate percentage of area of courtyard without lawn with the new design, $N(x)$.

31. A stone arch bridge has a span of 50 metres. The shape of the curve AB can be modelled using a quadratic equation.

a. Taking A as the origin $(0, 0)$ and given that the maximum height of the arch above the water level is 4.5 metres, show using algebra, that the shape of the arch can be modelled using the equation $b(x) = -0.0072x^2 + 0.36x$, where $b(x)$ is the vertical height of the bridge, in metres, and x is the horizontal distance, in metres.

b. A floating platform p metres high is towed under the bridge. Given that the platform needs to have a clearance of at least 30 centimetres on each side, explain why the maximum value of p is 10.7 centimetres.





To test your understanding and knowledge of this topic, go to your learnON title at www.jacplus.com.au and complete the **post-test**.

Answers

Topic 9 Non-linear relationships

9.1 Pre-test

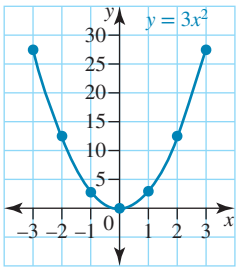
1.

x	-3	-2	-1	0
y	9.5	5	1.5	-1

2. $x = -1$
3. B
4. A
5. D
6. $(-4, -1)$
7. $b = 20$
8. D
9. D
10. $(-0.79, 0.62)$ and $(0.79, 0.62)$
11. B
12. C
13. a. 2 b. 1
14. D
15. B

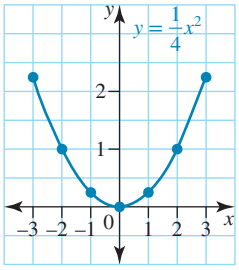
9.2 Plotting parabolas from a table of values

1. a.



$x = 0, (0, 0)$

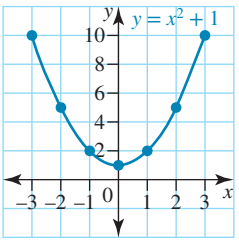
b.



$x = 0, (0, 0)$

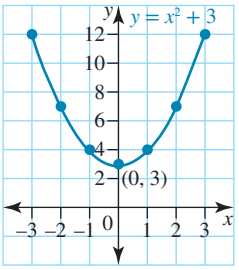
2. Placing a number greater than 1 in front of x^2 makes the graph thinner. Placing a number greater than 0 but less than 1 in front of x^2 makes the graph wider.

3. a.



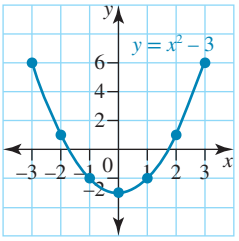
$x = 0, (0, 1), 1$

b.



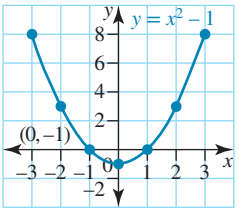
$x = 0, (0, 3), 3$

c.



$x = 0, (0, -3), -3$

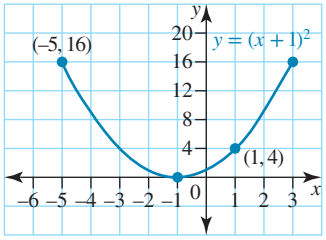
d.



$x = 0, (0, -1), -1$

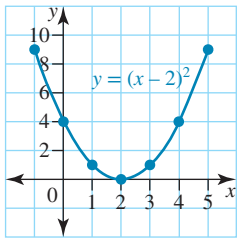
4. Adding a number raises the graph of $y = x^2$ vertically that number of units. Subtracting a number lowers the graph of $y = x^2$ vertically that number of units.

5.



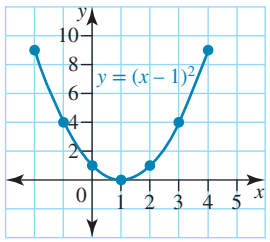
$x = -1, (-1, 0), 1$

6.



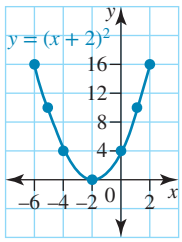
$x = 2, (2, 0), 4$

7.



$x = 1, (1, 0), 1$

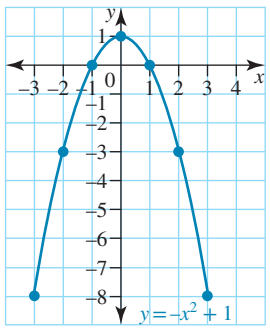
8.



$x = -2, (-2, 0), 4$

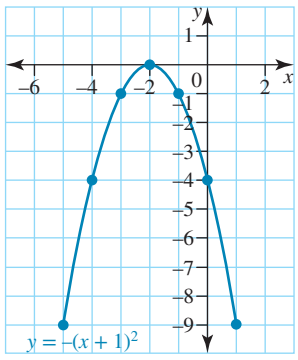
9. Adding a number moves the graph of $y = x^2$ horizontally to the left by that number of units. Subtracting a number moves the graph of $y = x^2$ horizontally to the right by that number of units.

10.



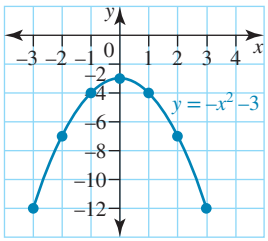
$x = 0, (0, 1), 1$

11.



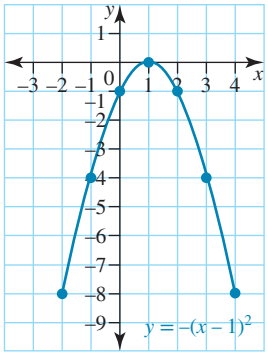
$x = -2, (-2, 0), -4$

12.



$x = 0, (0, -3), -3$

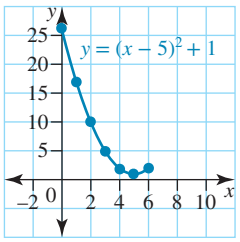
13.



$x = 1, (1, 0), -1$

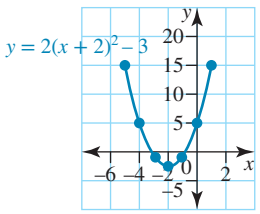
14. The negative sign inverts the graph of $y = x^2$. The graphs with the same turning points are: $y = x^2 + 1$ and $y = -x^2 + 1$; $y = (x - 1)^2$ and $y = -(x - 1)^2$; $y = (x + 2)$ and $y = -(x + 2)^2$; $y = x^2 - 3$ and $y = -x^2 - 3$. They differ in that the first graph is upright while the second graph is inverted.

15. a.



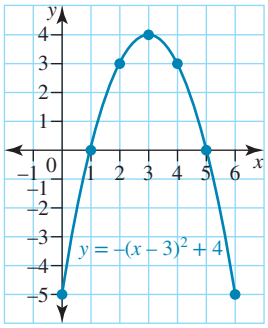
b. $x = 5$ **c.** (5, 1), min **d.** 26

16. a.



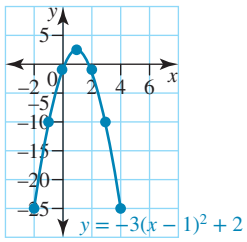
b. $x = -2$ **c.** $(-2, -3)$, min
d. 5

17. a.



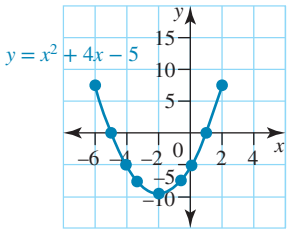
b. $x = 3$ **c.** (3, 4), max **d.** -5

18. a.



b. $x = 1$ **c.** $(1, 2)$, max **d.** -1

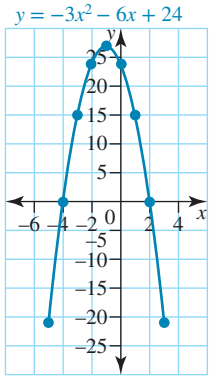
19. a.



b. $x = -2$ **c.** $(-2, -9)$, min

d. -5

20. a. $y = -3x^2 - 6x + 24$



b. $x = -1$ **c.** $(-1, 27)$, max

d. 24

21. a. If the x^2 term is positive, the parabola has a minimum turning point. If the x^2 term is negative, the parabola has a maximum turning point.

b. If the equation is of the form $y = a(x - b)^2 + c$, the turning point has coordinates (b, c).

c. The equation of the axis of symmetry can be found from the x-coordinate of the turning point. That is, $x = b$.

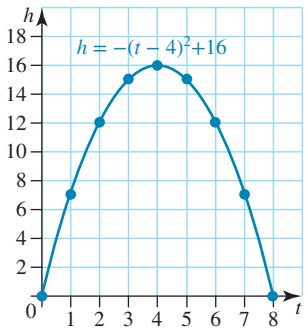
22. C

23. B

24. C

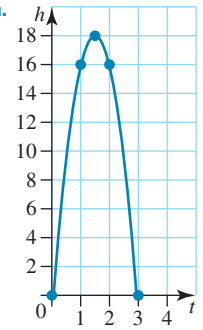
25. A

26. a.



b. i. 16 m **ii.** 8 s

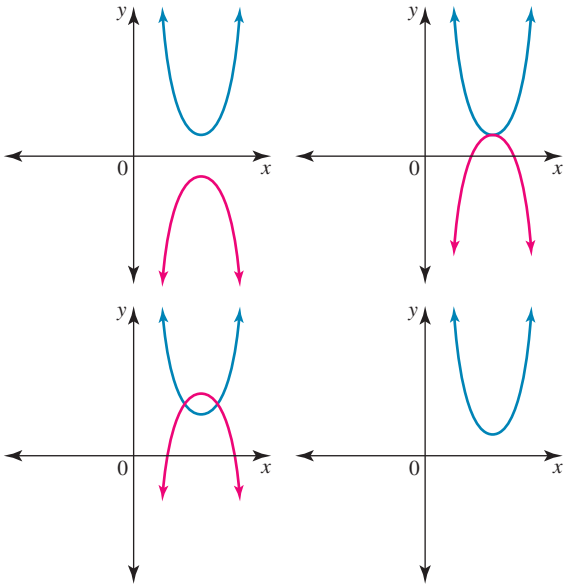
27. a.



b. i. 18 m **ii.** Yes, by 3 m

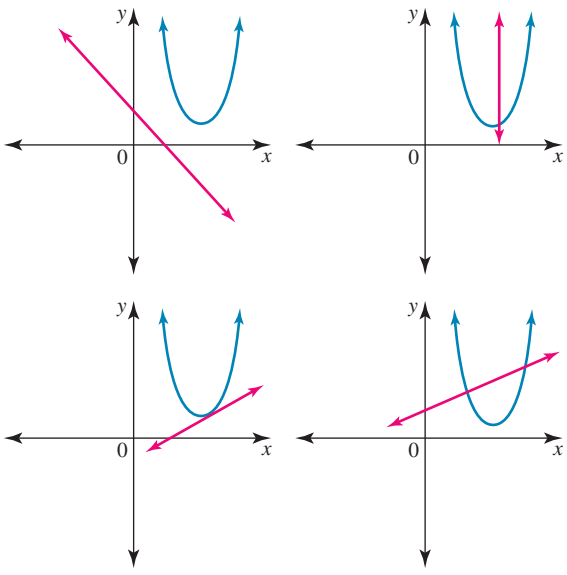
iii. 1.5 s **iv.** 3 s

28. a.



b. An infinite number of points of intersection occur when the two equations represent the same parabola, with the effect that the two parabolas superimpose. For example $y = x^2 + 4x + 3$ and $2y = 2x^2 + 8x + 6$.

c. It is possible to have 0, 1 or 2 points of intersection.

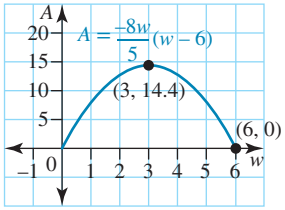


29. a.

w cm	-1	0	1	2	3	4	5	6	7
A cm^2	-11.2	0	8	12.8	14.4	12.8	8	0	-11.2

b. The length of a rectangle must be positive, and the area must also be positive, so we can discard $w = -1, 0, 6$ and 7.

c.



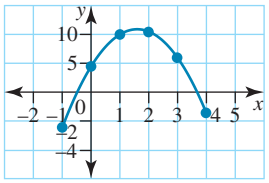
d. The maximum area is 14.4 cm^2.

e. The dimensions of this rectangle are 3 cm $\times$ 4.8 cm.

30. a.

x	-1	0	1	2	3	4
y	-2.4	1.8	4	4.2	2.4	-1.4

b.



c. x cannot equal -1 as this would put the ball behind her; at $x = 4$, the ball is under ground level.

d. The maximum height reached is 4.36 m.

e. The ball will hit the ground 3.688 m from the player.

31. $y = 2x^2 - 3x + 1$

9.3 Sketching parabolas using transformations

1. a. Narrower, TP (0, 0) **b.** Wider, (0, 0)

2. a. Narrower, TP (0, 0) **b.** Narrower, TP (0, 0)
c. Wider, TP (0, 0)

3. a. Wider, TP (0, 0) **b.** Narrower, TP (0, 0)
c. Narrower, TP (0, 0)

4. a. Vertical 3 up, TP (0, 3)
b. Vertical 1 down, TP (0, -1)

5. a. Vertical 7 down, TP (0, -7)

b. Vertical $\frac{1}{4}$ up, TP $\left(0, \frac{1}{4}\right)$

c. Vertical $\frac{1}{2}$ down, TP $\left(0, -\frac{1}{2}\right)$

6. a. Vertical 0.14 down, TP (0, -0.14)
b. Vertical 2.37 up, TP (0, 2.37)

c. Vertical $\sqrt{3}$ up, TP $\left(0, \sqrt{3}\right)$

7. a. Horizontal 1 right, (1, 0)
b. Horizontal 2 right, (2, 0)

8. a. Horizontal 10 left, $(-10, 0)$
b. Horizontal 4 left, $(-4, 0)$

c. Horizontal $\frac{1}{2}$ right, $\left(\frac{1}{2}, 0\right)$

9. a. Horizontal $\frac{1}{5}$ left, $\left(-\frac{1}{5}, 0\right)$

b. Horizontal 0.25 left, $(-0.25, 0)$

c. Horizontal $\sqrt{3}$ left, $\left(-\sqrt{3}, 0\right)$

10. a. (0, 1), max **b.** (0, -3), min

11. a $(-2, 0)$, max **b** (0, 0), min **c** (0, 4), max

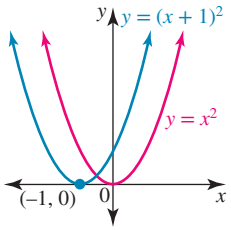
12. a. (0, 0), max **b.** (5, 0), min **c.** (0, 1), min

13. a. Narrower, min **b.** Narrower, max

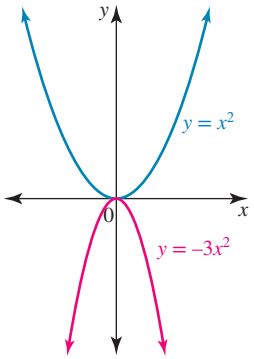
14. a. Wider, min **b.** Wider, max **c.** Narrower, max

15. a. Wider, min **b.** Narrower, min
c. Wider, max

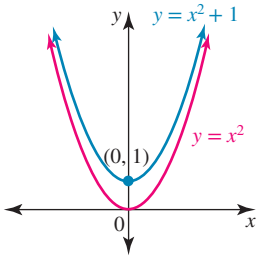
16. a. i. Horizontal translation 1 left
ii. $(-1, 0)$
iii.



b. i. Reflected, narrower (dilation)
ii. (0, 0)
iii.

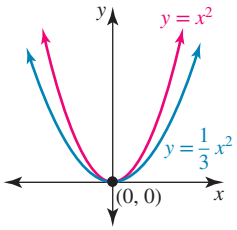


c. i. Vertical translation 1 up
ii. (0, 1)
iii.



17. a. i. Wider (dilation)
ii. (0, 0)

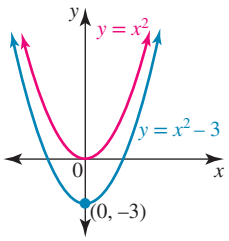
iii.



b. **i.** Vertical translation 3 down

ii. $(0, -3)$

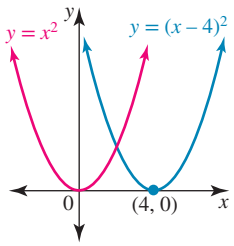
iii.



18. a. **i.** Horizontal translation 4 right

ii. $(4, 0)$

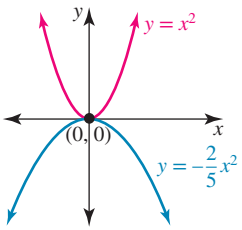
iii.



b. **i.** Reflected, wider (dilation)

ii. $(0, 0)$

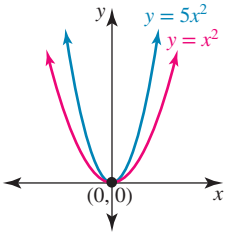
iii.



c. **i.** Narrower (dilation)

ii. $(0, 0)$

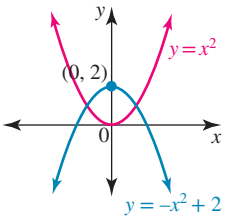
iii.



19. a. **i.** Reflected, vertical translation 2 up

ii. $(0, 2)$

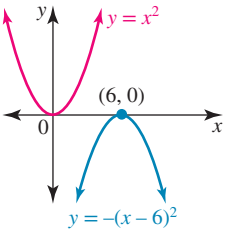
iii.



b. **i.** Reflected, horizontal translation 6 right

ii. $(6, 0)$

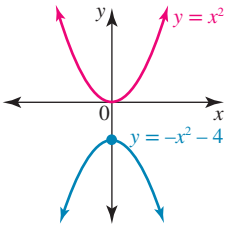
iii.



20. a. **i.** Reflected, vertical translation 4 down

ii. $(0, -4)$

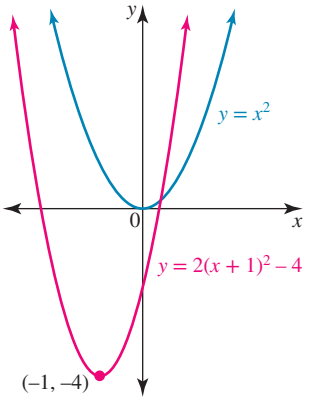
iii.



b. **i.** Narrower (dilation), horizontal translation 1 left, vertical translation 4 down

ii. $(-1, -4)$

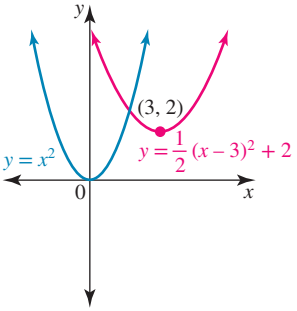
iii.



c. **i.** Wider (dilation), horizontal translation 3 right, vertical translation 2 up

ii. $(3, 2)$

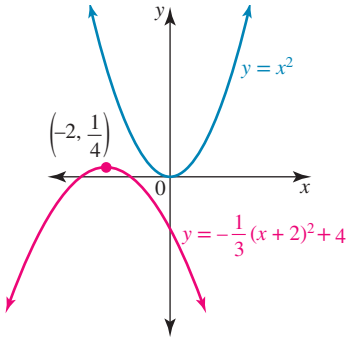
iii.



21. a. i. Wider (dilation), reflected, horizontal translation 2 left, vertical translation $\frac{1}{4}$ up

ii. $\left(-2, \frac{1}{4}\right)$

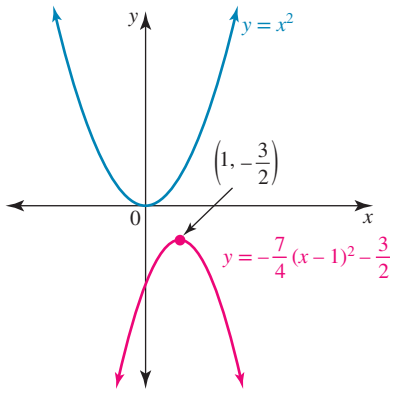
iii.



b. i. Narrower (dilation), reflected, horizontal translation 1 right, vertical translation $\frac{3}{2}$ down

ii. $\left(1, -\frac{3}{2}\right)$

iii.



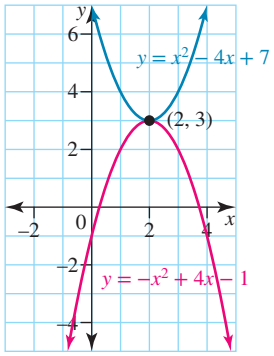
22. a. 10 cm **b.** 5 cm **c.** 5 cm
d. $y = (x - 5)^2$

23. a. 40 m
b. 5 m
c. $h = -\frac{1}{10}(d - 70)^2 + 40$
d. $h = -\frac{3}{10}(d - 70)^2 + 120$, this is a dilation by a factor of 3 from the d axis.

24. a. and **c.**
See figure at the bottom of the page.*
b. 12.25 m **d.** 12.75 m

25. a. $y = -(x - 2)^2 + 3 = -x^2 + 4x - 1$

b.

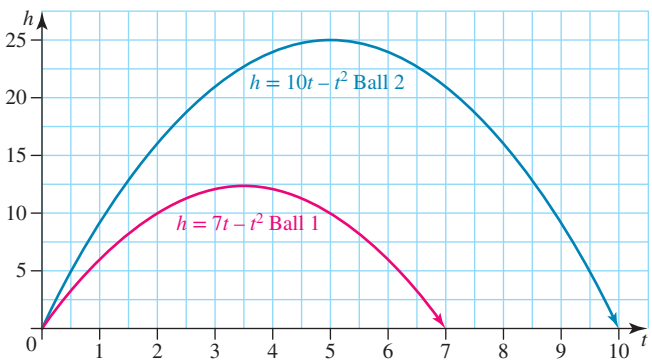


26. a. $(2, -7)$ and $(0, 5)$

b. In the following order: translate 2 units left, translate 7 units up, reflect in the x-axis then dilate by a factor of $\frac{2}{3}$ from the x-axis.

---

*24. a. and c.



27. a. $Y = -(x-3)^2 + 3$

b. Reflected in x-axis, translated 3 units to the right and up 3 units. No dilation.

c. $(3, 3)$

d. See figure at the bottom of the page.*

28. $v = -2h^2 + 40h$, where v is the vertical distance h is the horizontal distance.

9.4 Sketching parabolas using turning point form

1. a. $(1, 2)$, min

b. $(-2, -1)$, min

c. $(-1, 1)$, min

2. a. $(2, 3)$, max **b.** $(5, 3)$, max **c.** $(-2, -6)$, min

3. a. $\left(\frac{1}{2}, -\frac{3}{4}\right)$, min **b.** $\left(\frac{1}{3}, \frac{2}{3}\right)$, min

c. $(-0.3, -0.4)$, min

4. a. i. $(-3, -5)$ **ii.** Min **iii.** Narrower

b. i. $(1, 1)$ **ii.** Max **iii.** Same

5. a. i. $(-2, -4)$ **ii.** Max **iii.** Narrower

b. i. $(3, 2)$ **ii.** Min **iii.** Wider

6. a. i. $(-1, 7)$ **ii.** Max **iii.** Wider

b. i. $\left(\frac{1}{5}, -\frac{1}{2}\right)$ **ii.** Min **iii.** Wider

7. a. $y = (x+1)^2 - 3$, graph iii

b. $y = -(x-2)^2 + 3$, graph i

c. $y = -x^2 + 1$, graph ii

8. a. $y = (x-1)^2 - 3$, graph iii

b. $y = -(x+2)^2 + 3$, graph i

c. $y = x^2 - 1$, graph ii

9. A

10. C

11. B

12. C

13. B

14. a. i. -3 **ii.** $-3, 1$

b. i. 12 **ii.** 2

15. a. i. -18 **ii.** No x-intercepts

b. i. -5 **ii.** $-1, 5$

16. a. i. 4 **ii.** No x-intercepts

b. i. 4

ii. $-3 - \sqrt{5}, -3 + \sqrt{5}$ (approx. $-5.24, -0.76$)

17. a. i. $(4, 2)$

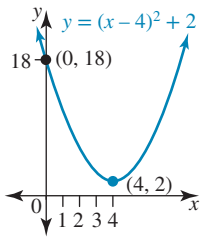
ii. Min

iii. Same width

iv. 18

v. No x-intercepts

vi.



b. i. $(3, -4)$

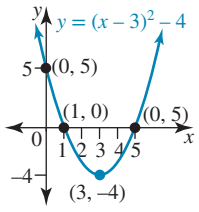
ii. Min

iii. Same width

iv. 5

v. $1, 5$

vi.



c. i. $(-1, 2)$

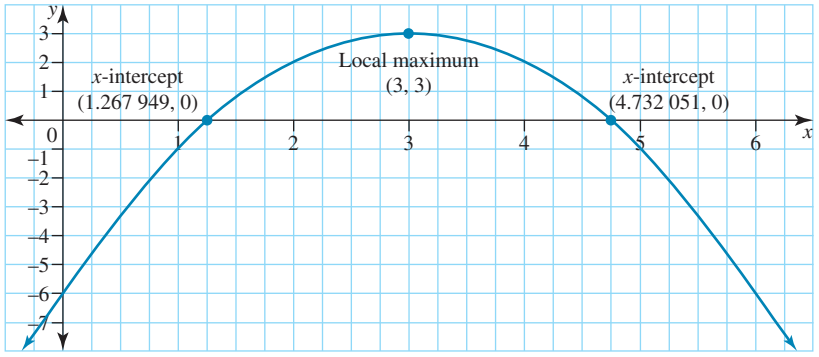
ii. Min

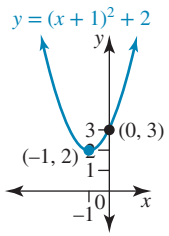
iii. Same width

iv. 3

v. No x-intercepts

*27. d.



vi. 

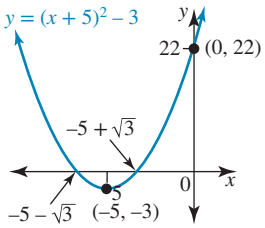
18. a. i. $(-5, -3)$

ii. Min

iii. Same width

iv. 22

v. $-5 - \sqrt{3}, -5 + \sqrt{3}$ (approx. $-6.73, -3.27$)

vi. 

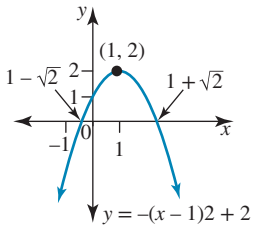
b. i. $(1, 2)$

ii. Max

iii. Same width

iv. 1

v. $1 - \sqrt{2}, 1 + \sqrt{2}$ (approx. $-0.41, 2.41$)

vi. 

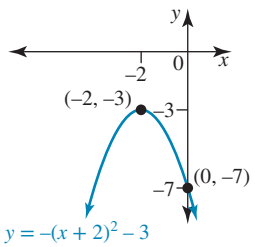
c. i. $(-2, -3)$

ii. Max

iii. Same width

iv. -7

v. No x-intercepts

vi. 

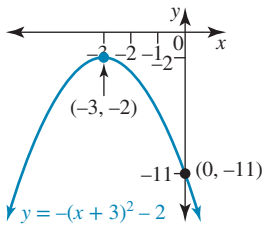
19. a. i. $(-3, -2)$

ii. Max

iii. Same width

iv. -11

v. No x-intercepts

vi. 

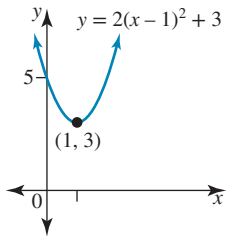
b. i. $(1, 3)$

ii. Min

iii. Narrower

iv. 5

v. No x-intercepts

vi. 

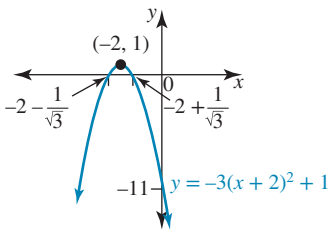
c. i. $(-2, 1)$

ii. Max

iii. Narrower

iv. -11

v. $-2 - \frac{1}{\sqrt{3}}, -2 + \frac{1}{\sqrt{3}}$ (approx. $-2.58, -1.42$)

vi. 

20. a. $2\left(x - \frac{3}{4}\right)^2 - \frac{73}{8} = 0$

b. $x = \frac{3}{4} \pm \frac{\sqrt{73}}{4}$

c. $\left(\frac{3}{4}, -\frac{73}{8}\right)$, minimum

21. a. $y = -\frac{2}{3}(x + 4)^2 + 6$

b. $(-7, 0)$

22. a. $y = -x^2$

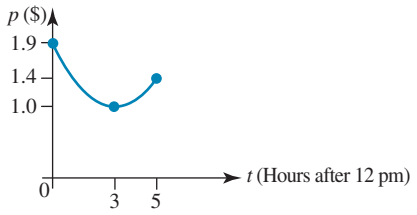
b. $y = 7x^2$

c. $y = (x + 3)^2$

d. $y = x^2 + 6$

e. $y = -\frac{1}{4}(x - 5)^2 - 3$

23. a.



b. $1.90 **c.** $1 **d.** 3 pm **e.** $1.40

24. a. 0.5 m

b. $\left(15 + 4\sqrt{15}\right)$ m

c. Maximum height is 8 metres when horizontal distance is 15 metres.

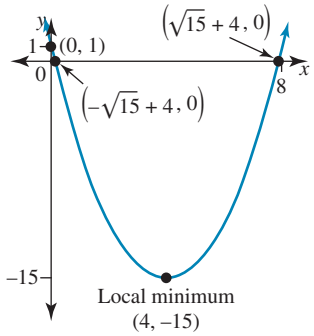
25. a. Sample responses can be found in the worked solutions in the online resources.
An example is $y = (x - 2)^2 + 6$.

b. $y = -\frac{1}{2}(x - 2)^2 + 6$

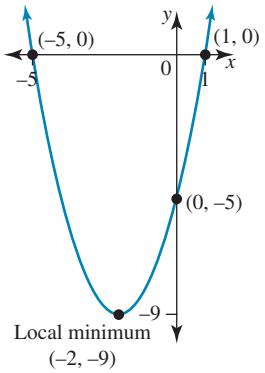
26. a. Sample responses can be found in the worked solutions in the online resources.
An example is $y = (x - p)^2 + q$.

b. $y = \left(\frac{r - q}{p^2}\right)(x - p)^2 + q$

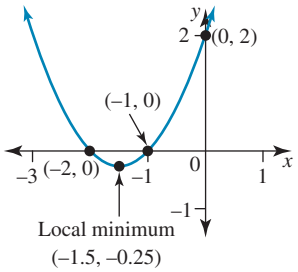
27. a. $y = (x - 4)^2 - 15$



b. $y = (x + 2)^2 - 9$



c. $y = \left(x + \frac{3}{2}\right)^2 - \frac{1}{4}$



28. a. $y = 3(x - 2)^2 - 8$

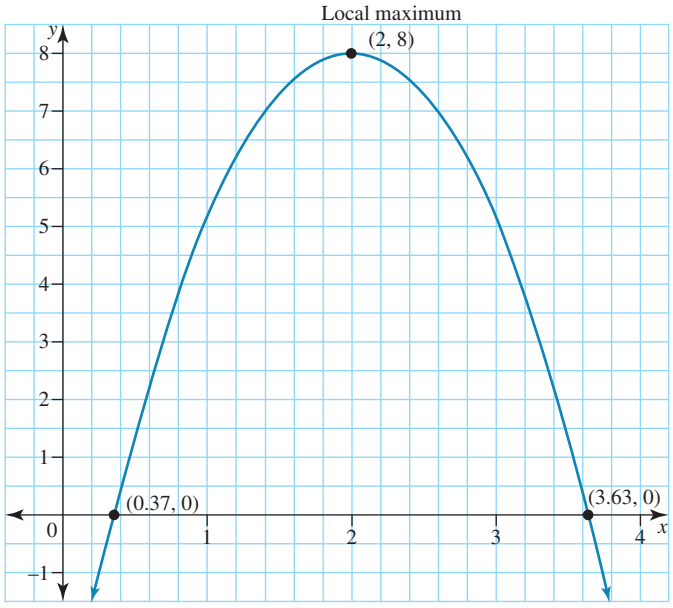
b. Dilated by a factor of 3 parallel to the y-axis or from the x-axis as well as being translated 2 units to the right and down 8 units.

c. $y = -3(x - 2)^2 + 8$

d. See figure at the bottom of the page.*

---

*28. d.

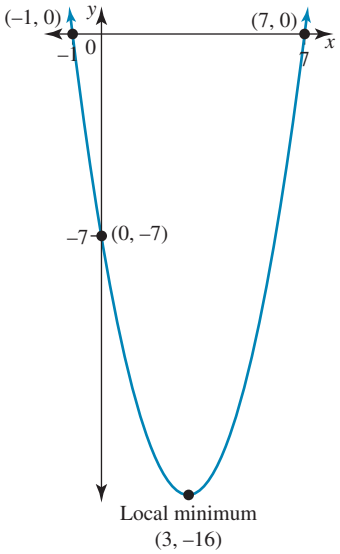


29. a. $y = -\frac{1}{2}(x+3)^2 + 8$

b. In the following order: a dilation of a factor of $\frac{1}{2}$ from the x-axis, a reflection in the x-axis, a translation of 3 units left and 8 units up.

c. x- intercepts: $(-7, 0)$ and $(1, 0)$
y- intercept: $\left(0, \frac{7}{2}\right)$

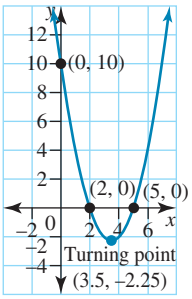
d. The new rule is $y = (x-3)^2 - 16$



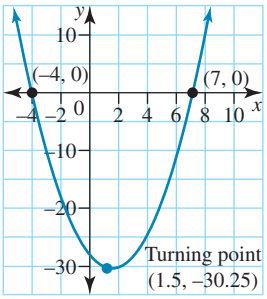
9.5 Sketching parabolas in expanded form

1. You need the x-intercepts, the y-intercept and the turning point to sketch a parabola.

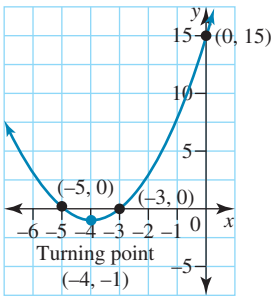
2. a.



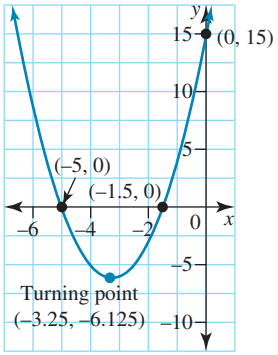
b.



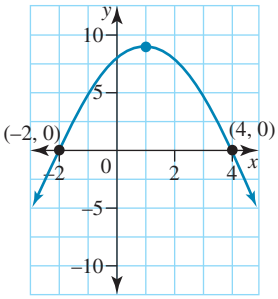
3. a.



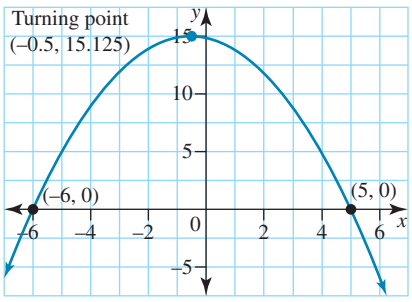
b.



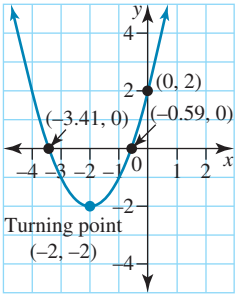
4. a.



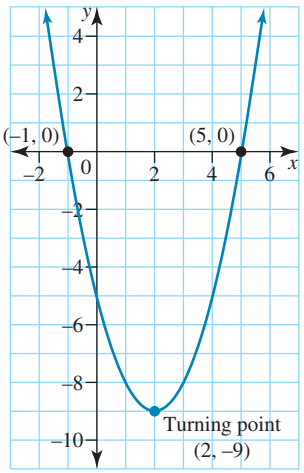
b.



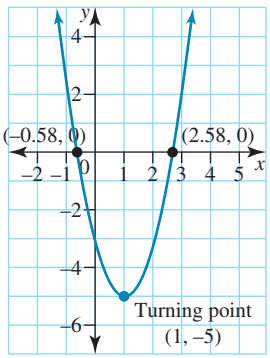
5. a.



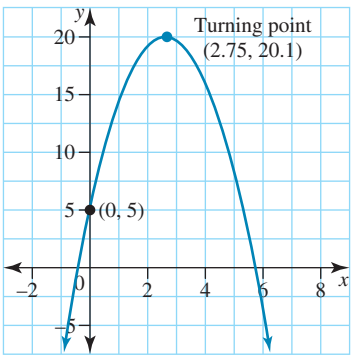
b.


(–1, 0)
(5, 0)
Turning point
(2, –9)

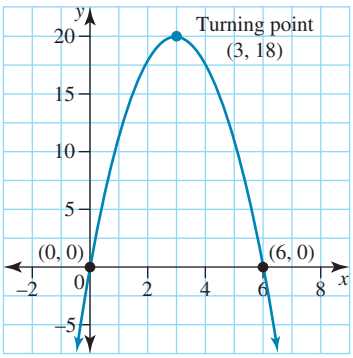
c.


(–0.58, 0)
(2.58, 0)
Turning point
(1, –5)

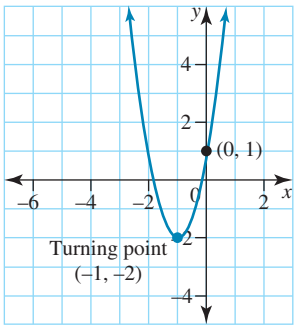
6. a.


Turning point
(2.75, 20.1)
(0, 5)

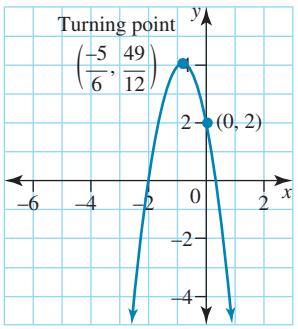
b.


Turning point
(3, 18)
(0, 0)
(6, 0)

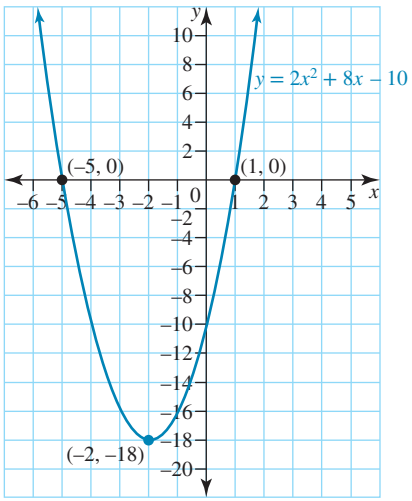
c.


(0, 1)
Turning point
(–1, –2)

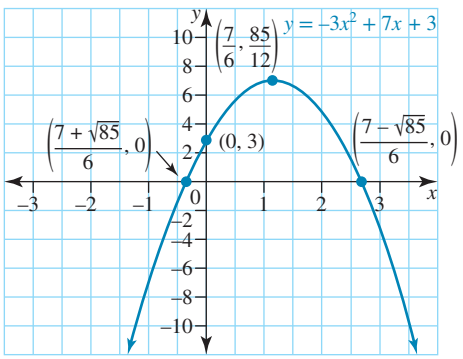
7. a.


Turning point
(–5/6, 49/12)
(0, 2)

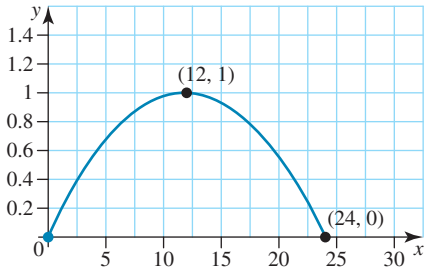
b.


y = 2x² + 8x – 10
(–5, 0)
(1, 0)
(–2, –18)

c.


y = –3x² + 7x + 3
(7/6, 85/12)
((7 + √85)/6, 0)
(0, 3)
((7 – √85)/6, 0)

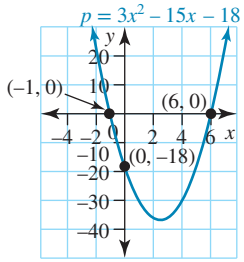
8. a.



b. 24 m

c. 1 m

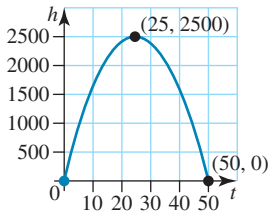
9. a.



b. 6th month

c. The company breaks even at the end of the 6th month. The first month in which a profit is made is the 7th month.

10. a.



b. 0

c. 2500 m

d. 25 seconds

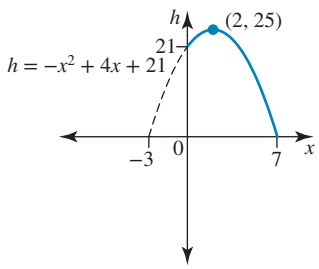
e. 50 seconds

11. 200

12. $-(m+n)$

13. $y = -\frac{1}{2}x^2 + 2x + 9$

14. a.

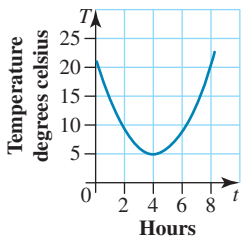


b. 25 m

c. 2 m

d. 7 m

15. a.



b. 21 °C

c. Decreasing

d. Increasing

e. 5 °C after 4 hours

f. 21 °C

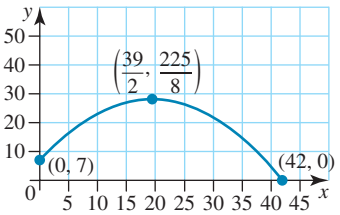
16. a. $a = -\frac{1}{18}, b = \frac{13}{6}, c = 7$

b. 7 metres

c. $\frac{225}{8}$ metres

d. 42 metres

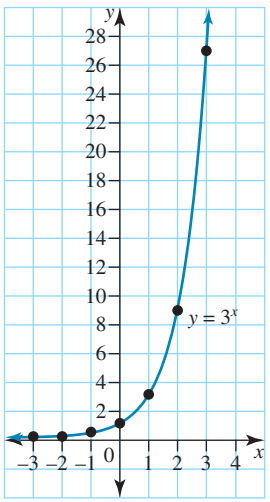
e.



9.6 Exponential graphs

1.

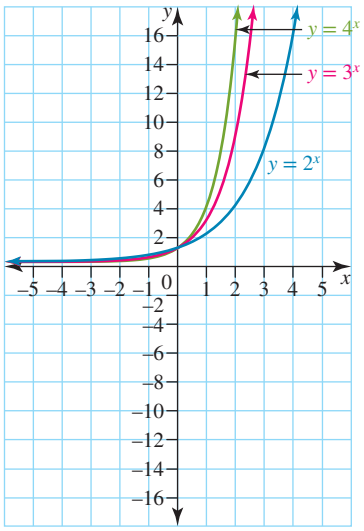
x	-3	-2	-1	0	1	2	3
y	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27



2. a. 2 **b.** 3 **c.** 4

3. a. 10 **b.** a

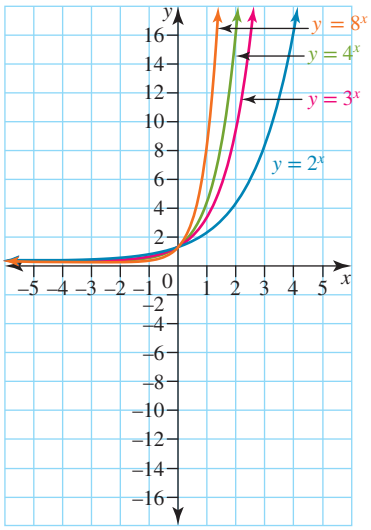
4.



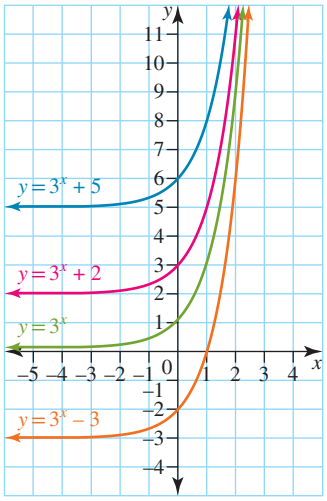
a. The graphs all pass through $(0, 1)$. The graphs have the same horizontal asymptote, $(y = 0)$. The graphs are all very steep.

b. As the base grows larger, the graphs become steeper.

c.



5.



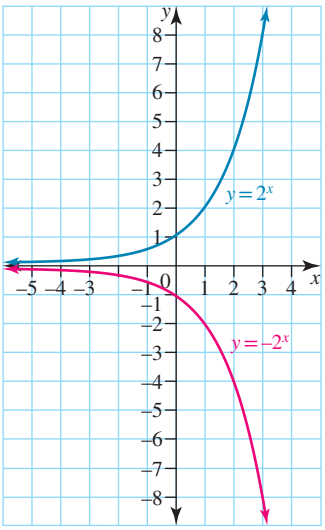
a. The shape of each graph is the same.

b. Each graph has a different y- intercept and a different horizontal asymptote.

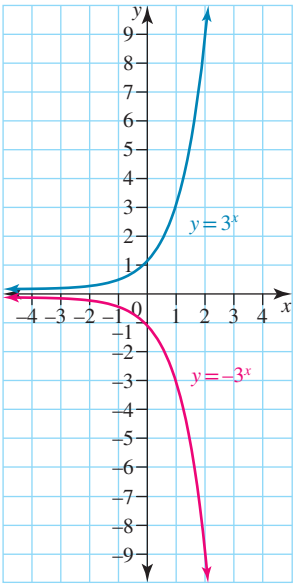
c. **i.** $(0, 11)$

ii. $y = 10$

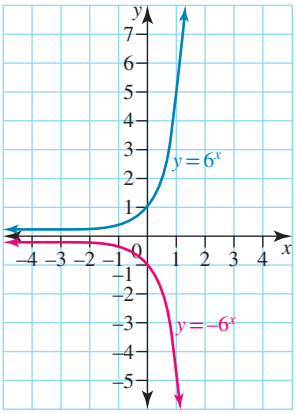
6. a.



b.

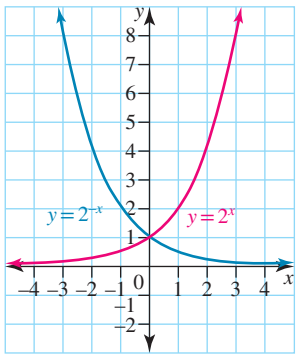


c.

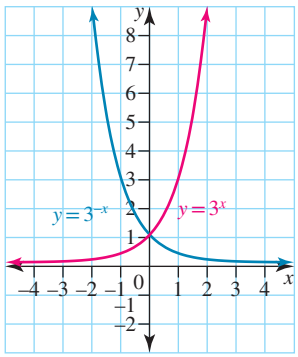


d. In each case the graphs are symmetric about the x-axis.

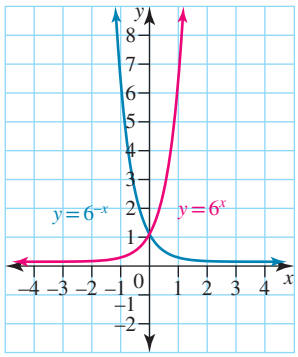
7. a.



b.

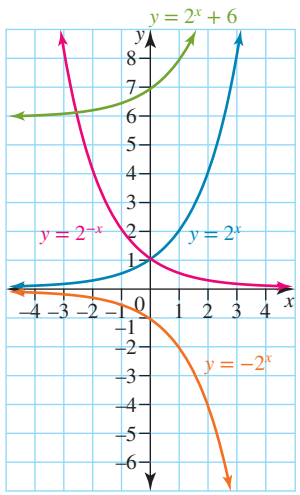


c.

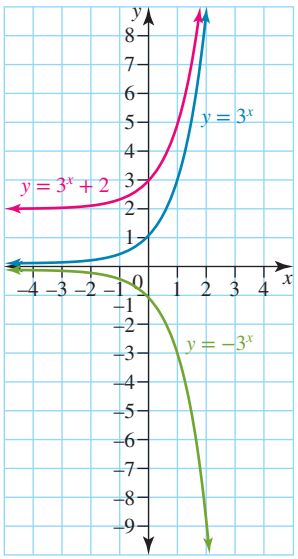


d. In each case the graphs are symmetric about the y-axis.

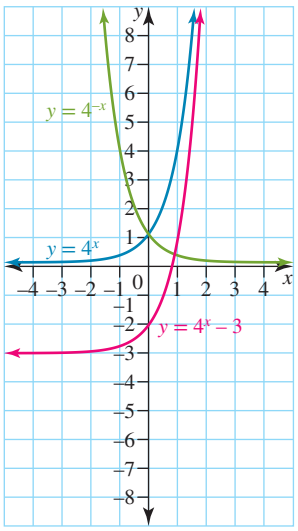
8. a–c.



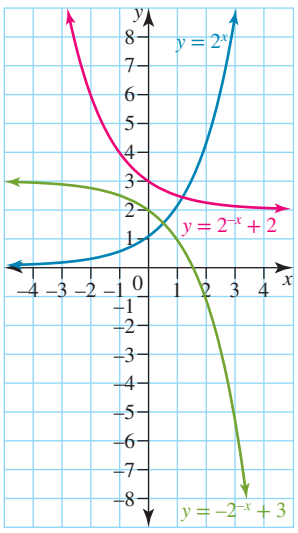
9. a, b



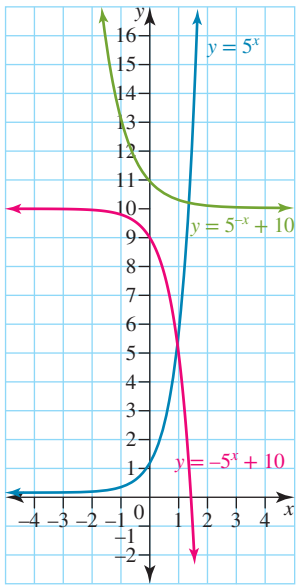
10. a, b



11. a, b



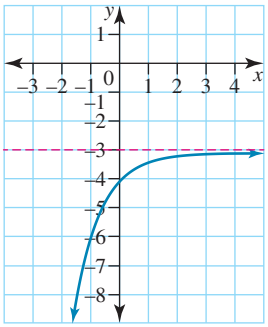
12. a, b



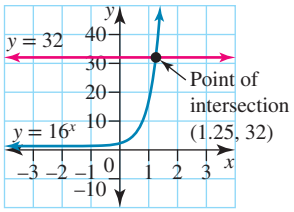
13. a. B **b.** C
c. D **d.** A

14. a. B **b.** D
c. A **d.** C

15.

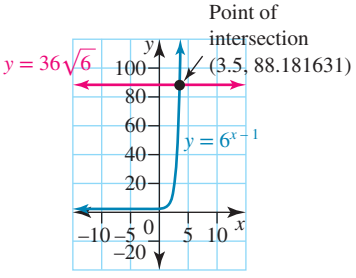


16.



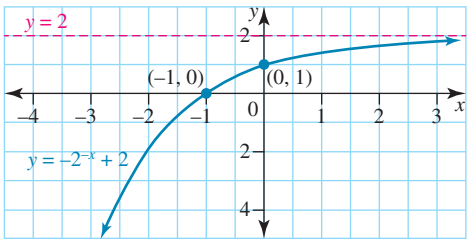
$x = 1.25$

17.



$x = 3.5$

18.



19. a. 10 000
b. i. 1111
ii. 41
iii. 0

20. a. Yes
b. There is a constant ratio of 1.3.
c. 30%
d. 3.26 million
e. 30.26 million

9.7 Inverse proportion

1. a. $s = \frac{k}{t}$ or $t = \frac{k}{s}$ **b.** No

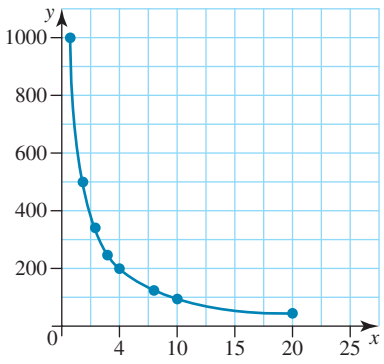
c. $t = \frac{k}{p}$ or $p = \frac{k}{t}$ **d.** $L = \frac{k}{p}$ or $p = \frac{k}{L}$

e. No **f.** $t = \frac{k}{n}$ or $n = \frac{k}{t}$

2. Sample responses can be found in the worked solutions in the online resources.

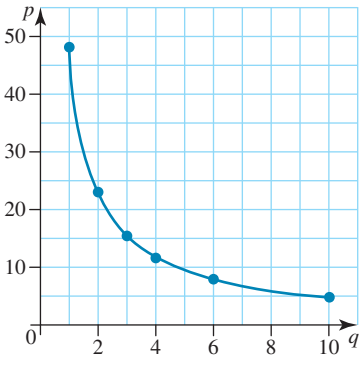
3. a. $k = 1000, y = \frac{1000}{x}$

b.



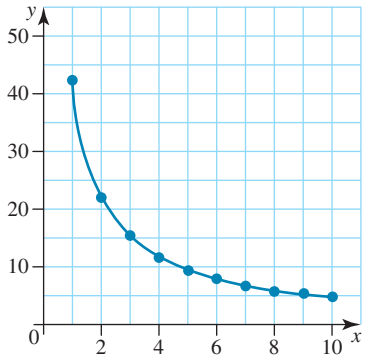
4. a. $k = 48, p = \frac{48}{q}$

b.



5. a. $k = 42, y = \dfrac{42}{x}$

b.



6. a. 4000 **b.** $a = \dfrac{4000}{m}$

c. $20\ \text{m/s}^2$ **d.** $4\ \text{m/s}^2$

7. a. 500 **b.** 2500 pencils

c. 1000 pencils

8. a. 337.5 **b.** 3.97 hours = 3 h 58 min

c. 96.4 km/h **d.** 337.5 km

9. a. 17 500 **b.** $233.33

c. 70 people **d.** $17 500

10. a. 200

b. 1 ampere

c. 13.3 ohms

d. Sample responses can be found in the worked solutions in the online resources.

11. a. k = 15 **b.** $V = 20L$

c. $P = 2.5$ atmospheres

12. The constant of proportionality represents the distance of the trip; therefore, when this value is smaller, the time taken to complete the trip at the same speed is also smaller.

13. a. 10 **b.** $T = \dfrac{10}{P}$

c. 1 hour 40 minutes

14. a. 100.048 **b.** $T = \dfrac{100.048}{V}$ **c.** 9.58 seconds

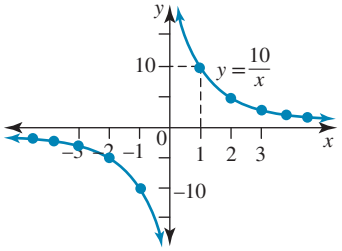
d. 15.00 seconds

15. a. $C = \dfrac{1350}{n}$ **b.** $96.43 per person

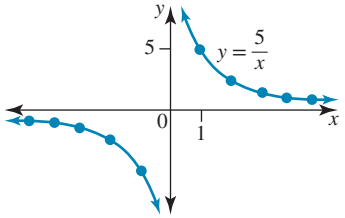
c. $30 per person

9.8 Sketching the graph of a reciprocal

1. See table at the bottom of the page.*

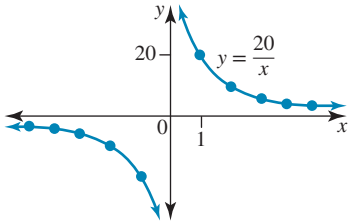


2. a.



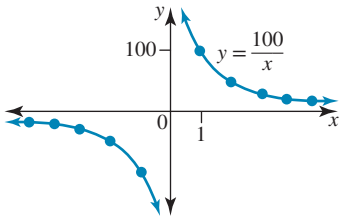
b. $x = 0, y = 0$

3. a.



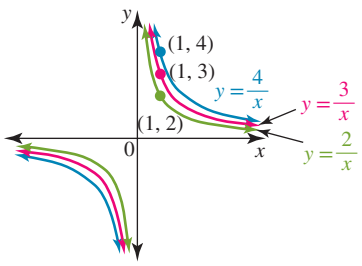
b. $x = 0, y = 0$

4. a.



b. $x = 0, y = 0$

5.

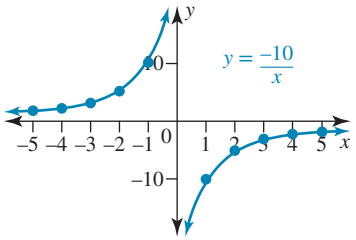


6. It increases the y-values by a factor of k and hence dilates the curve by a factor of k.

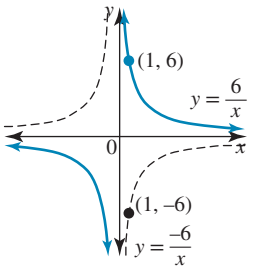
*1.

x	−5	−4	−3	−2	−1	0	1	2	3	4	5
y	−2	−2.5	−3.3	−5	−10	Undef.	10	5	3.3	2.5	2

7.

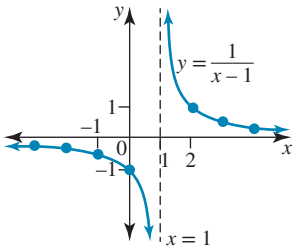


8.



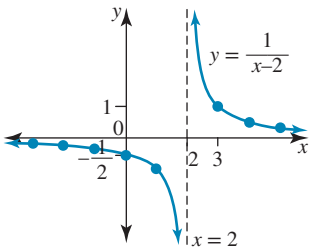
9. The negative reflects the curve $y = \frac{k}{x}$ in the x-axis.

10. See table bottom of the page.*

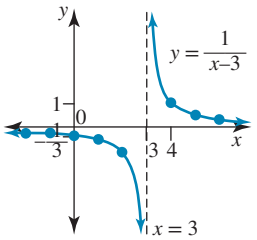


Equation of vertical asymptote is $x = 1$.

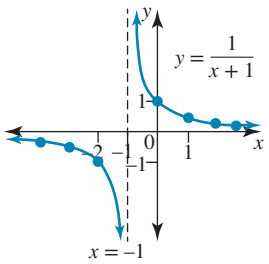
11. a.



b.



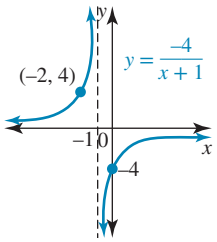
12.



13. The a translates the graph left or right, and $x = a$ becomes the vertical asymptote.

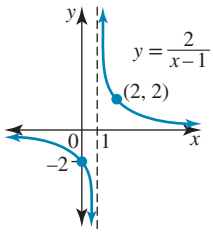
14. a. i. $x = -1, y = 0$

ii.



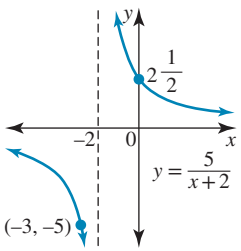
b. i. $x = 1, y = 0$

ii.



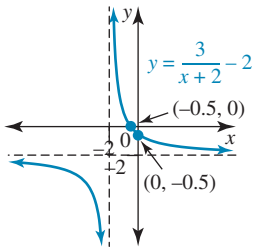
15. a. i. $x = -2, y = 0$

ii.



b. i. $x = -2, y = -2$

ii.

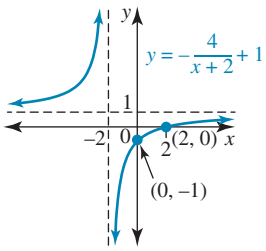


*10. a.

x	−3	−2	−1	0	1	2	3	4
y	−0.25	−0.33	−0.5	−1	Undef.	1	0.5	0.33

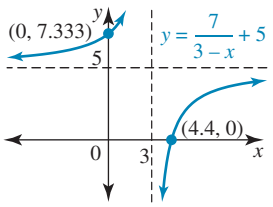
16. a. i. $x = -2, y = 1$

ii.



b. i. $x = 3, y = 5$

ii.



17. Sample responses can be found in the worked solutions in the online resources. Possible answers:

a. $y = \frac{1}{x-3}$

b. $y = \frac{1}{x+10}$

18. a. Any equation of the form $y = \frac{a}{x-2} + 3$.

b. $y = \frac{-14}{x+2} + 4$

19. $y = \frac{-2}{x+3} - 1, x = -3, y = -1$

20. a.

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y	$\frac{1}{4}$	1	4	4	1	$\frac{1}{4}$

See figure at the bottom of the page.*

b. $x = 0$ **c.** $y = -\frac{1}{x^2}$ **d.** $y = \frac{1}{x^2}$

21. a. 100 °C

b. 33.1 °C

c. 17.85 minutes ≈ 18 minutes

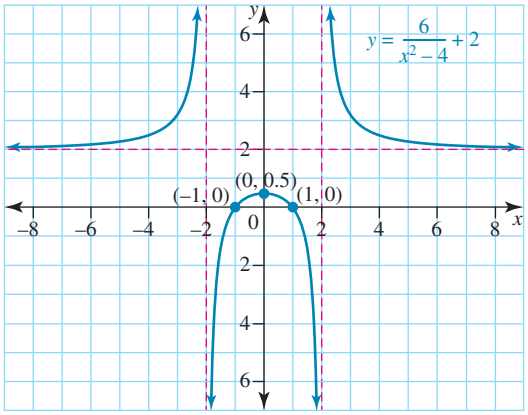
d. No. The horizontal asymptote is 22 °C, so the temperature will never drop below this value.

22. a. y-intercept: $\left(0, \frac{1}{2}\right)$

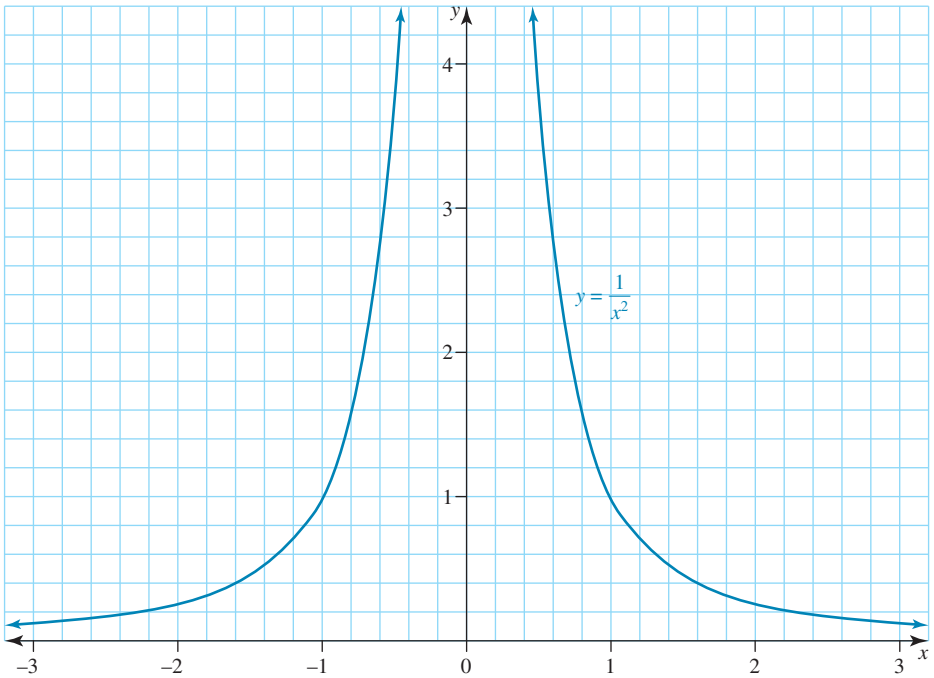
x-intercepts: $(1, 0)$ and $(-1, 0)$

b. $x = 2, x = -2$ and $y = 2$

c.



*20. a.

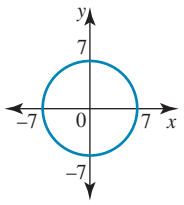


$x = 0, y = 0$

23. $y = -\dfrac{3}{(x+1)^2} + 2, x = -1, y = 2$

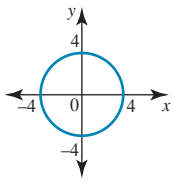
9.9 Sketching the circle

1. a.



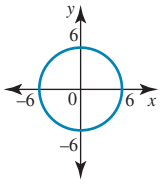
Centre $(0, 0)$, radius 7

b.



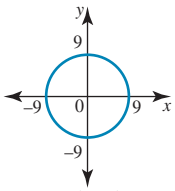
Centre $(0, 0)$, radius 4

2. a.



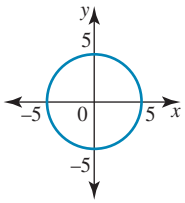
Centre $(0, 0)$, radius 6

b.



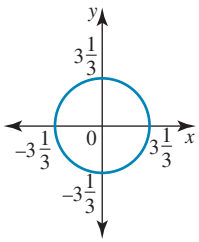
Centre $(0, 0)$, radius 9

3. a.



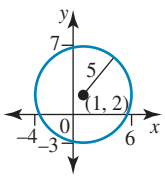
Centre $(0, 0)$, radius 5

b.

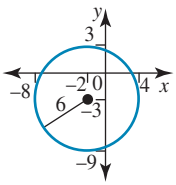


Centre $(0, 0)$, radius $\dfrac{10}{3}$

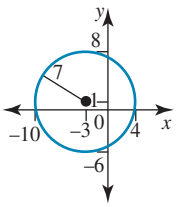
4. a.



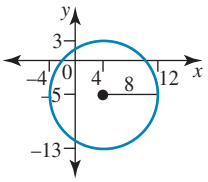
b.



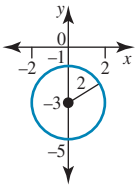
5. a.



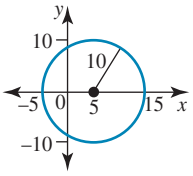
b.



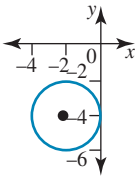
6. a.



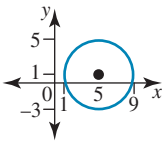
b.



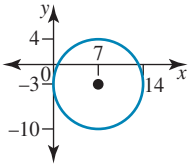
7. a. $(x+2)^2 + (y+4)^2 = 2^2$



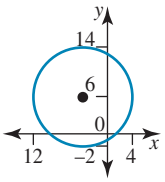
b. $(x-5)^2 + (y-1)^2 = 4^2$



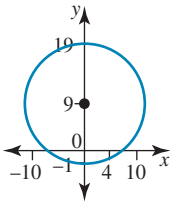
8. a. $(x-7)^2 + (y+3)^2 = 7^2$



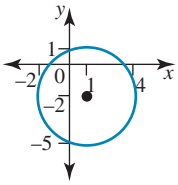
b. $(x+4)^2 + (y-6)^2 = 8^2$



9. a. $x^2 + (y-9)^2 = 10^2$



b. $(x-1)^2 + (y+2)^2 = 3^2$



10. D

11. B

12. D

13. $(x-5)^2 + (y-3)^2 = 16$

14. a. 2 cm, 13.8 cm

b. 3.9 cm/s

15. $(-2, 0)$

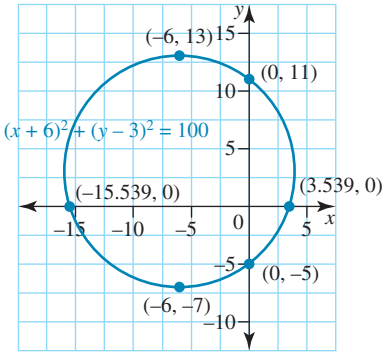
16. a. See the figure at the bottom of the page.*

b. i. $(0.707, 0.707)$ and $(-0.707, -0.707)$

ii. $(0, 0)$ and $(1, 1)$

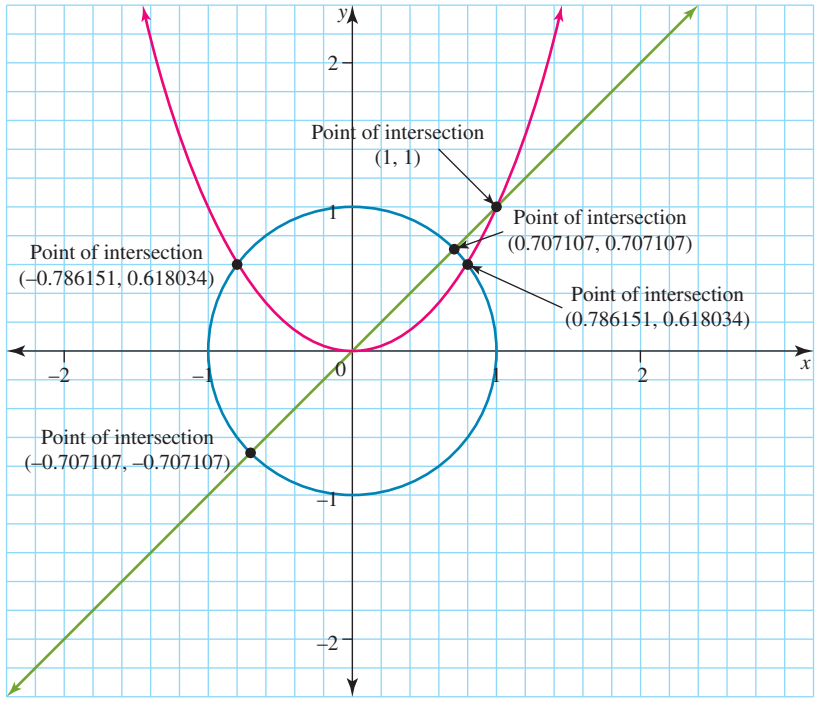
iii. $(0.786, 0.618)$ and $(-0.786, 0.618)$

17.



18. $(-3, 4)$, $(3, 4)$, $(0, 5)$ and $(0, -5)$

*16. a.



19. $(x-1)^2 + (y-1)^2 = 4$ centre at $(1, 1)$ and radius of 2 units.
$(x-4)^2 + (y-1)^2 = 1$ centre at $(4, 1)$ and radius of 1 unit.
The circles intersect (touch) at $(3, 1)$.
See figure at the bottom of the page.*

20. $(x-2)^2 + (y-5)^2 = 4$ centre at $(2, 5)$ and radius of 2 units.

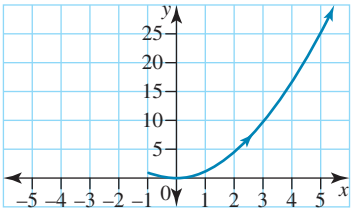
Project

1.

t	x	y
0	0	0
1	1	1
2	2	4
3	3	9
4	4	16
5	5	25

2.

Parametric equations
$x = t$ and $y = t^2$



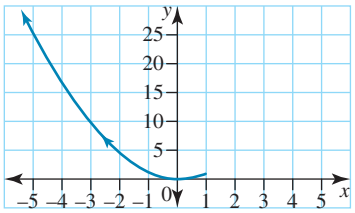
3. Sample response: Yes the graph is different. $Y = x^2$ is a parabola and therefore, the graph appears in the I and the II quadrants. Where as in Q2 the graph only appears in the I quadrant. Other sample responses can be found in the worked solutions in the online resources.

4.

t	x	y
0	1	1
1	0	0
2	−1	1
3	−2	4
4	−3	9
5	−4	16

Sample responses can be found in the worked solutions in the online resources.
$t = 1, 0, -1, -2, -3, -4$.

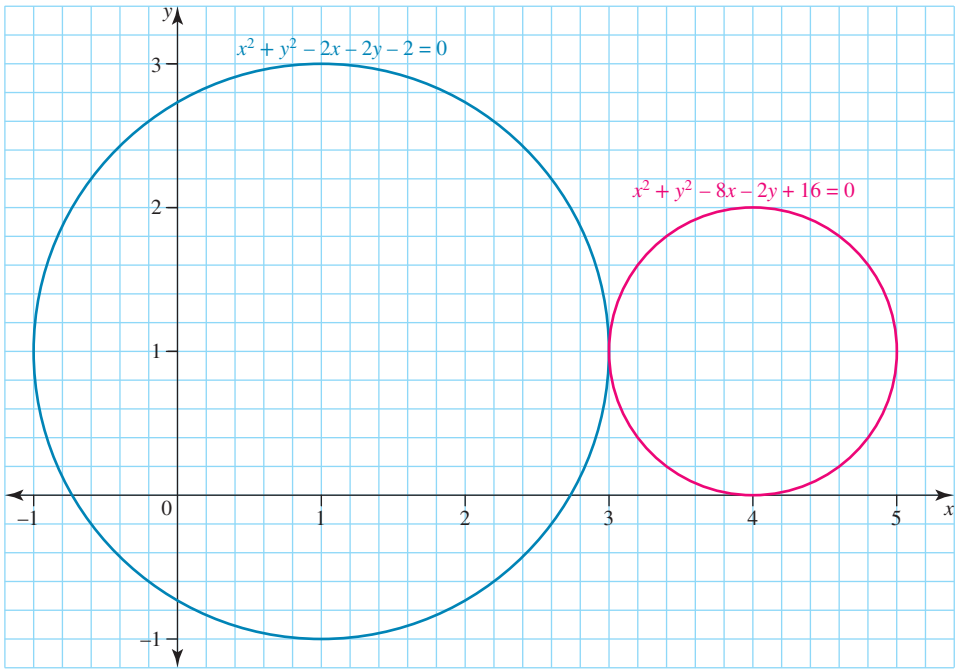
Parametric equations
$x = 1 - t$ and $y = (1 - t)^2$



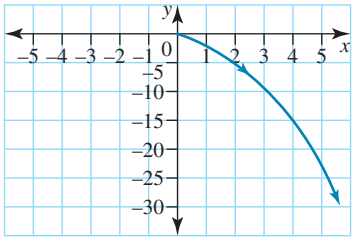
5.

t	x	y
0	0	0
1	1	−1
2	2	−4
3	3	−9
4	4	−16
5	5	−25

*19.

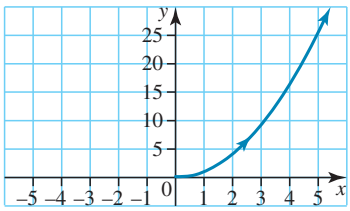


Parametric equations
$x = t$ and $y = -t^2$



6. Sample response: The shape of the graph will be the same as shown in Q5 but it will be in quadrant III (reflection of the graph in Q5). Other sample responses can be found in the worked solutions in the online resources.

Parametric equations
$x = t$ and $y = t^2$



9.10 Review questions

1. D

2. A

3. D

4. A

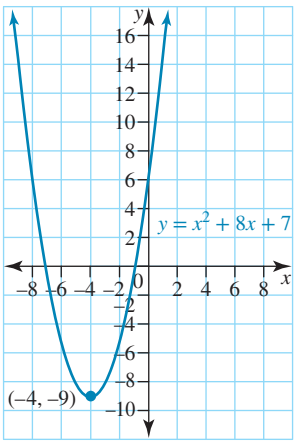
5. C

6. D

7. B

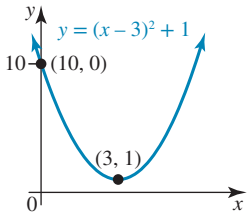
8. a. $(4, -15)$ **b.** $(-2, -9)$

9. See table at the bottom of the page.*

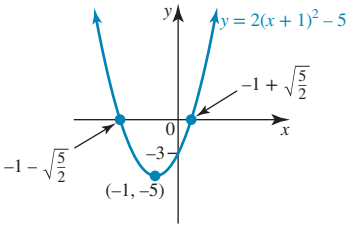


TP $(-4, -9)$; x-intercepts: -7 and -1

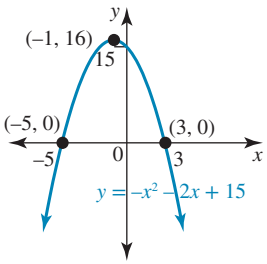
10. a. TP $(3, 1)$; no x-intercepts; y-intercept: $(0, 10)$



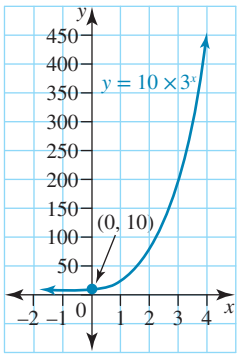
b. TP $(-1, -5)$; x-intercepts: $-1 - \sqrt{\frac{5}{2}}, -1 + \sqrt{\frac{5}{2}}$; y-intercept: $(0, -3)$



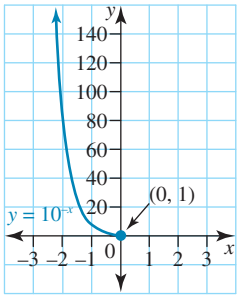
11. TP $(-1, 16)$; x-intercepts: -5 and 3; y-intercept: $(0, 15)$



12.



13.

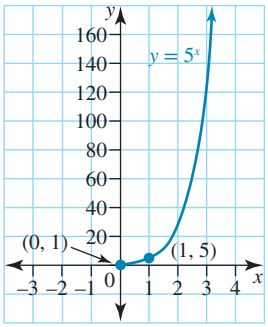


*9.

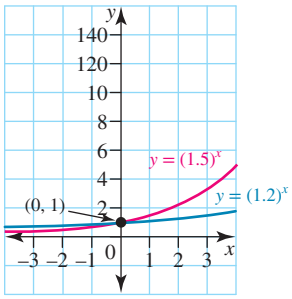
x	−9	−8	−7	−6	−5	−4	−3	−2	−1	0	1
y	16	7	0	−5	−8	−9	−8	−5	0	7	16

14. a. See table at the bottom of the page.*

b.

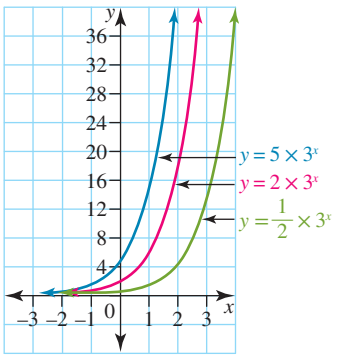


15. a.



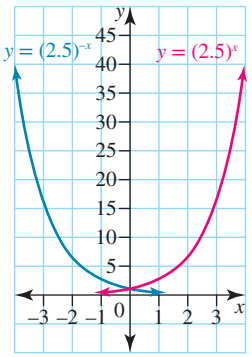
b. Increasing the value of a makes the graph steeper for positive x-values and flatter for negative x-values.

16. a.



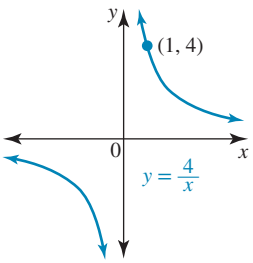
b. Increasing the value of k makes the graph steeper.

17. a.

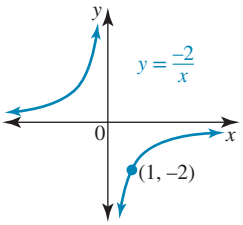


b. Changing the sign of the index reflects the graph in the y-axis.

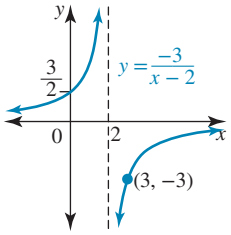
18. a.



b.

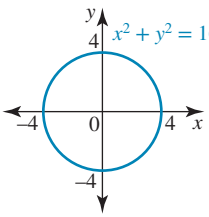


19.

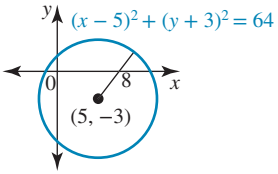


20. Sample responses can be found in the worked solutions in the online resources. Possible answer is $y = \frac{1}{x+3}$.

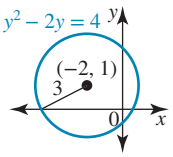
21. a.



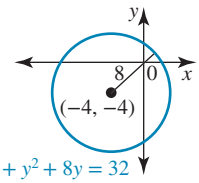
b.



22. a. $x^2 + 4x + y^2 - 2y = 4$



b.



$x^2 + 8x + y^2 + 8y = 32$

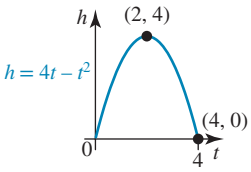
---

*14. a.

x	−3	−2	−1	0	1	2	3
y	0.008	0.04	0.2	1	5	25	125

23. $x^2 + y^2 = 36$

24. a.

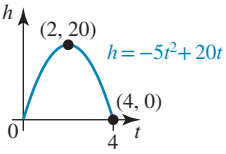


b. 4 m

c. 2 seconds

d. 4 seconds

25. a.



b. 4 seconds

c. 2 seconds $(1 < t < 3)$

d. The ball is never above a height of 20 m.

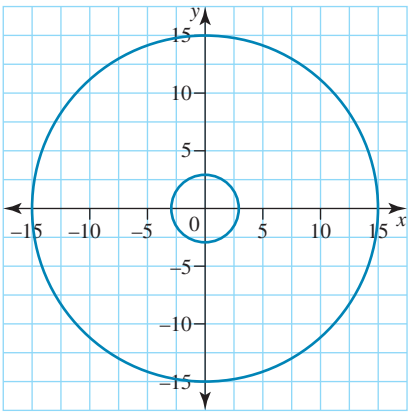
26. a. $[0, 12]$

b. 32 m

c. 11:41 am to 6:19 pm.

27. a. First ripple's radius is 3 cm, second ripple's radius is 15 cm.

b.



c. 2.4 cm/s

d. 1 minute 22.1 seconds after it is dropped.

28. a. 0.14 g/ml **b.** 0.7 g/ml

29. $(-0.77, 0.59), (2, 4), (4, 16)$
By using the zoom function, the graph of $y = 2^x$ rises more steeply than the graph of $y = x^2$, so there are no further points of intersection.

30. a. $P \leq -x^2 + 5x, 0 \leq x \leq 5$

b. 6.25 m

c. i. Sample responses can be found in the worked solutions in the online resources.

ii. Dilation by a factor of 0.56 parallel to the y-axis.

d. 66.7%

31. a. Sample responses can be found in the worked solutions in the online resources.

b. When $x = 0.3$, $b = 10.7$. Therefore if p is greater than 10.7 cm the platform would hit the bridge.